Dollars and Sense: Self-Selection in Financial Literacy Education*

Elaine Shen† Junru Lyu Laila Voss

October 30, 2026

Click **here** for the most recent version

Abstract

Standard Roy models of human capital investment predict that individuals self-select into education based on its expected returns. However, the individuals who stand to benefit most may face frictions that discourage investment. This paper studies selection in the context of financial literacy education and documents an inverse-Roy selection pattern. First, we combine evidence from administrative records on more than 140,000 UC Berkeley undergraduates, linked to credit panel outcomes, with a student financial decision-making survey to document positive selection into financial education: demand for financial education is positively correlated with baseline financial knowledge and pre-class credit outcomes. Second, to test whether there is selection with respect to marginal returns, we move to a controlled online experiment ($n = 1{,}049$). We show that individuals systematically mispredict the returns to education; expected gains are approximately flat across baseline knowledge, while actual gains decline sharply with baseline knowledge. Finally, we experimentally vary instructional format to test whether technology-backed education can change selection patterns by reducing perceived cognitive costs. We find that AI- and video-backed education increase demand for financial education relative to traditional lecture-style instruction but fail to change selection patterns. Our findings highlight a disconnect between educational returns and demand, suggesting that financial education interventions may fail to close knowledge gaps without addressing underlying frictions.

*IRB of record: UC Berkeley Protocol #2024-04-17372 (Observational Data), 2024-03-17249 (Lab Experiment); AEA Trial Registration #AEARCTR-0016395.
We thank Ned Augenblick, Sydnee Caldwell, Stefano DellaVigna, Pat Kline, Ulrike Malmendier, Dmitry Taubinsky, and Chris Walters for their immensely valuable feedback. We further thank Luisa Cefala, Terry Odean, Gautam Rao, Matteo Saccarola, Afras Sial, Sharath Sonti, Karthik Tadepalli, and Carly Urban for helpful comments. We are also grateful to Andrew Eppig and Jennifer Lynn Hogg for their assistance with the data linkage and for providing valuable context on the Registrar and UC-CCP data. Elaine is especially grateful to the Institute for Business and Social Impact (IBSI), the UC Berkeley Experimental Social Science Laboratory, and the National Endowment for Financial Education (NEFE) for generous financial support. All errors are our own.

†Shen: UC Berkeley (elaineshen@berkeley.edu), Lyu: UC Berkeley (jrlyu823@berkeley.edu), Voss: UC Berkeley (lailavoss@berkeley.edu).

1 Introduction

Standard economic models often assume that individuals make investment decisions with complete information and rational expectations; this implies that individuals with the greatest potential benefits are most likely to invest in education (Willis and Rosen, 1979; Card, 1999). Several frictions, however, may prevent individuals who have the *potential* to benefit from education from investing optimally. For example, individuals may overestimate their existing knowledge, misjudge the returns to education, or misperceive the costs associated with acquiring information. These frictions may allow gaps in education to persist, particularly if individuals with lower baseline knowledge are more likely to under-invest in education.

This paper studies self-selection in financial education; that is, which individuals choose to invest in education, how that decision relates to their baseline knowledge and their potential returns, and how easy it is to change selection patterns. We study these questions in the context of financial literacy education, where the costs and benefits of education can be quantified in monetary, welfare-relevant terms. Specifically, we aim to answer three questions: 1) is there selection into financial literacy education on baseline ability; 2) is there selection into financial literacy education on treatment effects; and 3) can education formats that lower the cognitive cost of learning, such as AI-based instruction, increase demand for education and change selection patterns?

We use two complementary approaches to study self-selection into financial literacy education. The first uses administrative data on more than 140,000 UC Berkeley undergraduates to document selection into a large introductory personal finance class. The class is an especially good setting to study financial literacy education; more than 500 students per semester have completed the class on average since 2010, it often has a long wait-list, is very highly rated, and is taught by a highly experienced finance professor. We link student records from the UC Berkeley Registrar's Office to individual-level credit records from the University of California Consumer Credit Panel (UC-CCP), and supplement them with survey measures of financial literacy. We then compare the later credit outcomes of students who take the class with those of students who never take it, reweighting never-takers on observable characteristics with inverse propensity score weights. Because enrollment is not random, we interpret these comparisons as adjusted associations rather than causal effects of the class. Finally, we measure selection on the intensive margin using the time students spend on the course website, which captures how much attention and effort students devote to the class once enrolled, a dimension of selection that is typically unobserved.

To study the mechanisms behind self-selection and to test for selection on treatment effects in a broader adult population, we conduct an online experiment with $n = 1{,}049$ US participants on Prolific, a large online survey platform. In the experiment, each participant completes two sets of incentivized financial decision-making exercises. Between the two exercises, participants state

their willingness to pay for three versions of financial education material, and a randomly selected subset is assigned to complete one of them. We also elicit each participant's beliefs about their own score with and without each version of the material, which allows us to compare perceived and actual gains for the same individuals. We then examine whether the true benefits from education are positively correlated with WTP, as standard models of investment in education would predict, or if individuals *mispredict* the benefits of education. Finally, we vary the format of the material (a traditional lecture, a video, and an AI-based learning assistant) to test whether formats designed to lower the cognitive cost of learning change who demands education.

In both settings, demand for financial education does not fall with baseline knowledge: individuals select into education on their level of knowledge rather than on their gain from it. Students who enroll in a large financial literacy class (UGBA 135) tend to be positively selected on family background, quantitative skills, financial decision-making ability, and credit report outcomes: before the class, students who enroll have average credit scores 7.3 points higher than students who never take it, controlling for graduation cohort, GPA, quantitative coursework, and family background. In the online experiment, willingness to pay is flat in the baseline score: each additional baseline point changes average WTP by \$0.03 (s.e. = 0.07).

The treatment effect of the education material is largest for participants with the lowest baseline scores, yet their willingness to pay (WTP) is no higher than that of participants with the highest baseline scores, for whom the treatment effect is small. The controlled lab setting allows us to measure beliefs and decompose the misprediction. We find that participants' perceived gains from education do not vary systematically with their baseline score (a slope of 0.05 points per baseline point, s.e. = 0.04). In contrast, the actual gain, measured as the difference in endline scores between participants randomly assigned to the education material and those assigned to the control group, declines sharply with the baseline score: it falls from about 1.7 points for participants with baseline scores of 5 or below to about zero for those with the highest possible baseline score. As a result, the belief error, defined as the perceived minus the actual gain, increases by 0.30 points per baseline point. Low-ability participants also overestimate their baseline score, by 0.7 points on average among those scoring 5 or below, while participants scoring 9 or 10 underestimate theirs by 2.5 points. The AI and video formats raise demand relative to the traditional lecture, but they do not change how demand varies with baseline knowledge, potentially suggesting that factors other than cognitive costs limit take-up among those with the largest returns.

This paper relates to a large strand of literature on mistakes in financial decision-making by showing how mistaken beliefs about the returns to financial education can limit information acquisition. Studies have shown that individuals with lower levels of financial literacy, especially low-income individuals and minorities, have lower rates of retirement savings, higher costs of borrowing, and lower rates of investment in high-return assets (van Rooij et al., 2011; Huston,

2012; Lusardi and de Bassa Scheresberg, 2013; Mitchell and Lusardi, 2015; Lusardi and Mitchell, 2023). Despite the seemingly large benefits of financial literacy, several studies have pointed out difficulties encouraging participation in financial education (Bruhn et al., 2014; Lara Ibarra et al., 2017). We show that perceived gains from financial education are largely unrelated to baseline knowledge, whereas actual gains decline with it, so demand does not concentrate among the individuals with the largest returns.

Our findings also relate to the literature on selection in education and the difficulty targeting educational resources to individuals with the greatest potential returns (Willis and Rosen, 1979; Card, 1999; Walters, 2018). Bhattacharya et al. (2012) document patterns of positive selection into free investment advice. Lusardi et al. (2017) predict that higher-income, higher-literacy individuals may derive the greatest expected benefit from certain financial education resources. Gemmo et al. (2023) find a similar pattern while studying take-up of an investment information resource. Many of these papers study advice or education related to portfolio allocation decisions. We focus instead on a simple, foundational financial concept, compound interest, and find that treatment effects are larger for individuals with lower baseline knowledge. Our results highlight that selection patterns may vary depending on the nature of the educational investment.

Finally, our selection results may help explain the mixed results surrounding studies that measure the efficacy of financial literacy education (Willis, 2011; Hastings et al., 2013; Fernandes et al., 2014; Miller et al., 2015; Urban et al., 2020; Kaiser et al., 2022). Self-selection on attention may limit the effectiveness of financial literacy education even when it is mandatory: if the students who engage most with a required class are those with the highest baseline knowledge, the class may have limited effects for students with the lowest baseline knowledge. Because we randomize assignment after eliciting demand, we can estimate treatment effects for participants with both high and low willingness to pay. The treatment effect does not rise with willingness to pay (0.00 points per dollar, s.e. = 0.03), so demand does not sort the participants who would benefit most into education. The results suggest the true potential benefits of financial literacy education could be larger if selection patterns were corrected.

The selection patterns we identify may also have distributional consequences, especially as states increasingly adopt high school financial literacy mandates and as financial literacy education becomes more prevalent on college campuses (Luedtke and Urban, 2023). If students with less prior knowledge invest less time and effort in financial education, then publicly funded programs, including mandates, may not be as effective.[1]

This paper proceeds as follows. Section 2 provides a simple Roy selection conceptual framework to fix ideas and generate predictions. Section 3 describes the registrar and credit bureau data

[1]As of June 2025, at least 29 US states either currently have or plan to adopt a financial literacy mandate that requires students to take a financial literacy class before graduating from high school.

and documents selection on observables. We also use the credit bureau data to show that the adjusted credit-score gap between takers and never-takers varies with baseline characteristics such as quantitative major, underrepresented minority status, and pre-class financial strength, a pattern consistent with, but not proof of, selection on treatment effects. Section 4 describes the online experiment, tests for selection on treatment effects, and examines the role of beliefs and costs. Finally, Section 5 concludes with a discussion of areas for future research.

2 Conceptual Framework

Three patterns organize the paper. The gain from the education material is largest for participants with the lowest baseline scores; willingness to pay for the material rises with the baseline score; and formats that lower the cost of the material raise demand at every baseline score without changing which participants demand it. This section states the model that connects the three. It is the generalized Roy model of selection into treatment (Roy, 1951; Willis and Rosen, 1979; Heckman and Vytlacil, 2005; Heckman et al., 2006) with one change: participants choose on elicited beliefs about their gain rather than on the gain itself. The model defines the objects the empirical sections estimate and states the predictions they test.

2.1 Outcomes

Participant i has financial knowledge K_i. The baseline score $Y_{i,0}$, the number of correct decisions out of 10 before the education stage, measures it with noise: $Y_{i,0} = K_i + \varepsilon_{i0}$. Let $Y_i(1)$ and $Y_i(0)$ be the participant's endline scores with and without the education material. Without the material, knowledge does not change between the two assessments, so $Y_i(0) = K_i + \varepsilon_{i1}$, with $\mathbb{E}[\varepsilon_{i0}] = \mathbb{E}[\varepsilon_{i1}] = 0$. In expectation, the endline score without education therefore equals the baseline score: $\mathbb{E}[Y_i(0) \mid K_i] = \mathbb{E}[Y_{i,0} \mid K_i] = K_i$. Conditional on the observed baseline score, the two differ by regression to the mean, so the empirical sections estimate the outcome without the material from the control group rather than from the baseline; in the control group, the mean score moves from 6.48 to 6.80. The observed endline score is $Y_{i,1} = T_i Y_i(1) + (1 - T_i) Y_i(0)$, where T_i indicates random assignment to the material. The gain from the material is

$$\tau_i = Y_i(1) - Y_i(0), \qquad \tau(y) \equiv \mathbb{E}[\tau_i \mid Y_{i,0} = y]. \tag{1}$$

The three formats teach identical content, so the gain does not depend on the format. Section 4.2 cannot reject equal treatment effects across formats, which is consistent with this restriction. The average treatment effect τ in Equation (11) is the mean of τ_i, and the coefficient ϕ in the same equation is the slope of $\tau(y)$.

Whether $\tau(y)$ rises or falls with the baseline score is the question of whether existing knowledge

and the material are complements or substitutes in the production of skill (Cunha and Heckman, 2007). Education in investing and portfolio choice is complementary with wealth and financial sophistication: the return to learning rises with the amount of capital the knowledge is applied to (Lusardi et al., 2017). Material that teaches a fixed body of content is a substitute for existing knowledge: a participant gains only from the part of the content they do not already know, so participants who know more gain less and $\tau'(y) < 0$. The Rule of 72 material is of the second kind.[2] The experiment estimates the sign and size of $\tau'(y)$ from the randomization rather than assuming them.

2.2 Beliefs

Before random assignment, participant i reports an expected endline score without the education material, $\hat{Y}_i(0)$, and an expected endline score with each format m, $\hat{Y}_i(m)$, where m is the AI, video, or lecture format. The perceived gain from format m is

$$\hat{\tau}_{im} = \hat{Y}_i(m) - \hat{Y}_i(0). \tag{2}$$

The four forecasts are not incentivized (Section 4.1). Because the perceived gain is the difference between two forecasts, a tendency to report every forecast too high or too low cancels; only an error that differs between the forecast with the material and the forecast without it enters $\hat{\tau}_{im}$. Noise in the forecasts attenuates the estimated relationships between the perceived gain and willingness to pay, and Section 4.2 treats the estimates accordingly. The belief error separates into two parts, each measured in the experiment:

$$\hat{\tau}_{im} - \tau_i = \Big(\hat{Y}_i(m) - Y_i(1)\Big) - \Big(\hat{Y}_i(0) - Y_i(0)\Big). \tag{3}$$

The first term is the error in the forecast of the educated outcome. The second is the error in the forecast of the uneducated outcome: a negative value means the participant expects to do worse without the material than they would, which raises the perceived gain one for one. Section 4.2 measures both terms at the level of conditional means: the second from the control group and the first from the treated group.

Separately, participants also misjudge their baseline score, which does not enter the perceived gain. If participants observe their own ability with noise, and the noise is larger for participants who know less, Bayesian updating toward the population mean produces forecasts that rise less than one for one with the true score. Participants below the mean then overestimate their baseline

[2] In the experiment the score is bounded at 10 and the material teaches a single skill, so $\tau'(y) < 0$ holds mechanically. Appendix A.1.1 writes the lab as a special case in which the material closes a share of the gap between the participant's score without education and the maximum score. Nothing in this section depends on the bound.

score and overestimation is largest for individuals with the lowest baseline scores. Appendix A.1.2 states this model and its two further implications, that the dispersion of self-assessment errors falls with the baseline score and that stated confidence predicts accuracy more closely for high scorers, and lists the tests.

2.3 Choice

Payment for the education material is made in the same state of the world as the endline bonus, so one point on the endline score is worth one dollar to the participant. Participant i's willingness to pay for format m and the decision to buy it at price p are

$$W_{im} = \hat{\tau}_{im} - c_{im}, \qquad D_{im} = \mathbb{1}\{W_{im} \geq p\}, \tag{4}$$

where c_{im} is the dollar cost to participant i of the time and effort format m requires. The cost is negative for a participant who enjoys the material. Because W_{im} and $\hat{\tau}_{im}$ are both elicited, the cost $c_{im} = \hat{\tau}_{im} - W_{im}$ is observed for every participant and format.

Equation (4) nests two benchmarks. In the Roy model, participants know their potential outcomes, so $\hat{\tau}_{im} = \tau_i$ and participants choose on the true gain net of cost. Under rational expectations, $\hat{\tau}_{im} = \mathbb{E}[\tau_i \mid \mathcal{I}_i]$ for some information set $\mathcal{I}_i$ that contains the baseline score, so participants choose on an unbiased forecast of the gain. The model in this paper leaves $\hat{\tau}_{im}$ unrestricted and measures it.

2.4 Predictions

The conceptual framework yields three predictions. Prediction 1 concerns selection across baseline scores, Prediction 2 concerns selection on the gain among participants with the same baseline score, and Prediction 3 concerns the education formats.

Prediction 1: selection on the baseline score. Let $\beta(y) = \mathbb{E}[\hat{\tau}_{im} - \tau_i \mid Y_{i,0} = y]$ be the mean belief error and $\bar{c}(y) = \mathbb{E}[c_{im} \mid Y_{i,0} = y]$ the mean cost at baseline score y. Taking the conditional mean of Equation (4),

$$\mathbb{E}[W_{im} \mid Y_{i,0} = y] = \tau(y) + \beta(y) - \bar{c}(y). \tag{5}$$

In the Roy benchmark and under rational expectations, $\beta(y) = 0$ at every y. If the mean cost is the same at every baseline score, mean willingness to pay then has the same slope as the mean gain: it falls with the baseline score when the material is a substitute for existing knowledge and rises when the material is a complement. A willingness to pay that is flat or rising in the baseline score while $\tau'(y) < 0$ requires the mean belief error to rise with the baseline score, the mean cost to fall with it, or both.

One pattern of beliefs produces a rising belief error without any further assumption. If participants perceive the material as adding a roughly fixed number of points wherever they start, the mean perceived gain is flat in the baseline score while the mean gain declines, so $\beta'(y) > 0$: beliefs become more optimistic relative to the gain as the baseline score rises, so that low scorers underestimate their gain relative to high scorers. Section 4.2 tests this pattern by comparing two slopes in the baseline score, the slope of $\tau(y)$ from the randomization and the slope of the mean perceived gain $\mathbb{E}[\hat{\tau}_{im} \mid Y_{i,0} = y]$ from the belief data. Equation (5) then reports how much of the slope of willingness to pay each of the three terms accounts for.

Prediction 2: selection on the gain for a given baseline score. Prediction 1 compares participants across baseline scores. Prediction 2 asks whether, among participants with the same baseline score, willingness to pay also tracks the part of the gain that the baseline score does not predict. Among participants with the same baseline score, the Roy benchmark implies

$$\mathrm{Cov}(\tau_i, W_{im} \mid Y_{i,0} = y) = \mathrm{Var}(\tau_i \mid Y_{i,0} = y) - \mathrm{Cov}(\tau_i, c_{im} \mid Y_{i,0} = y). \tag{6}$$

The covariance is positive if the gain varies across participants with the same baseline score and the cost is uncorrelated with the gain. More generally, the covariance is positive whenever the variance of the gain exceeds its covariance with the cost. Given either condition, participants with larger gains should, on average, be willing to pay more than participants with the same baseline score and smaller gains. Under rational expectations, $\mathbb{E}[\tau_i \mid \hat{\tau}_{im}, Y_{i,0}] = \hat{\tau}_{im}$: the gain rises one for one with the perceived gain. The coefficient κ in Equation (12), the change in the treatment effect for each additional dollar of willingness to pay holding the baseline score fixed, tests the first statement. Replacing willingness to pay with the perceived gain in the same specification tests the second. Both tests are identified because assignment to the material did not depend on willingness to pay or on beliefs, so the gain is estimated at every level of both.

Prediction 3: education formats. The gain is common across formats, so the premium participant i places on format m over the lecture is

$$W_{im} - W_{iL} = \left(\hat{\tau}_{im} - \hat{\tau}_{iL}\right) - \left(c_{im} - c_{iL}\right), \tag{7}$$

where L denotes the lecture. The premium consists of a belief term and a cost term and contains no true gain. A premium that is the same at every baseline score raises willingness to pay at every baseline score and leaves the slope of willingness to pay in the baseline score unchanged: the format brings in additional participants of the same kind. A format changes which participants select in only if its premium is larger for low scorers than for high scorers. Section 4.2 tests whether the slope of willingness to pay in the baseline score is equal across formats, and Section 4.2.3 compares the baseline scores and gains of the participants the AI and video formats bring in with those of

the rest of the sample.

3 Observational Data

3.1 Data and Empirical Strategy

We use two complementary approaches to understand the mechanisms behind self-selection in financial education. The first combines observational data from three sources to document selection into a large personal finance class, and to compare the later credit outcomes of students who do and do not take it, at UC Berkeley: Intro to Personal Finance (UGBA 135). The class is taught by Professor Terrance Odean and enrolled about 750 to 800 students per semester at the end of term in 2024 and 2025. The class is highly rated, available to any UC Berkeley student, and has no finance or math prerequisites. The class has been offered since Fall 2008, as UGBA 196 and UGBA 96 before it became UGBA 135 in Fall 2019, and is often fully enrolled.

To build the dataset, we combine observational data from three sources: 1) course records from the UC Berkeley Registrar's Office, 2) credit bureau data from the University of California Consumer Credit Panel (UC-CCP), and 3) survey data from a personal finance survey fielded through the Associated Students of the University of California (ASUC).

Data from the Registrar's office includes anonymized individual-level degree, admissions, and course data for all undergraduate entry cohorts from 2007-2025.[3] This includes college application data (parental education level, the applicant's self-reported family annual income, University of California Application Fee Waiver status, and intended major), degree date (major/GPA at graduation, graduation date), and demographic data (gender, race/ethnicity, and age), transfer student status, course enrollment data, and student's GPA for each semester.

We also measure the attention and effort students devote to the class, although these are chosen by the student. We use the total amount of time (in minutes) that each student spends on the course website as a proxy for attention and effort. Notably, time spent on the course website is an especially relevant measure of attention for this class: the weekly homework assignment for the class involves watching pre-recorded videos and completing online multiple choice quizzes on the course website. The instructors also regularly post recordings of the in-class lectures, lecture notes, and budgeting and other financial resources on the course website.

To measure financial literacy with measures similar to those in our online experiment, we partnered with the ASUC Financial Literacy Task Force to survey UC Berkeley students. The survey was fielded in two waves, from November 2024 to March 2025 and from late April to early May 2025. Respondents were entered into a drawing for gift cards that did not depend on their

[3]Transfer student data is from 2016-2025.

answers. Our sample has 667 students, keeping each student's first response. In the survey we measure financial knowledge and stress and attitudes towards personal finances. To measure financial knowledge, we ask the Big Three financial literacy questions and have students answer three hypothetical decision-making questions. Finally, we ask students for their hypothetical willingness to pay for UGBA 135. Students who plan to take the class report how much they would pay for an identical UC Berkeley Extension course if they could not get a spot; students who do not plan to take it report how much they would pay to waive the class if it were a graduation requirement, which we record as a negative willingness to pay.

The hypothetical decision-making questions ask students to make decisions about investing (index funds), debt, and compound interest in scenarios where there is a financially dominant choice. Students are told to maximize the amount of money they have in the scenario. These scenarios give us a measure of financial decision-making ability against an objective benchmark and allow us to measure financial mistakes as the amount of money the participants lose by choosing a dominated option. The financial decision-making score (0–100) sums performance on the three tasks, each worth up to about 33 points; the Big Three questions are not part of the score. While both the willingness to pay elicitation and the financial decision-making questions are unincentivized, these questions are similar to those used in our experiment, and give us another measure of selection patterns.

Finally, to study long-term financial outcomes, we link course information from the Registrar's office with the University of California Consumer Credit Panel (UC-CCP). The UC-CCP is a longitudinal panel of approximately 40 million consumers starting in 2004 and continuing quarterly. The sample comprises anonymized credit records of a nationally representative 2% sample of U.S. adult consumers with credit records along with a full sample of 100% of Californians with credit histories, allowing us to observe outcomes for the majority of students at UC Berkeley.

3.2 Selection on Baseline Ability – Observational Data

Standard human capital accumulation theories suggest a negative correlation between baseline ability and demand for education, as those with the lowest starting knowledge typically have the lowest marginal cost (i.e. opportunity cost) and highest marginal benefits from education. However, both our results from the observational data and lab setting contradict this prediction.

We first document selection into the personal finance class at UC Berkeley. Using individual-level data, we compare the demographic and major characteristics of UGBA 135, the large personal finance class at UC Berkeley, against the broader undergraduate population. We focus on demographic variables and parents' education and a proxy for family income: whether the student qualified for a University of California application fee waiver, which applicants receive based on the family size and income they report on the application.

Table 1 shows that this selection extends to credit outcomes measured before the class. In the quarter before the class term, students who enroll in UGBA 135 have an average credit score of 692, compared with 672 for students who never take the class. Within graduation cohorts, the difference in average credit scores is 19.6 points; it falls to 7.3 points after additionally controlling for pre-class GPA, prior quantitative coursework, gender, ethnicity indicators, parental college attendance, fee-waiver status, and residency, and remains statistically significant. Students who enroll are also less likely to have negative marks on their credit files: 0.20 percent have a bankruptcy, compared with 0.79 percent of students who never take the class, and 0.60 percent have an account in collections, compared with 1.8 percent. A joint test rejects that the two groups have the same credit outcomes ($p < 0.001$). Panel B shows that the selection on observables we document for the class of 2023 in Appendix Table A3 holds across all cohorts: students who enroll have higher GPAs, have taken more quantitative courses, and are less likely to be underrepresented minorities, first-generation college students, or recipients of a UC fee waiver.

We find strong evidence of positive selection on observables. As shown in Appendix Table A3, students who enroll in the financial literacy course are significantly different from non-enrollees across several dimensions. Specifically, enrollees are 17 percentage points more likely to come from quantitative majors (e.g., math, economics, and data science) (77% vs. 60%, $p < 0.01$), tend to have higher cumulative GPAs, and are significantly less likely to be underrepresented minorities (12% vs. 24%, $p < 0.01$). Furthermore, students from higher socioeconomic backgrounds appear more likely to invest in financial education; enrollees are less likely to have qualified for an application fee waiver, our proxy for low family income, and are less likely to come from families without a college degree when compared to the broader population of Berkeley students.

We complement these descriptive comparisons with three regressions examining whether student characteristics predict course completion after accounting for other observed differences. The first specification includes demographic characteristics, pre-class GPA, prior quantitative coursework, and graduation-cohort indicators; the second adds pre-class major indicators; and the third also includes pre-class credit score. All three specifications use the same 68,315 students, comprising 8,833 course completers and 59,482 never-takers with the required observed baseline information. This common sample differs from the samples used in the preceding descriptive comparisons. Appendix Table A6 reports the specifications and sample restrictions.

In the most adjusted specification (Panel A of Appendix Table A6, column 3), a one-point increase in pre-class GPA is associated with a 3.14-percentage-point increase in the probability of course completion (SE 0.40), one additional quantitative-department-by-term enrollment record with a 0.99-percentage-point increase (SE 0.06), and a 100-point increase in baseline credit score with a 0.41-percentage-point increase (SE 0.09). These associations persist after controlling for pre-class major and the other included characteristics. They describe selection into course completion

	Takers	Never-takers	Difference	
			Graduation-year FE	+ Covariates
Panel A. Credit bureau records, quarter before the class term				
Has a credit file (%)	87.0	85.2	2.1***	1.5***
			(0.36)	(0.37)
Credit score	692	672	19.6***	7.3***
			(1.62)	(1.65)
Any bankruptcy (%)	0.20	0.79	−0.60***	−0.24**
			(0.093)	(0.094)
Any delinquency (%)	0.24	0.92	−0.70***	−0.32***
			(0.101)	(0.104)
Number of delinquencies	0.003	0.013	−0.010***	−0.004**
			(0.0017)	(0.0018)
Any account in collections (%)	0.60	1.8	−1.3***	−0.46***
			(0.143)	(0.146)
Total debt ($)	7,620	9,820	−2,370***	−970**
			(394)	(404)
Joint test, p-value			<0.001 (F = 74.1)	
Students linked to the UC-CCP	10,742	77,607		
with a credit file	9,348	66,163		
Panel B. Student characteristics before the class term				
Cumulative GPA	3.45	3.36	0.12***	0.09***
			(0.0047)	(0.0047)
Quantitative major (%)	55.3	44.1	11.4***	−0.86**
			(0.43)	(0.37)
Quantitative courses taken	5.97	3.90	1.70***	1.37***
			(0.035)	(0.035)
Male (%)	48.8	46.1	2.5***	0.26
			(0.44)	(0.44)
Underrepresented minority (%)	13.2	23.0	−9.4***	−7.3***
			(0.36)	(0.31)
First-generation college student (%)	16.8	20.6	−3.9***	1.0***
			(0.36)	(0.32)
Application fee waiver (%)	13.2	17.9	−4.5***	−1.2***
			(0.32)	(0.29)
Age at credit-file entry	18.98	19.45	−0.49***	−0.28***
			(0.035)	(0.035)
Joint test, p-value			<0.001 (F = 529.1)	
Students in the Registrar data	14,852	125,390		

Table 1: Pre-Class Credit Records and Student Characteristics: UGBA 135 Takers vs. Never-Takers

Notes: Panel A compares takers and never-takers in the quarter before the actual or assigned comparison class term. Never-takers receive randomly drawn comparison terms to align baseline measurement. Panel A uses students linked to the UC-CCP; rows other than "Has a credit file" require a credit file in that quarter. Panel B uses Registrar records regardless of linkage, except age at credit-file entry, which requires linkage and enters the Panel A joint test. Columns report group means and taker-indicator coefficients controlling for graduation-year fixed effects, then additionally for baseline covariates excluding the row's outcome. Parentheses contain conventional, unclustered OLS standard errors. For percentage outcomes, differences and standard errors are in percentage points. Samples vary across rows and specifications because of missing outcomes or covariates; displayed counts describe the broader groups. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

rather than the effect of taking the course.

On the same sample, the pre-class credit-score difference falls from 20.08 points after controlling for graduation cohort to 7.73 points after additionally accounting for pre-class major and other student characteristics. Appendix Figure A.3 and Panel B of Appendix Table A6 report these comparisons. Holding the sample fixed ensures that this reduction reflects covariate adjustment rather than changes in which students are included.

Using data from the time spent on the UGBA 135 course website, we find evidence of selection on attention. Asian students are more likely, and White and male students less likely, to spend more time on the course website; the results are shown in Appendix Table A4. Students who spend more time on the course website are more likely to have a higher grade in UGBA 135 and a higher semester GPA overall. The differences highlight selection on attention as another important dimension that could help explain the limited efficacy of financial literacy mandates.

We also provide evidence of selection on financial decision-making ability. Using the ASUC survey data, we examine the relationship between the hypothetical willingness to pay (WTP) for the course and performance on hypothetical financial decision-making questions (e.g., maximizing returns in compound interest questions and debt repayment tasks). Figure 1 plots these results. We find that WTP is positively correlated with financial decision-making scores, although the relationship is noisy: each 10-point increase in the score is associated with about \$6 more WTP (robust s.e. = \$3.2, $p = 0.06$, $N = 666$), and the relationship is driven by the larger second wave. On this measure, students with the highest demand for the course are arguably students who already possess high levels of financial knowledge. While the decisions in the survey are unincentivized, the correlation raises further concerns that students who stand to benefit the most from financial literacy education – students starting with lower levels of financial knowledge or from less privileged backgrounds – may not be the ones self-selecting into the class.

Taken together, the observational evidence suggests strong positive selection into financial education along observable characteristics and financial decision-making ability. However, these patterns are inherently descriptive. In the administrative data, demand for education is jointly determined by expected returns, perceived costs, institutional constraints, peer effects, and information frictions, making it difficult to isolate the underlying mechanism driving selection. In particular, we cannot observe individuals' beliefs about their own baseline ability or the returns to education, nor can we measure how these beliefs relate to actual learning outcomes. Our controlled lab experiment, described below in Section 4, helps us address these limitations and isolate the relationship between baseline ability, expected returns, and demand for education.



Figure 1: WTP for UGBA 135 versus Financial Decision-Making Score

Notes: Figure 1 shows the results from the ASUC survey conducted at UC Berkeley. We plot the relationship between the students' score on a set of financial decision-making questions and the hypothetical demand for UGBA 135. The sample is the 666 respondents with a willingness to pay, pooling both survey waves. The financial decision-making score (0–100) sums performance on three hypothetical tasks on index-fund fees, debt repayment, and compound interest. Willingness to pay is in dollars; negative values are the amount a student who does not plan to take the class would pay to waive it if it were required. Both the decision-making questions and the willingness-to-pay question are hypothetical and unincentivized; the results provide suggestive evidence of the positive selection we observe in the administrative data.

3.3 Credit Outcomes of Takers and Never-Takers

In this section we compare the consumer-credit record outcomes of students who completed the UGBA 135 personal finance course, whom we refer to as the "takers," versus students who do not take the course, whom we refer to as "never-takers." Students who dropped the course and never completed it are considered separately. The comparisons below describe the differences in credit outcomes associated with course participation.

For student i, let $D_i = 1$ indicate course completion and $D_i = 0$ indicate never taking the course. We measure follow-up outcomes three years after graduation (Appendix Table A9 reports outcomes one and five years after graduation). We also observe outcomes separately at ages 25 and 30 in the Appendix. Baseline credit outcomes refer to the quarter before the course begins. For never-takers, a comparison date is drawn using the observed interval between university entry and course enrollment among takers in the same graduation-cohort and transfer-status group, with broader groups used when necessary. The random draw is held fixed so that the same comparison dates are used across specifications. These dates determine which characteristics are measured

before potential course participation; the graduation-based follow-up dates are still determined by graduation. Assigned graduation dates are used when a degree date is unavailable, and students enter each comparison only when the required outcome and covariates are observed.

For each outcome and measurement date, we estimate

$$Y_{ih} = \alpha_h + \tau_h D_i + X_i'\beta_h + \lambda_{c(i),h} + \mu_{m(i),h} + \varepsilon_{ih}. \tag{8}$$

Here Y_{ih} is student i's outcome at measurement horizon h, such as credit score three years after graduation; α_h is an intercept; and τ_h is the adjusted difference between takers and never-takers. The vector X_i contains an indicator for male students, the recorded ethnicity indicators, indicators for no parent having attended college and for an application fee waiver, pre-class GPA, prior quantitative coursework, and an indicator for non-US residency; β_h contains their regression coefficients. Prior quantitative coursework counts distinct quantitative-department-by-term enrollment records before the course, rather than individual catalog courses. The terms $\lambda_{c(i),h}$ and $\mu_{m(i),h}$ are indicators for graduation cohort $c(i)$ and major category $m(i)$, respectively; they allow mean outcomes to differ across cohorts and majors. The error ε_{ih} represents outcome variation not explained by the regressors. Major categories in these overall comparisons can reflect graduation or the latest census record and therefore need not precede the course. Standard errors allow observations to be correlated within each graduation-cohort-by-major group.

Three years after graduation, takers have credit scores 26.96 points higher than never-takers before adjustment (SE 1.26; N = 58,631). Controlling for student characteristics, graduation cohort, and major reduces the gap to 13.55 points (SE 1.02; N = 56,684). Because the two specifications use different samples, this reduction reflects both adjustment and changes in sample composition. The adjusted gaps are 12.41 points one year after graduation (SE 1.05) and 12.81 points five years after graduation (SE 1.15); these comparisons also use different available samples across horizons. Appendix Figure A.4 and Table A7 report the full set of comparisons.

Re-weighting by baseline observables. Table 2 reports the differences in credit score, total debt, delinquency, and collections for takers versus never-takers three years after graduation under different reweighting specifications. Column (1) shows the unweighted results. Column (2) reweights the never-takers so their baseline characteristics better match the baseline characteristics of the takers. Column (3) reweights both takers and never-takers. Missing numeric predictors in the participation model are replaced by their sample medians; outcomes are not imputed. Appendix A.3 describes the predictors, weighting procedures, and diagnostic checks.

The raw credit-score difference is 26.96 points (SE 1.26), compared with 13.34 points under taker-targeted weighting (SE 0.92) and 13.71 points under overlap weighting (SE 0.95). The delinquency and collections differences also narrow after weighting, although takers continue to

have lower rates of both outcomes. The weighted total-debt differences are imprecisely estimated. Thus, accounting for measured baseline characteristics reduces several observed differences without eliminating them.

Outcome	(1) Takers & Never-Takers Unweighted	(2) Only Never-Takers Reweighted	(3) Takers & Never-Takers Reweighted	N
Credit score (points)	26.96***	13.34***	13.71***	58,631
	(1.26)	(0.92)	(0.95)	
Total debt (dollars)	−278	2,009	1,533	57,466
	(1,429)	(1,370)	(1,311)	
Any delinquency (pp)	−0.953***	−0.353***	−0.375***	58,682
	(0.109)	(0.084)	(0.088)	
Any collections (pp)	−1.615***	−0.629***	−0.663***	58,682
	(0.168)	(0.120)	(0.126)	

Table 2: Unweighted and Weighted Outcome Differences Three Years after Graduation

Notes: Each column shows the difference in the outcome variable of interest for takers minus never-takers with standard errors in parentheses. Column (1) shows the unweighted results. Column (2) reweights the never-takers so their baseline characteristics better match the baseline characteristics of the takers. Column (3) reweights both takers and never-takers. All outcomes are measured three years after graduation. "pp" denotes percentage points. The reported observation counts match across the three estimates within each outcome. Missing numeric participation-model predictors are median-imputed; outcomes are not imputed. Standard errors are clustered by graduation-cohort-by-major group; weighted standard errors treat fitted weights as fixed. Estimates describe observational differences. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

3.3.1 Differences by Baseline Characteristics

We next ask whether the adjusted difference in outcomes between takers and never-takers varies by gender, baseline credit score, academic preparation, and other measured characteristics. For each characteristic Z_i, we estimate a separate regression at three years after graduation:

$$Y_{i,g+3} = \alpha + \tau D_i + \theta Z_i + \delta D_i Z_i + (X \backslash Z)_i' \gamma + \lambda_{c(i)} + u_i. \tag{9}$$

The subscript $g + 3$ means three years after graduation, rather than three years after the course. The coefficient θ describes the relationship between Z_i and the outcome among never-takers, while δ measures how the taker–never-taker gap changes with Z_i. For a binary characteristic, the adjusted gaps are τ when $Z_i = 0$ and $\tau + \delta$ when $Z_i = 1$. Continuous characteristics are expressed in sample-standard-deviation units, so δ is the change in the gap for a one-standard-deviation increase in that characteristic. The vector $(X \backslash Z)_i'$ contains the other student controls, excluding Z_i itself; γ contains their coefficients, $\lambda_{c(i)}$ denotes graduation-cohort indicators where $c(i)$ is the cohort of student i, α is an intercept, and u_i is the error. These regressions include cohort indicators but not the major indicators in Equation (8).

Figure 2 shows how the adjusted taker versus never-taker difference in credit score varies across student characteristics. Appendix Figures A.5–A.7 report total debt and two measures of delinquency. For credit score, the gap is 6.16 points smaller for a one-standard-deviation increase in pre-class credit score (SE 1.13) and 5.42 points smaller for a one-standard-deviation increase in prior quantitative coursework (SE 0.78). It is 9.52 points larger for students classified as underrepresented minorities (SE 2.10) and 13.94 points smaller for students in quantitative majors before the course (SE 1.36). These four interactions, along with the interaction with age, remain statistically distinguishable from zero at the 5% level after Benjamini–Hochberg adjustment across all 40 outcome–characteristic comparisons. Each coefficient measures a difference between adjusted gaps, rather than the gap for a particular subgroup. Each regression uses students with the required outcome and characteristics available, so samples can differ across comparisons. Appendix Table A11 reports the credit-score coefficients, standard errors, and sample sizes.

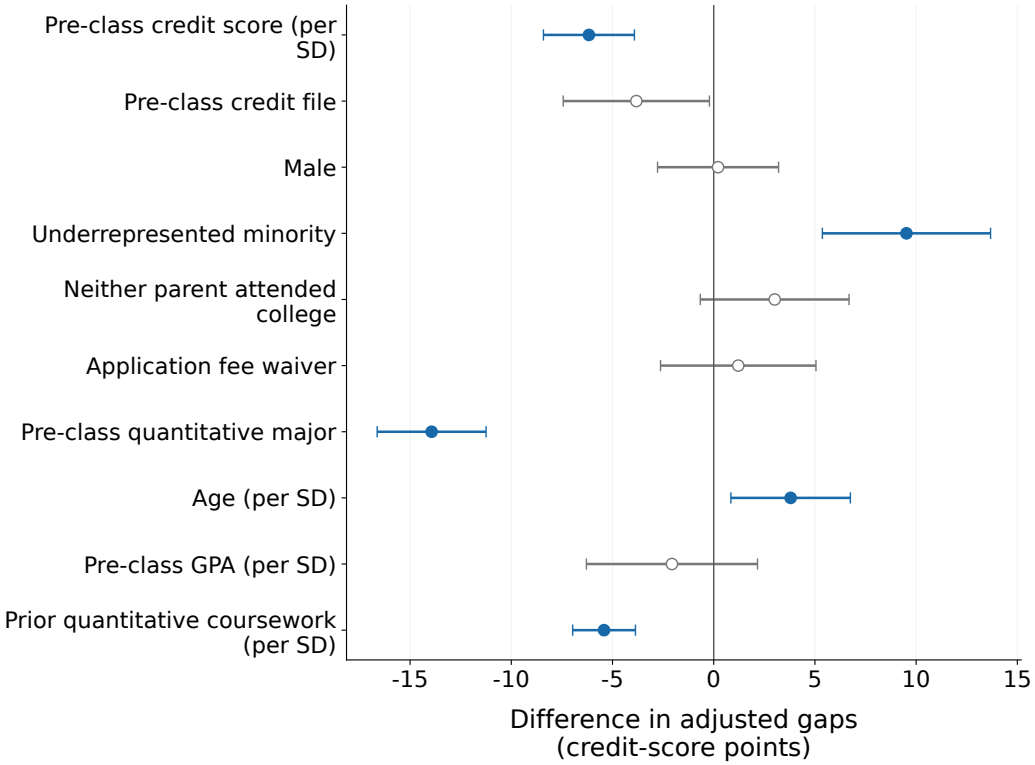


Figure 2: Adjusted Credit-Score Differences across Student Characteristics

Notes: Each point is the interaction coefficient δ from Equation (9) for credit score three years after graduation: the change in the taker–never-taker gap associated with the indicated characteristic. Continuous characteristics are in standard-deviation units. Bars are 95% pointwise confidence intervals. Filled markers pass the 5% Benjamini–Hochberg threshold across all 40 outcome–characteristic comparisons.

Credit-score differences by starting credit score. We use credit score as a detailed illustration because it is observed both before the course and at follow-up, allowing us to compare subsequent outcomes across students' starting credit circumstances on a common scale. Figure 3

shows the adjusted taker–never-taker difference in follow-up credit score across pre-class credit scores. Results for additional credit outcomes appear in Table 2 and Appendix Figures A.5–A.7; Appendix A.3 reports additional estimates and corresponding credit-score comparisons by pre-class GPA and prior quantitative coursework. Appendix Figures A.8 and A.9 show the observed follow-up means and the adjusted follow-up scores for takers and never-takers with the same distribution of other characteristics. In the linear specification, the predicted gap falls from 29.78 points (SE 2.09) at a pre-class score of approximately 599 to 3.32 points (SE 0.99) at a score of approximately 766. The estimation sample contains 39,690 students with observed outcomes and covariates within the baseline range represented in both groups. This is a more restricted sample than the separate interaction regressions above. The downward slope shows that students with lower starting scores have a larger change in post-class credit score.

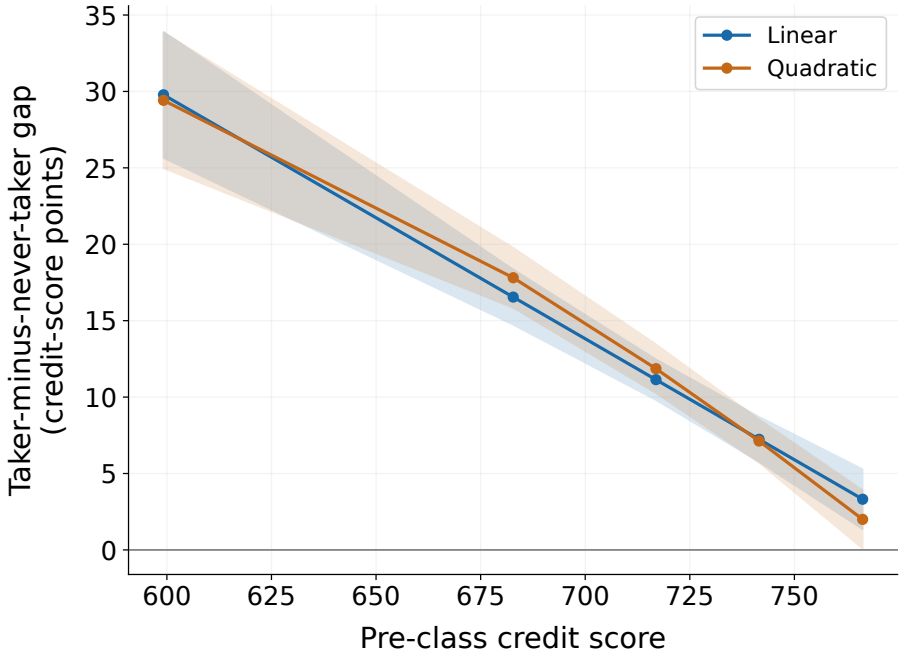


Figure 3: Adjusted Credit-Score Gap by Pre-Class Credit Score

Notes: The outcome is credit score three years after graduation, measured after the actual or assigned course-completion date. Lines show the adjusted taker–never-taker difference at each pre-class credit score from a linear interaction and a quadratic sensitivity specification fitted to the same students. Shaded bands are 95% pointwise confidence intervals clustered by graduation-cohort-by-major group. The displayed baseline range is shared by takers and never-takers.

Credit-score differences by academic background. The same comparisons using pre-class GPA and prior quantitative coursework also show smaller adjusted credit-score gaps among students with greater academic preparation. After additionally controlling for pre-class credit score, the linear predicted gap decreases from 18.98 points (SE 1.54) at a GPA of 2.94 to 9.86 points (SE 1.39) at a GPA of 3.87, in a sample of 38,422 students. Across prior quantitative coursework,

the corresponding gap decreases from 22.54 points (SE 1.42) at approximately 0.64 quantitative-department-by-term records to 5.59 points (SE 1.14) at approximately 9.56 records, in a sample of 42,153 students. The fractional values are group averages of an integer count. Figures A.10a and A.10b present these comparisons. These models use observed pre-class GPA and a common baseline range, so they differ in sample and controls from the broader one-characteristic-at-a-time regressions.

Changes around course enrollment. A complementary event study compares changes in credit outcomes around the course-start quarter with changes among eligible comparison students. The pooled post-course credit-score coefficient is 2.83 points (SE 2.28; 95% confidence interval $[-1.64, 7.30]$), based on 21,568 students. The corresponding total-debt coefficient is \$4,081 (SE \$999), and the coefficient for any delinquency is -0.529 percentage points (SE 0.100 percentage points). Appendix A.3 reports the specification, Appendix Table A12 reports the estimates, and Appendix Figures A.11 and A.12 show the time paths. These estimates use different samples, timing, and comparisons from the graduation-based regressions; their difference is not an estimate of selection bias.

Comparison with students who dropped the course. Students who enrolled initially but did not complete the course provide a separate observational comparison. Their mean pre-class GPA is 3.50, compared with 3.45 among students who completed the class; the corresponding sample sizes are 2,475 and 14,852. Appendix Table A5 reports baseline comparisons. Droppers and takers have similar pre-class credit records (joint test $p = 0.54$) but differ on Registrar characteristics such as GPA (joint test $p < 0.001$). A student counts as having dropped the course if they initially enrolled in the course at the start of the semester, but are not enrolled in the course at the end of the semester and have no grade in the class. We do not use an exact withdrawal date. Appendix Figures A.1 and A.2 show standardized differences in baseline characteristics between takers and never-takers and between takers and droppers.

4 Lab Experiment

4.1 Data and Empirical Strategy

To demonstrate that the selection patterns we observe are not limited to classroom settings and to better understand mechanisms, we turn to our lab experiment. We recruit Prolific participants who are 18 years or older and located in the United States. 1,159 participants complete the survey and correctly answer two comprehension questions about the instructions. We then exclude three groups, in this order. First, for a random subset of participants we implement their own price-list choice, so whether they receive an education material is not randomly assigned; we drop these 52

participants. Second, we drop 48 participants with inconsistent responses on two or more of the three willingness-to-pay price lists.[4] Third, we drop 10 participants who used AI to write a free-text response. The question asking participants to explain their choice of preferred education material contains a sentence in transparent text, invisible on screen, that asks for the phrase "invisible text recovered" to be included in the answer. A participant reading the page does not see the sentence, but an AI tool given the question text does, so an answer containing the phrase indicates that it was generated by AI. The final sample has $n = 1{,}049$ participants. Results that use willingness to pay for a single format use every participant whose price list for that format is consistent, and results that use the average across formats use the 967 participants whose three lists are all consistent. Participants in our experiment have an average age of about 43 years old, are 66% white, and have an average income of about $65,000. The key demographic characteristics of our final sample are presented in Table 3.

	Obs.	Mean	SD	P(25)	P(75)
Age (yrs)	1,049	42.77	12.48	33.00	51.00
Male	1,049	0.43	0.50	0.00	1.00
Time (in minutes)	1,049	37.34	17.41	25.73	44.82
Race - White	1,049	0.66	0.47	0.00	1.00
Race - Hispanic	1,049	0.07	0.25	0.00	0.00
Race - Asian	1,049	0.17	0.37	0.00	0.00
Race - Black	1,049	0.06	0.24	0.00	0.00
Income ($ est)	1,049	65085.80	49591.65	30000.00	87500.00
College Degree	1,049	0.69	0.46	0.00	1.00

Table 3: Summary Statistics for Experiment Participants

Notes: Table 3 presents the summary statistics of the demographics of the participants in the lab experiment. Participants are recruited through Prolific, a large online survey platform commonly used by researchers. All participants must be 18 years or older and located in the United States. We exclude participants with inconsistent responses on two or more of the three willingness-to-pay price lists, participants who are flagged by our invisible text as having used AI to complete their written responses, and participants who have their education material choices implemented.

In the experiment, each participant completes two sets of incentivized financial decision-making exercises. In each financial decision-making exercise, participants choose between two financial options: a fixed dollar amount, or an investment in which the starting amount and growth rate are

[4]Each price list consists of 11 separate decisions between skipping the material and completing it, with a payment attached to one of the two options. In the first decision, skipping pays $10 and completing pays nothing; the payment for skipping falls in $2 steps to zero, and the payment for completing then rises in $2 steps to $10 in the last decision. Because completing becomes relatively more attractive down the list, a participant chooses to skip in the early decisions and switches at most once to completing, and the decision at which they switch identifies their willingness to pay. A list is inconsistent when the participant switches between the two options more than once, or switches only from completing to skipping, choosing the material when it pays less but not when it pays more. For the 82 participants with one inconsistent list, we treat the willingness to pay for that format as missing and keep their other two. Studies that elicit several price lists differ in how they treat such responses: some exclude the participant, and others keep the responses and model them as noise (Jack et al., 2022; Engel and Kirchkamp, 2019). Appendix Table A15 shows that the main willingness-to-pay results are similar when average WTP is instead computed over the consistent lists of participants with one or two inconsistent lists.

varied. We then elicit demand for each version of the education material. About 5% of participants have their own price-list choice implemented; all other participants are randomly assigned to one of four arms: 1) AI education, 2) video education, 3) university lecture education, or 4) control. Finally, all participants complete an endline set of financial decisions and are asked multiple choice questions about their experience with financial education and their use of AI in financial decisions. We also collect information on age, gender, race, and income. Figure 4 provides an overview of the experiment.

Demand for education, or willingness to pay (WTP), is measured with three multiple price lists, one for each format, in the spirit of the Becker–DeGroot–Marschak mechanism (Becker et al., 1964): one row of one list may be selected and implemented. Each list is the 11-decision list described in footnote 4, in which the participant chooses between skipping the material with a payment and completing it with a payment. WTP can therefore be negative: a participant who chooses to skip until completing the material pays \$4 has a WTP of about −\$3.

If participants receive the education material, they must fully complete the material by passing attention checks and correctly answering three multiple choice comprehension check questions before they are able to move on in the experiment.

We then test whether these selection patterns are in part driven by the perception that traditional versions of financial literacy education may be unengaging or cognitively difficult. We offer three different formats of the education material: 1) a classroom style university lecture format, 2) a video education format, and 3) an interactive AI-powered learning assistant. The video and AI based learning formats allow us to test whether supply-side policies making education more accessible and less difficult can change selection patterns. All versions of the education material teach the exact same material; the Rule of 72, a simple mental shortcut for calculating the value of the investment options. We intentionally try to hold all versions of the education material constant by keeping the definitions, comprehension check questions, and explanations to the multiple choice questions exactly the same. We also emphasize to participants that all of the materials are based on the same content and vary only in presentation format. The university lecture format is framed as the most difficult version while the AI-powered learning assistant is framed as the most beginner friendly, interactive, and easy to understand version. We describe the video education version using similar language and framing as popular online learning platforms like Khan Academy and Coursera.[5] Appendix Figures A.13, A.14, and A.15 show screenshots of the three versions of the education material.

We also measure each participant's beliefs about their baseline score (their estimated baseline score), their forecast of their endline score if they receive no education material, their forecast of their endline score with each version of the education material, and the reasons they prefer

[5]See, for example, Khan Academy's personal finance course and Coursera's personal finance courses.

each version. The forecasts are not incentivized, so that a participant cannot raise their payoff by adjusting their effort on the endline questions to match a forecast; the estimate of the baseline score and the assessment of the endline score that participants give after answering the endline questions are incentivized. Each incentivized estimate pays an extra \$1 to a randomly selected 5% of participants if it is within \$1 of the bonus the participant actually earns on those questions. In the notation of Section 2, the forecast with no education material is $\hat{Y}_i(0)$ and the forecast with format m is $\hat{Y}_i(m)$. We use two differences between these forecasts. Our main measure of beliefs about the return to the education material is the perceived gain from format m, $\hat{\tau}_{im} = \hat{Y}_i(m) - \hat{Y}_i(0)$: the respondent's forecast of their endline score with the material minus the forecast without the material. It compares forecasts of the same two outcomes, the endline score with and without the education material, that the experiment compares to estimate the actual gain. Appendix Table A14 shows that willingness to pay moves more with the perceived gain than with the forecast with the material minus the estimated baseline score. Eliciting expectations also allows us to measure how overconfident participants are in their own knowledge. Finally, we ask the participants a set of five unincentivized multiple choice questions on compound interest from Ambuehl et al. (2022), as well as questions about their experience with financial literacy education, personal finances, and AI usage.

To make the price lists incentive-compatible, a randomly selected 10% of participants are paid a bonus based on one of their two sets of financial decisions. For the half of these participants whose endline decisions are selected, about 5% of the sample (52 participants), one row of one randomly chosen price list is implemented, so their own choice determines whether they receive the education material; we exclude them from the analysis. All other participants are randomly assigned: 25% to no education material and 75% split evenly across the AI, video, and lecture materials. Randomly assigning participants with both positive and negative WTPs to receive financial literacy education allows us to identify whether the true benefits from education are positively correlated with demand, as standard models of investment in education would predict, or if individuals actually *mispredict* the benefits of education. The price-list payment is made only if the participant's endline decisions are selected for the bonus, so the cost or benefit of the education material is realized in the same state of the world as the payoff from what the participant learns. Table 4 shows the balance table for the four treatment arms.

One challenge the financial education literature has often faced is the question of how to measure financial literacy in objective, welfare-relevant terms. This stems from the fact that financial literacy encompasses a wide variety of skills, including numeracy, statistics and probability, attention to detail, knowledge of financial terms and definitions, and the ability to read and process complex financial contracts. Additionally, in many cases these shortcomings can be overcome through exerting effort (for example, looking up the concepts online, or following the advice of knowledgeable friends or experts). This raises the question of whether teaching financial literacy is

	Control		AI Edu		Video Edu		Lecture Edu	
	Mean	(SD)	Mean	(SD)	Mean	(SD)	Mean	(SD)
Male	0.46	(0.50)	0.44	(0.50)	0.42	(0.49)	0.39	(0.49)
Age	42.34	(12.45)	41.89	(12.21)	43.56	(12.41)	43.22	(12.82)
Time (mins)	28.44	(15.01)	41.51	(18.81)	39.59	(17.54)	39.62	(15.03)
Race - White	0.68	(0.47)	0.65	(0.48)	0.62	(0.49)	0.70	(0.46)
Race - Black	0.05	(0.23)	0.07	(0.25)	0.08	(0.27)	0.05	(0.22)
Income ($ est)	65830.08	(49237.02)	62264.71	(46427.95)	69813.43	(55008.19)	62351.85	(46974.71)
College	0.66	(0.47)	0.69	(0.46)	0.73	(0.45)	0.66	(0.47)
Observations	256		255		268		270	

Table 4: Balance Table by Treatment Arm

Notes: Table 4 presents the balance table of the key participant demographics by treatment arm. Treatment status is randomly assigned by the computer at the individual level.

simply increasing motivation or teaching individuals to exert effort. Finally, there is the question of how knowledge of financial concepts is translated into real-world behavior that improves welfare, as measured against an objective benchmark.[6] In fact, there is evidence that being able to *recall* financial concepts is not the same as being able to *apply* these concepts and translate them into improvements in financial decision-making (Ambuehl et al., 2022).

Acknowledging this difficulty, we choose to focus on what we believe is one of the core components of financial literacy: numeracy. Specifically, our questions test for an understanding of interest rates and compound interest. We focus on these concepts for two main reasons. The first is that we believe that quantitative principles are a core component of making good financial decisions. In fact, "The Big Three," the widely-used, standard measures of financial literacy developed by Lusardi and Mitchell (2011), are at their core questions about numeracy.[7] The second benefit is that focusing on numeracy allows us to measure financial knowledge in a way that directly maps to monetary gains and losses. This allows us to measure and quantify the benefits of our financial education intervention using an objective, welfare-relevant metric.

Our question is based on the methodology developed in Ambuehl et al. (2022). We ask participants to choose between two financial instruments: one that is a complexly-framed dollar amount and one that is a simply-framed dollar amount. Participants are told that there is a chance $1 will be added to their bonus for each decision where they choose the financial instrument that is worth more money in 72 days. Since there are ten questions in total, the potential bonus ranges from $0 to $10. We also tell participants that there is no interest rate or investment risk and no uncertainty. Figure 5 shows an example of the question.

[6]Early papers focus on outcomes such as saving which may not be universally welfare improving. See Bernheim and Taubinsky (2018) for a discussion.

[7]Two of the Big Three questions test for an understanding of interest rates (one directly and one indirectly by asking about inflation). The third question asks about diversification, a statistical concept that is closely related to numeracy.

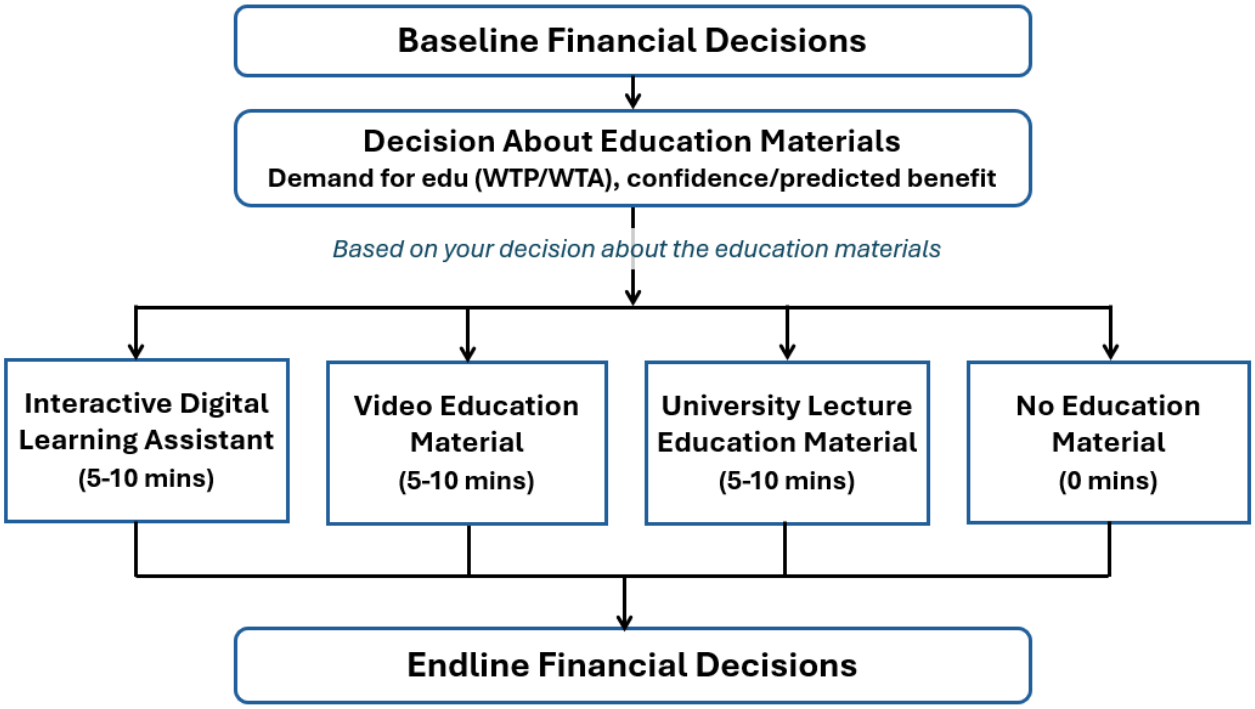


Figure 4: Lab Experiment Overview

Notes: The diagram in Figure 4 provides an overview of the experimental design for the lab experiment. Participants were first asked to answer a set of 10 baseline financial decision-making questions that test the participant's knowledge of compound interest. We then ask participants to predict how they performed on the 10 baseline questions, elicit a willingness to pay for each version of the education material, and ask participants to predict how they would do on the endline questions *if they were to receive the specified version of the education material*. 90% of participants are then randomized into one of four experimental arms: the AI education material, video education material, lecture education material, or control (no education material). Finally, all participants answer a set of 10 endline financial decision-making questions, which test the same concepts and are very similar in structure and difficulty to the baseline questions.

Of the two financial instruments, there is one option that maximizes the amount of money the participant will receive at the end of the study. Since one option is framed as a compound interest calculation, participants should calculate the dollar value of the complexly framed option, thus testing their knowledge of compound interest. As in Ambuehl et al. (2022), the complexly framed amount can be calculated quickly using the Rule of 72, a mental shortcut that allows you to quickly estimate how long it would take a dollar amount to double at a fixed interest rate. We then define the change in score as the endline score (0–10) minus the baseline score (0–10).

4.2 Selection on Baseline Ability and Treatment Effects

We first seek to replicate the selection patterns we see in the observational data in our lab setting. One potential concern is that a controlled setting may attenuate the selection mechanisms we observe in the real world. Beyond the ability to correctly predict the costs and benefits of education, real-world selection patterns may encompass factors such as present focus, uncertainty about the

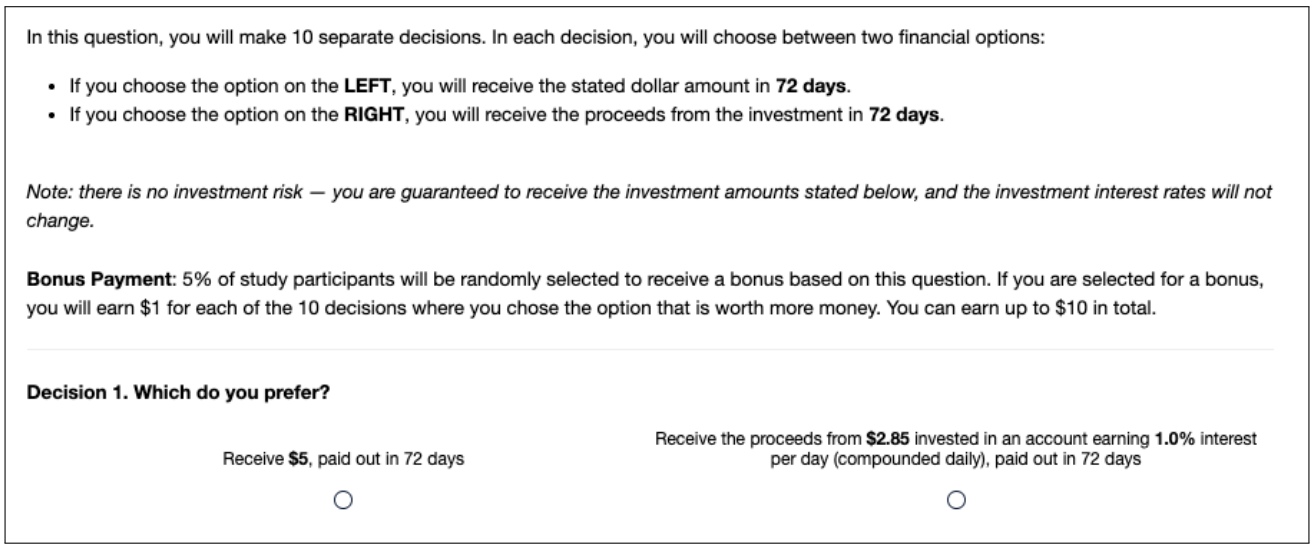


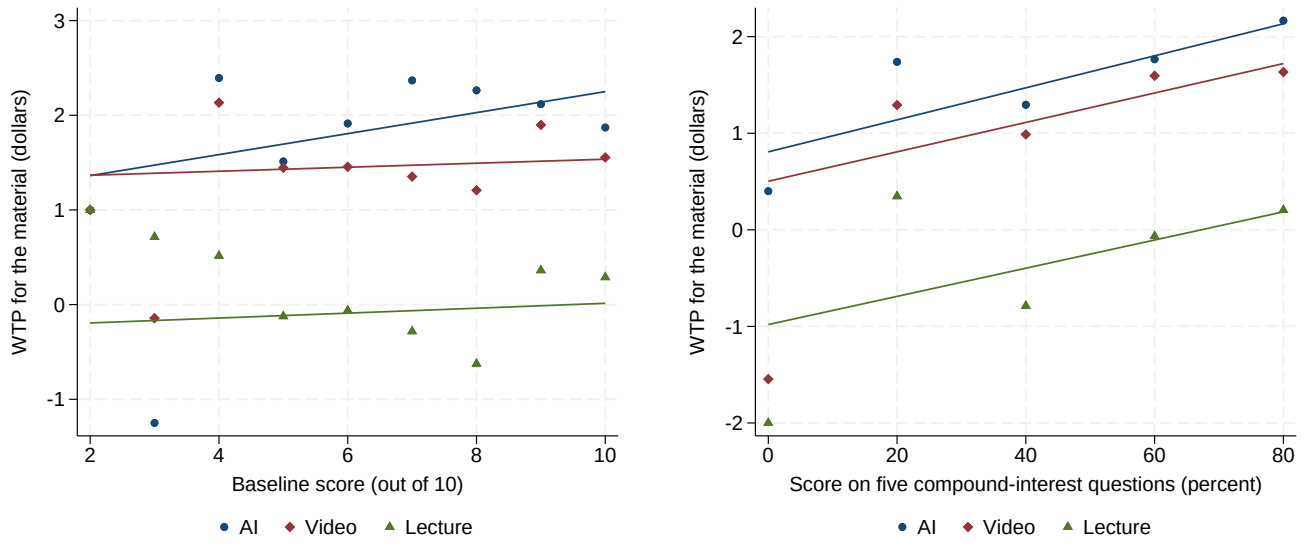
Figure 5: Financial Decision-Making Question

Notes: Figure 5 shows the financial decision-making question used in the lab experiment. The question is based on a core concept in personal finance; understanding compound interest and exponential growth and is based on a similar question in Ambuehl et al. (2022).

future, and concerns about the decay of knowledge that we control for in the lab. For example, we remove any uncertainty about the potential benefits of education by telling participants they will use the material immediately after (in the endline questions). We may also reduce uncertainty about the quality of the education material by providing it as a part of the experiment, essentially signaling its usefulness. Finally, we remove any concerns about knowledge decaying over time, thereby increasing the potential benefits of the education material, by asking the endline questions immediately after the education material. Thus, it would be unsurprising if any selection patterns are more muted, or even non-existent, in a lab setting.

Despite these limitations, demand for education does not fall with baseline financial knowledge in our lab setting. Figures 6(a) and 6(b) plot WTP for each format against the score on the baseline "Rule of 72" decision-making questions and against the score on five financial literacy multiple choice questions about compound interest. Holding the demographic controls fixed, each additional point on the baseline score changes average WTP by \$0.03 (s.e. $= 0.07$), and the format-specific slopes are between \$0.00 (lecture) and \$0.10 (AI) per baseline point (Section 4.2.2); none is statistically different from zero at the 10% level. Under the Roy benchmark of Section 2, the slope of WTP in the baseline score would equal the slope of the gain, which the next paragraphs estimate at -0.25 points per baseline point.

To estimate the treatment effect of the education material, we run the following specification,



(a) Rule of 72 Questions

(b) Compound Interest Multiple Choice Questions

Figure 6: Selection on Baseline Ability

Notes: Figures 6(a) and 6(b) show selection on baseline financial decision-making ability. Figure 6(a) shows the relationship between WTP for each version of the education material and baseline financial decision-making ability, measured as the baseline score on the Rule of 72 questions. The baseline score ranges from 0 to 10, one point for each correct decision. Figure 6(b) shows the relationship between demand for education and the participant's score on five multiple choice questions about compound interest. These five questions are taken from Ambuehl et al. (2022) and are unincentivized. We elicit demand for each of the 3 different versions of the education material: 1) the AI version, 2) the video version, and 3) the university lecture version.

which corresponds to Column (1) of Table 5:

$$Y_{i,1} = \alpha + \beta_1 T_i + \beta_2 Y_{i,0} + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i \tag{10}$$

where i indexes participants; $Y_{i,1}$ is the endline score and $Y_{i,0}$ is the baseline score, each the number of correct decisions out of 10; T_i equals one if participant i is assigned to education material and zero if participant i is assigned to the control group; and $\mathbf{X}_i$ is a vector of baseline demographic controls: indicators for being college educated, male, and white, the midpoint of the participant's income bracket, and age. In Equation (10), α is the intercept; β_1 is the average treatment effect of assignment to the education material; β_2 is the coefficient on the baseline score, which the specification restricts to be the same in the treated and control groups; $\boldsymbol{\gamma}$ is the vector of coefficients on the demographic controls; and ε_i is the error term. Columns (2) through (4) of Table 5 estimate the same specification separately for each education material type, comparing participants assigned to that format with the control group; in these columns, T_i equals one if participant i is assigned to that format.

Table 5 reports estimates of Equation (10). In all four columns, assignment to education

material raises the endline score, and every estimate is statistically significant at the 1% level. Pooling across formats, the average treatment effect is 1.29 points (s.e. = 0.12), which equals 0.68 standard deviations of the endline score in the control group (control-group mean = 6.80, standard deviation = 1.89). Estimating each format separately gives effects of a similar magnitude, with the largest gain for the AI material and the smallest for the lecture material, but the differences between the three format-specific treatment effects are not statistically significant at the 5% level. These results establish that the interventions are effective.

The coefficient on the baseline score in Column (1) of Table 5, 0.42, does not measure the persistence of test scores, because it averages two different slopes. Test–retest persistence is the slope of the endline score on the baseline score in the control group, 0.61; among treated participants, the slope is 0.36. Because the slopes differ, Table 6 reports the fully interacted regression of Lin (2013), which interacts the treatment indicator with every baseline covariate, each centered at its sample mean:

$$Y_{i,1} = \alpha + \tau T_i + \rho \tilde{Y}_{i,0} + \phi\, T_i \tilde{Y}_{i,0} + \tilde{\mathbf{X}}_i' \boldsymbol{\theta} + T_i \tilde{\mathbf{X}}_i' \boldsymbol{\lambda} + \varepsilon_i \tag{11}$$

where $\tilde{Y}_{i,0}$ and $\tilde{\mathbf{X}}_i$ are the baseline score and the demographic controls minus their sample means, and $Y_{i,1}$, T_i, and the demographic controls are defined as in Equation (10). Because every covariate is centered, α is the predicted endline score of a control participant whose covariates equal the sample means, and τ is the average treatment effect: the average, over all participants, of the treatment effect the model predicts for each participant. The parameter ρ is the slope of the endline score on the baseline score in the control group and $\rho + \phi$ is the corresponding slope in the treated group, so ϕ is the change in the treatment effect for each additional point on the baseline score, holding the demographic controls fixed. The vector $\boldsymbol{\theta}$ contains the coefficients on the centered demographic controls in the control group, the vector $\boldsymbol{\lambda}$ contains the differences between the treated-group and control-group coefficients on those controls, and ε_i is the error term. In Column (1) of Table 6, the average treatment effect is 1.27 points (s.e. = 0.11). This estimate differs from the ANCOVA estimate in Column (1) of Table 5 by 0.02 points, and its standard error is smaller (0.11 versus 0.12). Each additional baseline point lowers the treatment effect by 0.25 points (s.e. = 0.06). The treatment effect of the education material is therefore larger for participants with lower baseline scores.

Table 8 decomposes the slope of WTP in the baseline score according to Equation (5). Averaged across formats, the slope of WTP is 0.03 dollars per baseline point (s.e. = 0.07). It equals the slope of the gain, -0.25 (s.e. = 0.07), plus the slope of the belief error, 0.31 (s.e. = 0.08), minus the slope of the recovered cost $c_{im} = \hat{\tau}_{im} - W_{im}$, 0.04 (s.e. = 0.07). The belief error offsets the decline in the gain, and the cost does not vary with the baseline score. The same pattern holds within each format: for the AI material the belief-error slope is 0.43 (s.e. = 0.09) and the cost slope is -0.01 (s.e. = 0.08), and for the lecture they are 0.17 (s.e. = 0.09) and 0.03 (s.e. = 0.09).

Standard errors are from 500 bootstrap replications over participants.

	Endline score (0–10)			
	(1) All formats vs. control	(2) AI vs. control	(3) Video vs. control	(4) Lecture vs. control
Assigned any material	1.292***			
	(0.118)			
Assigned AI material		1.414***		
		(0.153)		
Assigned video material			1.281***	
			(0.149)	
Assigned lecture material				1.158***
				(0.153)
Baseline score	0.418***	0.441***	0.476***	0.525***
	(0.029)	(0.041)	(0.039)	(0.040)
Observations	1,049	511	524	526
R^2	0.222	0.313	0.307	0.295
Demographic controls	Yes	Yes	Yes	Yes

Table 5: Treatment Effects of Education Material

Notes: This table reports treatment effects on the endline score (0–10) using an ANCOVA specification. Each column regresses the endline score on an indicator for assignment to the education material and on the baseline score. Column (1) pools all three formats into a single indicator for assignment to any material. Columns (2) through (4) estimate the same specification separately for the AI, video, and lecture materials, in each case comparing participants assigned to that format with the control group. All columns additionally control for demographics (college education, gender, race, income, and age). Robust standard errors are reported in parentheses.

4.2.1 Mechanisms: Belief Errors and Misprediction

Human capital theory also predicts that individuals invest in education until its expected marginal benefit equals its marginal cost, with expectations that are correct on average. The experiment allows us to test this prediction directly, because it measures three objects that administrative data do not: participants' baseline ability, their beliefs about the returns to education, and the causal returns themselves.

We first compare participants' willingness to pay (WTP) with the perceived gain and the actual gain from the education material (Figure 7). Figure 7(a) plots WTP against the perceived gain, the forecast of the endline score with the material minus the forecast without it. WTP increases with the perceived gain: pooling the three formats, a one-point increase in the perceived gain is associated with \$0.35 more WTP (s.e. = 0.05).

Figure 7(b) plots WTP against the actual gain, the difference in mean endline scores between treated and control participants with the same baseline score. WTP does not increase with the actual gain: pooling the formats, the slope is -0.11 dollars per point (s.e. = 0.18). WTP therefore tracks the perceived gain but not the actual gain, so the participants who would gain the most

	Endline score (0–10)				
	(1) All formats	(2) All formats	(3) AI format	(4) Video format	(5) Lecture format
Assigned material	1.273***	1.372***	1.454***	1.299***	1.180***
	(0.114)	(0.119)	(0.152)	(0.150)	(0.151)
Assigned material × baseline score	-0.250***	-0.259***	-0.330***	-0.266***	-0.145*
	(0.060)	(0.063)	(0.079)	(0.078)	(0.080)
Assigned material × WTP		0.002	0.021	0.017	-0.035
		(0.029)	(0.036)	(0.035)	(0.032)
Baseline score	0.607***	0.619***	0.615***	0.617***	0.599***
	(0.050)	(0.052)	(0.051)	(0.052)	(0.051)
WTP		0.016	0.013	0.014	0.012
		(0.023)	(0.019)	(0.021)	(0.020)
Control-group mean	6.80	6.79	6.81	6.80	6.77
Observations	1,049	967	496	509	516
R^2	0.241	0.260	0.352	0.339	0.331

Table 6: Heterogeneity in the Treatment Effect by Baseline Score and Willingness to Pay

Notes: Table 6 reports estimates of Equation (11) in Column (1) and of Equation (12) in Columns (2) through (5), on the randomized sample of treated and control participants. The dependent variable is the endline score (0–10). The treatment indicator is interacted with the baseline score and with each demographic control (college education, gender, race, income, and age), and every interacted covariate is centered at its sample mean, so the coefficient on the treatment indicator is the average treatment effect (Lin, 2013). Column (1) pools the three formats. Column (2) adds willingness to pay (WTP), averaged across the three formats and centered, and its interaction with treatment; the coefficient on the interaction is the change in the treatment effect for each additional dollar of WTP. Columns (3) through (5) estimate the Column (2) specification separately for the AI, video, and lecture materials, each comparing participants assigned to that format with the control group and using WTP for that format. Every participant states a WTP for every format before random assignment, so WTP is a pre-treatment covariate in both the treated and control groups. Columns (2) through (5) are restricted to participants with non-missing WTP for the relevant format or formats, so their samples are smaller than the sample in Column (1). Robust standard errors are reported in parentheses.

from the material are not more willing to pay for it.

The perceived gain in Section 2 is the forecast with the material minus the forecast without it. The forecast without the material is low: its mean is 4.17 points, against a mean estimated baseline score of 5.84 and a mean actual baseline score of 6.57, and 53% of participants expect a lower endline score without the material than their estimated baseline score. In the control group, the mean score rises from 6.48 at baseline to 6.80 at endline. Participants therefore underestimate the uneducated outcome by about 2.6 points on average, which raises the perceived gain: averaged across formats, the perceived gain is 1.99 points (2.42 for the AI material, 2.07 for the video, and 1.47 for the lecture), against a mean actual gain of 1.30 points.

Table 9 reports OLS regressions of WTP for each format on two beliefs about that format, both elicited before random assignment: the perceived gain from the format (the forecast of the endline score with the format minus the forecast without the material) and its perceived difficulty. All regressions also control for the perceived time needed to complete the material

Outcome	(1) Slope in baseline score	(2) Difference from slope of actual gain	(3) Observations
Panel A. Actual gain from the material			
Endline score, treated minus control	-0.250***		1,049
	(0.060)		
Panel B. Perceived gain: forecast with the material minus forecast without			
Average of the three formats	0.047	0.297***	1,049
	(0.043)	(0.079)	
AI format	0.096*	0.346***	1,049
	(0.053)	(0.083)	
Video format	0.038	0.288***	1,049
	(0.046)	(0.080)	
Lecture format	0.009	0.259***	1,049
	(0.046)	(0.082)	

Table 7: Slope of the Gain and of the Perceived Gain in the Baseline Score

Notes: Each row is one outcome. Column (1) reports the slope of that outcome in the baseline score, in points per baseline point. In Panel A, the slope is the coefficient on the interaction of the assigned-material indicator with the centered baseline score in Equation (11), the change in the actual gain per baseline point. In Panel B, the slope is the coefficient on the baseline score in a regression of the outcome on the baseline score. The perceived gain in Panel B is the forecast of the endline score with the format minus the forecast without the material; the first row averages it across the three formats. Column (2) reports the slope in Column (1) minus the slope of the actual gain, with standard errors from 500 bootstrap replications over participants. All regressions control for college education, gender, race, income, and age. Column (1) reports robust standard errors in parentheses. All 1,049 participants are included.

and for demographics. Appendix Tables A16, A17, and A18 add the actual change in score, and Appendix Table A22 reports Tobit estimates that account for the censoring of WTP at −\$11 and \$11.

WTP for each format increases with the perceived gain from that format; the coefficient is positive and statistically significant at the 1% level in all four columns. Averaged across formats, a one-point increase in the perceived gain is associated with an increase in average WTP of 28.2 cents (s.e. = 5.9). Participants pay more for material they expect to raise their score more, which validates our WTP measure.

WTP for each format decreases with the perceived difficulty of that format; the coefficient is negative and statistically significant at the 5% level in all four columns. Averaged across formats, a one-point increase in perceived difficulty is associated with a decrease in average WTP of 19.0 cents (s.e. = 6.9). Because the perceived gain and perceived difficulty are measured on different scales, we compare one-standard-deviation changes. In the average-WTP column, a one-standard-deviation increase in the perceived gain is associated with an increase in WTP of \$0.70, and a one-standard-deviation increase in perceived difficulty is associated with a decrease in WTP of \$0.39; the association with the perceived gain is the larger of the two. For the lecture, the decrease associated with a one-standard-deviation increase in perceived difficulty (\$0.85) exceeds the

	(1) Average WTP, all formats	(2) WTP for AI	(3) WTP for video	(4) WTP for lecture
Slope of willingness to pay, $\bar{W}'(y)$	0.028	0.114	0.026	-0.014
	(0.066)	(0.085)	(0.071)	(0.081)
Slope of the gain, $\tau'(y)$	-0.250	-0.324	-0.254	-0.155
	(0.065)	(0.076)	(0.078)	(0.078)
Slope of the belief error, $\beta'(y)$	0.313	0.431	0.286	0.168
	(0.082)	(0.092)	(0.096)	(0.086)
Slope of the cost, $\bar{c}'(y)$	0.035	-0.008	0.006	0.027
	(0.072)	(0.085)	(0.080)	(0.088)
Participants, WTP and beliefs	967	1,016	1,021	1,028
Participants, gain	1,049	511	524	526

Table 8: Decomposition of the Slope of Willingness to Pay in the Baseline Score

Notes: Table 8 reports the terms of Equation (5). Each entry is the slope of the row quantity in the baseline score from a regression with demographic controls (college education, gender, race, income, and age). The slope of the gain is the coefficient on the interaction of treatment with the baseline score in Equation (11), estimated on the format's treated participants and the control group. The slope of the belief error is the slope of the perceived gain minus the slope of the gain. The slope of the cost is the slope of the recovered cost $c_{im} = \hat{\tau}_{im} - W_{im}$. By construction, the first row equals the second plus the third minus the fourth. Column (1) averages WTP, the perceived gain, and the cost across the three formats and uses the 967 participants whose three price lists are all consistent; Columns (2) through (4) use the format's WTP, perceived gain, and cost for every participant whose price list for that format is consistent. The slope of the gain uses all 1,049 participants in Column (1) and, in Columns (2) through (4), the participants assigned to that format and the control group. Standard errors from 500 bootstrap replications over participants are reported in parentheses.

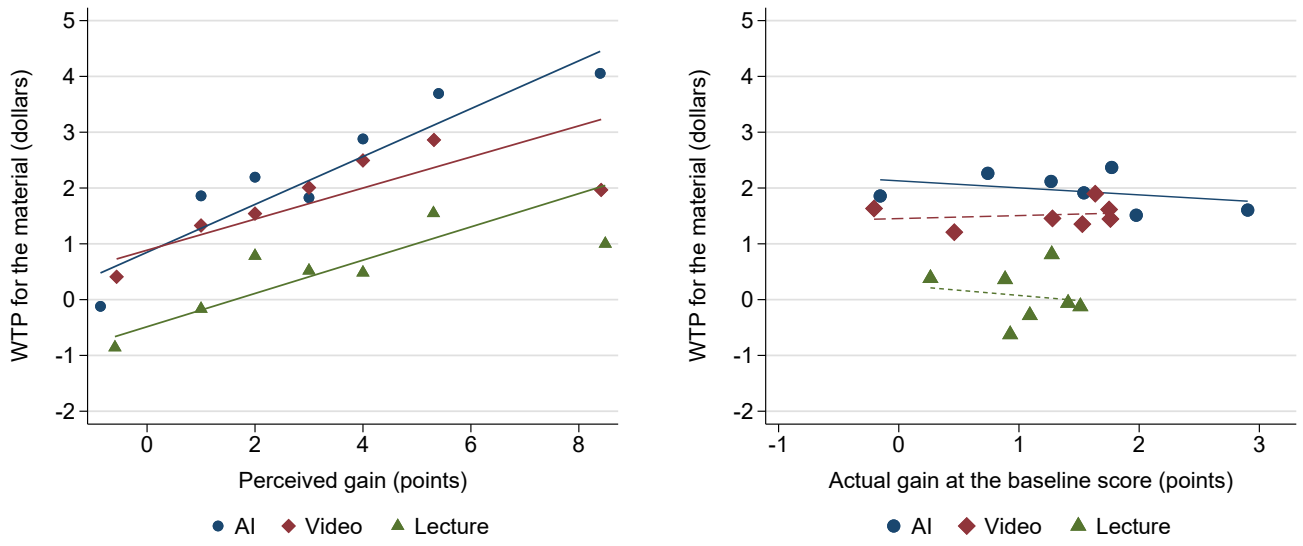
increase associated with a one-standard-deviation increase in the perceived gain ($0.76). Holding perceived difficulty fixed, a one-point increase in the perceived time needed to complete the material is associated with an increase in average WTP of $0.24 (s.e. = 0.10).

WTP does not predict the actual treatment effect of the material. Columns (2) through (5) of Table 6 add WTP and its interaction with the treatment indicator to Equation (11):

$$Y_{i,1} = \alpha + \tau T_i + \rho \tilde{Y}_{i,0} + \phi\, T_i \tilde{Y}_{i,0} + \psi \tilde{W}_i + \kappa\, T_i \tilde{W}_i + \tilde{\mathbf{X}}_i' \boldsymbol{\theta} + T_i \tilde{\mathbf{X}}_i' \boldsymbol{\lambda} + \varepsilon_i \tag{12}$$

where $\tilde{W}_i$ is participant i's WTP minus its sample mean, using WTP averaged across the three formats in Column (2) and WTP for the format being evaluated in Columns (3) through (5). The parameter ψ is the slope of the endline score on WTP in the control group, and κ is the change in the treatment effect for each additional dollar of WTP; both hold the baseline score and the demographic controls fixed. The remaining variables and parameters are defined as in Equation (11). In Column (2), the estimate of κ is 0.00 (s.e. = 0.03); in none of Columns (2) through (5) is the estimate of κ statistically significant at the 10% level.

Table 10 replaces WTP in Equation (12) with the perceived gain $\hat{\tau}_{im}$, which tests Prediction 2 of Section 2 directly. Under rational expectations the gain rises one for one with the perceived



(a) WTP versus perceived gain

(b) WTP versus actual gain

Figure 7: WTP versus Perceived and Actual Gain

Notes: Both panels plot willingness to pay (WTP, dollars) for each format. Panel (a) is a binned scatter plot, with ten quantile bins, of WTP for each format against the perceived gain from that format: the forecast of the endline score with the format minus the forecast without the material, both elicited before random assignment. Panel (b) has one point per format and baseline score, with scores of 3 and 4 pooled. The horizontal axis is the actual gain from the format at that baseline score: the coefficient on assignment to the format in a regression of the endline score on assignment and baseline-score fixed effects, among participants assigned to the format or to the control group. The vertical axis is mean WTP for the format among all participants with that baseline score. Lines are linear fits for each format. Both panels use every participant whose price list for the format is consistent. Pooling the three formats with format fixed effects and standard errors clustered by participant, the slope of WTP is \$0.35 per point of perceived gain (s.e. = 0.05) and -0.11 dollars per point of actual gain (s.e. = 0.18); the second standard error does not account for estimation error in the actual gain.

gain, so the coefficient on the interaction of treatment with the centered perceived gain equals one. Pooling the formats, the estimate is -0.03 (s.e. = 0.05); the hypothesis that it equals one is rejected ($p < 0.001$) and the hypothesis that it equals zero is not ($p = 0.473$). The format-specific estimates in Columns (2) through (4) are between −0.08 and 0.01. Participants who expect a larger gain from the material do not obtain one.

In a regression of WTP for the AI material on the AI Belief Index alone, a one-point increase in the index is associated with an increase in WTP of \$2.07 (s.e. = 0.25; Appendix Table A19, Column 7). The index is unrelated to the baseline score (Appendix Table A20). The AI Belief Index is the average of four standardized survey items on general AI use, AI use for personal finances, the belief that AI outperforms the participant, and trust in AI. The mean of the AI Belief Index is 0.13 points higher among college-educated participants than among participants without a college degree, a difference equal to 0.21 standard deviations of the index (Appendix Table A21). Participants answer the AI questions after the endline decisions, but the mean of

	(1) Average WTP, all formats	(2) WTP for AI	(3) WTP for video	(4) WTP for lecture
Perceived gain	0.282***	0.412***	0.277***	0.293***
	(0.059)	(0.061)	(0.059)	(0.065)
Perceived difficulty (0 to 10)	-0.190***	-0.214***	-0.173***	-0.320***
	(0.069)	(0.058)	(0.056)	(0.071)
Observations	967	1,016	1,021	1,028
R^2	0.049	0.093	0.043	0.066
Demographic controls	Yes	Yes	Yes	Yes

Table 9: Perceived Gain, Perceived Difficulty, and Willingness to Pay

Notes: Table 9 reports OLS regressions of willingness to pay (WTP, in dollars) for each format on two beliefs about that format, both elicited before random assignment. The perceived gain is the forecast of the endline score with the format minus the forecast without the material (points out of 10). Perceived difficulty is rated from 0 to 10, where 10 is very difficult. In Column (1), the dependent variable is the average WTP across the three formats and each belief is averaged across formats; the sample is the 967 participants whose three price lists are all consistent. Columns (2) through (4) report WTP for the AI, video, and lecture materials; each uses every participant whose price list for that format is consistent, so the samples differ slightly across columns. All columns also control for the perceived time needed to complete the material (0–10), college education, gender, race, income, and age. Robust standard errors are reported in parentheses.

the index does not differ between participants assigned to the AI material and the control group (difference = 0.00, s.e. = 0.06).

Table 11 shows how participants misjudge their baseline score. In Column (1), for each additional point (or question) answered correctly, participants increase their estimate of their baseline score by only 0.32 points. Part of this flattening is mechanical. Because the baseline score measures knowledge with noise (Section 2.1), even a participant who knew their own knowledge exactly would produce a slope equal to the reliability of the test rather than one. The slope of the endline score on the baseline score in the control group, 0.61, estimates that reliability. The slope of 0.32 is about half as large, so estimates rise with the baseline score much less than measurement noise alone would imply.

Column (2) reports the same pattern as an overestimate: participants with lower baseline scores systematically *overestimate* their baseline score. We refer to this as overconfidence, in the sense of overestimation of one's own performance (Moore and Healy, 2008). Participants with lower baseline scores overestimate by more: each additional point on the baseline score reduces the overestimate by 0.68 points. Figure 8 shows the share of participants who overestimate their baseline score and the distribution of the estimate at each baseline score, and Appendix Table A13 compares the estimated baseline score of participants with baseline scores of 5 or below and above 5.

Taken together with Tables 7 and 10, these results imply a misalignment between demand and returns. The gain from the material falls with the baseline score while the perceived gain does

	Endline score (0–10)			
	(1) All formats (average)	(2) AI format	(3) Video format	(4) Lecture format
Assigned material	1.275***	1.418***	1.277***	1.162***
	(0.114)	(0.150)	(0.146)	(0.150)
Assigned material × baseline score	-0.247***	-0.324***	-0.249***	-0.153*
	(0.061)	(0.078)	(0.076)	(0.079)
Assigned material × perceived gain	-0.033	0.009	-0.075	-0.039
	(0.046)	(0.050)	(0.057)	(0.066)
Baseline score	0.604***	0.605***	0.605***	0.605***
	(0.050)	(0.051)	(0.051)	(0.051)
Perceived gain	0.032	0.014	0.033	0.037
	(0.036)	(0.030)	(0.034)	(0.036)
p-value: interaction with perceived gain = 0	0.473	0.856	0.188	0.558
p-value: interaction with perceived gain = 1	0.000	0.000	0.000	0.000
Control-group mean	6.80	6.80	6.80	6.80
Observations	1,049	511	524	526
R^2	0.242	0.344	0.337	0.324

Table 10: Heterogeneity in the Treatment Effect by the Perceived Gain

Notes: Table 10 reports estimates of Equation (12) with the perceived gain in place of WTP. The dependent variable is the endline score (0–10). The assigned-material indicator is interacted with the baseline score, the perceived gain, and each demographic control (college education, gender, race, income, and age), and every interacted covariate is centered at its sample mean, so the coefficient on the assigned-material indicator is the average treatment effect (Lin, 2013). The perceived gain is the forecast of the endline score with the format minus the forecast without the material, both elicited before random assignment. Column (1) compares all participants assigned to any material with the control group and uses the perceived gain averaged across formats; Columns (2) through (4) compare participants assigned to the AI, video, and lecture materials with the control group and use the perceived gain for that format. The p-value rows test whether the coefficient on assigned material × perceived gain equals zero and whether it equals one, its value under rational expectations. Robust standard errors are reported in parentheses.

not, so the belief error rises with the baseline score. Participants with low baseline scores expect roughly the gain they obtain; participants with a perfect baseline score expect a gain of about two points and obtain none. Relative to the Roy benchmark of Section 2, in which WTP would fall with the baseline score at the rate of the gain, high scorers demand too much of the material and low scorers demand no more than high scorers.

4.2.2 Reducing the Costs of Education

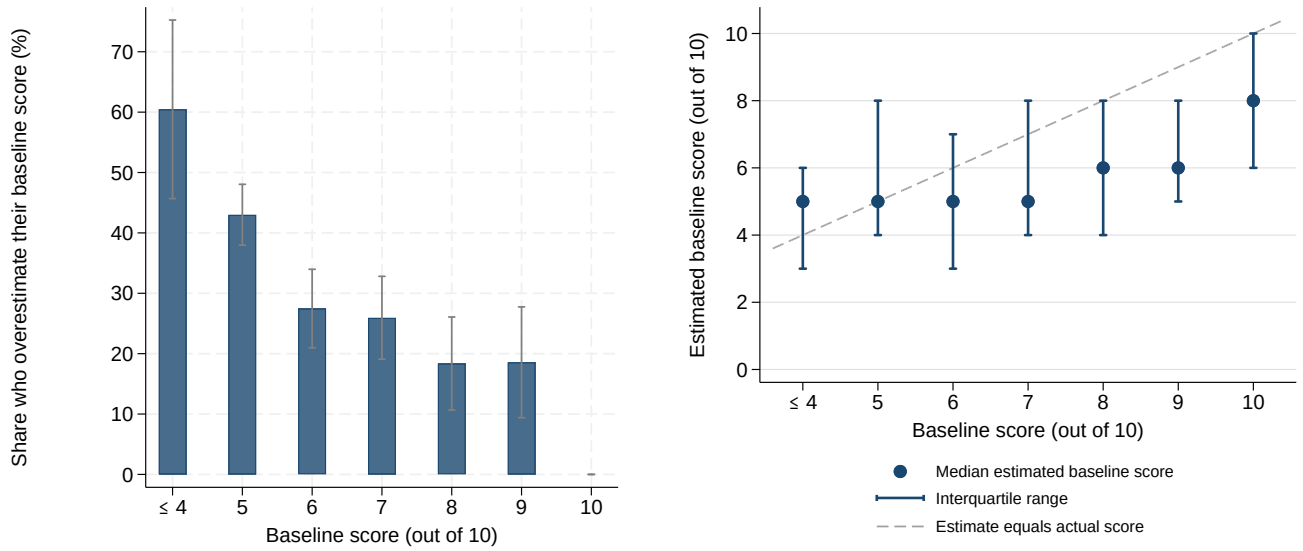
We test whether lowering the perceived cognitive cost of education can correct these selection distortions by offering participants different versions of the education material. The three formats are a university lecture (high cognitive cost), a video module, and an interactive AI learning assistant (low cognitive cost).

Figure 9 shows that we are successfully able to vary perceived difficulty of the education material. We elicit perceived difficulty, or perceived cognitive cost, using two measures (both elicited

	(1) Estimated baseline score	(2) Estimated minus actual baseline score
Baseline score	0.320***	-0.680***
	(0.044)	(0.044)
Observations	1,049	1,049
R^2	0.054	0.179

Table 11: Estimated and Actual Baseline Scores

Notes: Table 11 compares participants' estimates of their baseline score with their actual baseline score. The estimated baseline score is the participant's incentivized estimate of their own baseline score (0–10), elicited after the baseline questions and before the willingness-to-pay questions. Column (1) regresses participants' estimated baseline score on their actual baseline score. Column (2) regresses the estimated baseline score minus the actual baseline score on the actual baseline score; positive values indicate overestimation. The slope in Column (2) equals the slope in Column (1) minus one. Both regressions include controls for college education, gender, race, income, and age. Robust standard errors are reported in parentheses.



(a) Share who overestimate their baseline score

(b) Estimated baseline score

Figure 8: Estimated and Actual Baseline Scores

Notes: Both panels group participants by their actual baseline score, with scores of 4 or below pooled. Panel (a) plots the share of participants whose estimated baseline score is above their actual score; bars show 95% confidence intervals. Panel (b) plots the distribution of the estimated baseline score: the point is the median and the vertical bar spans the 25th to 75th percentiles. Points above the dashed 45-degree line are overestimates. The sample is all 1,049 participants.

before participants complete the education material): 1) how difficult participants perceive the education material to be, and 2) how much time they expect it will take them to complete and fully understand that version of the education material. On average, participants rate the AI material as the least difficult, the video material as the second most difficult, and the lecture as the most difficult. Participants who ranked a format first explained their choice in a free response and then classified that response into one or more listed reasons; the most common reason for preferring the AI and video materials is that they are easier to understand and clearer. The share of participants who chose each reason can be found in Appendix Figures A.16 and A.17.

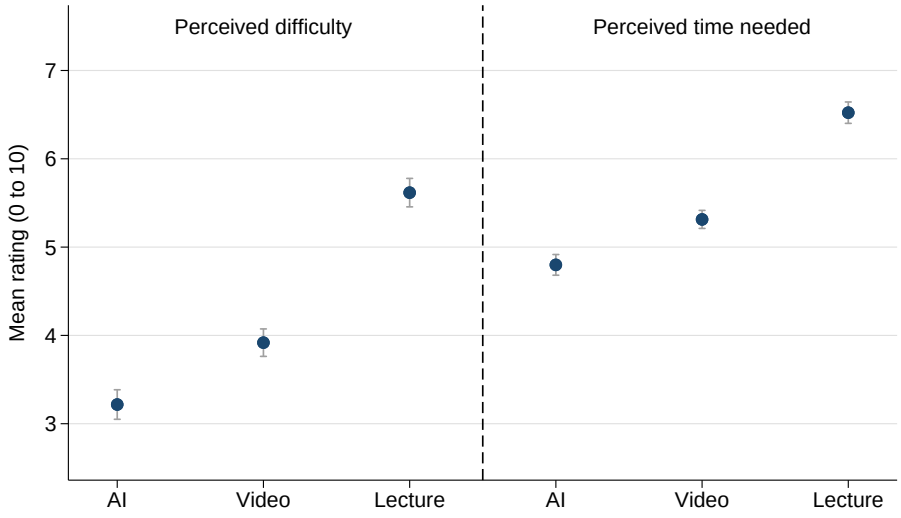


Figure 9: Perceived Cost of Education Material

Notes: Figure 9 plots the mean of two ratings for each format, elicited before random assignment: perceived difficulty and the expected time needed to complete the material, both on a 0–10 scale where 10 is the most difficult or the most time. Bars show 95% confidence intervals. Participants rate the AI material as the least difficult, the video as slightly more difficult, and the lecture as the most difficult, in line with how we described each format.

Figure 10 shows the perceived gain from each version of the material and Figure 6(a) shows how demand for education varies with each version. We find that the video and AI versions of the education material (the technology-backed versions designed to reduce cognitive difficulty) have a larger perceived gain than the lecture and increase demand across all ability levels. This suggests that reducing the cognitive cost of education does increase aggregate demand, however, lowering the cost of education does not change how demand varies with baseline knowledge.

While demand increased across the board, the relationship between baseline ability and WTP is the same for the AI and video formats as for the lecture. Table 12 reports the slope of WTP in the baseline score for each format from a regression stacked across formats, with standard errors clustered by participant: \$0.10 per baseline point for the AI material (s.e. $= 0.08$), \$0.02 for the video (s.e. $= 0.07$), and \$0.00 for the lecture (s.e. $= 0.08$); equality of the three slopes is not rejected ($p = 0.513$). The slopes of the perceived gain ($p = 0.202$) and of the recovered cost

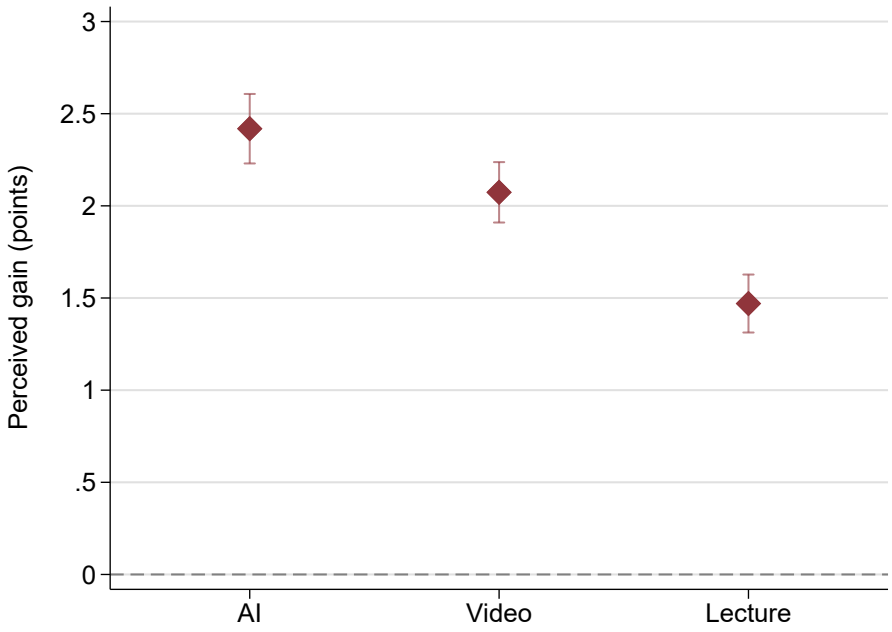


Figure 10: Perceived Gain, by Format

Notes: Figure 10 plots the mean perceived gain from each format: the forecast of the endline score with the format minus the forecast without the material, both elicited before random assignment, for all 1,049 participants. Bars show 95% confidence intervals. Participants expect the largest gain from the AI material, followed by the video and the lecture, the reverse of the order of perceived difficulty in Figure 9.

($p = 0.983$) are also not statistically different across formats. The AI and video formats shift demand up at every baseline score without changing its slope, which is Prediction 3 of Section 2: a format changes which participants select in only if its premium over the lecture is larger for low scorers, and it is not. Lowering the cognitive cost of the material therefore raises demand but does not make it track the gain from the material more closely. One possible explanation is the belief errors documented above: participants mispredict how much they will gain from the material and overestimate their baseline ability.

4.2.3 Technology and the Extensive Margin

While reducing cognitive costs does not change who is at the top of the demand distribution, it may still bring in participants who would not pay for the lecture. Because we elicit willingness to pay for all three formats from every participant, we can compare the set of participants who would pay for the lecture with the set who would pay for at least one format once the technology-backed formats are available. This comparison uses the 967 participants whose willingness to pay we observe for all three formats. It excludes 82 of the 1,049 participants in the final sample. On one of the three price lists, these participants switch more than once, or switch from completing the material to skipping it as its price falls, so the list does not identify their willingness to pay for

	(1) Willingness to pay (dollars)	(2) Perceived gain (points)	(3) Recovered cost
Slope in baseline score, AI format	0.101	0.089*	0.001
	(0.084)	(0.052)	(0.084)
Slope in baseline score, video format	0.023	0.038	0.006
	(0.073)	(0.046)	(0.078)
Slope in baseline score, lecture format	0.004	0.015	0.016
	(0.081)	(0.046)	(0.086)
p: equal slopes across formats	0.513	0.202	0.983
Observations (participant × format)	3,065	3,147	3,065

Table 12: Do the Formats Change the Slope in the Baseline Score?

Notes: Table 12 tests whether the slope in the baseline score of willingness to pay (WTP), of the perceived gain, and of the recovered cost differs across the AI, video, and lecture formats. If a format attracted relatively more low scorers, its WTP slope would be lower than the lecture's. The table stacks the three formats, so each participant contributes up to three observations, and regresses the column variable on format indicators, the interaction of each format indicator with the baseline score, and demographic controls (college education, gender, race, income, and age). Each row reports the slope of the column variable in the baseline score for one format. The perceived gain is the forecast of the endline score with the format minus the forecast without the material; the recovered cost is the perceived gain minus WTP. Columns (1) and (3) use every participant whose price list for the format is consistent; Column (2) uses all 1,049 participants. The p-value row reports the p-value of the hypothesis that the three slopes are equal. Standard errors clustered by participant are reported in parentheses.

that format. Of the 967 participants, 46.4% have a negative willingness to pay for the lecture. Of these lecture rejectors, 47.9% have a positive willingness to pay for the AI or video format, which is 22.2% of the sample (Table 13).

Figure 11 plots the share of participants willing to pay for each format at each price. The share willing to pay for at least one format is higher than the share willing to pay for the lecture at every price, so the technology-backed formats bring in additional participants across the full range of prices, not only when education is free. The AI format brings in more lecture rejectors than the video format: 18.3% of the sample would pay for the AI format but not the lecture, and 14.2% would pay for the video format but not the lecture (paired test of equal shares, $p < 0.001$). The two groups overlap: 10.2% of the sample would pay for both the AI and video formats but not the lecture, 8.1% would pay for the AI format alone, and 3.9% would pay for the video format alone.

Appendix Table A23 describes these groups. Among lecture rejectors, the 78 participants who would pay for the AI format alone rate their own financial decision-making ability lower relative to AI than the 38 participants who would pay for the video format alone. On a scale from 1 (much worse than AI) to 5 (much better than AI), the mean answer is 2.28 in the AI-only group and 2.84 in the video-only group ($p = 0.009$). The AI-only group is also less certain about its estimate of its own baseline score: on a scale from 0 to 100, mean certainty is 49 in the AI-only group and 63 in the video-only group ($p = 0.007$). The share female is 64.1% in the AI-only group and 55.3% in

	Participants	Share of the sample (%)	Share of lecture rejectors (%)
Would pay for the lecture	518	53.6	
Would not pay for the lecture (lecture rejectors)	449	46.4	100.0
Would pay for the AI or video format	215	22.2	47.9
Both the AI and video formats	99	10.2	22.0
AI format only	78	8.1	17.4
Video format only	38	3.9	8.5
Would pay for neither	234	24.2	52.1
Total	967	100.0	

Table 13: Participants Who Would Not Pay for the Lecture, by Willingness to Pay for the AI and Video Formats

Notes: Table 13 classifies participants by the sign of their willingness to pay for each education format. Willingness to pay takes odd values from −11 to 11 dollars and is never zero, so a negative value means the participant would not pay for the format and a positive value means the participant would. Lecture rejectors are participants with a negative willingness to pay for the lecture. The indented rows split the lecture rejectors by whether they would pay for both the AI and video formats, the AI format only, the video format only, or neither. The second column reports shares of the 967 participants whose willingness to pay we observe for all three formats, and the third column reports shares of the lecture rejectors. The sample excludes the 82 participants whose choices on one of the three price lists do not identify a willingness to pay.

the video-only group, a difference that is not statistically significant ($p = 0.37$). Pooling the two formats, the participants they bring in do not differ significantly from the rest of the sample in baseline score, gender, age, college education, income, race, or prior financial education (smallest p-value across the seven comparisons: $p = 0.14$).

The technology-backed formats increase the share of participants willing to pay for education without changing the relationship between baseline ability and demand. Within the AI and video formats, as within the lecture, willingness to pay does not vary with the baseline score, and the participants these formats bring in answer 6.67 of 10 baseline questions correctly on average, compared with 6.56 for the rest of the sample ($p = 0.42$). Among participants randomly assigned to an education format, the number of correct answers rises by 1.66 for the participants the technology-backed formats bring in and by 1.58 for the rest of the sample ($p = 0.66$). The participants the technology-backed formats bring in are therefore similar to the rest of the sample in both baseline knowledge and how much their score changes with the material.

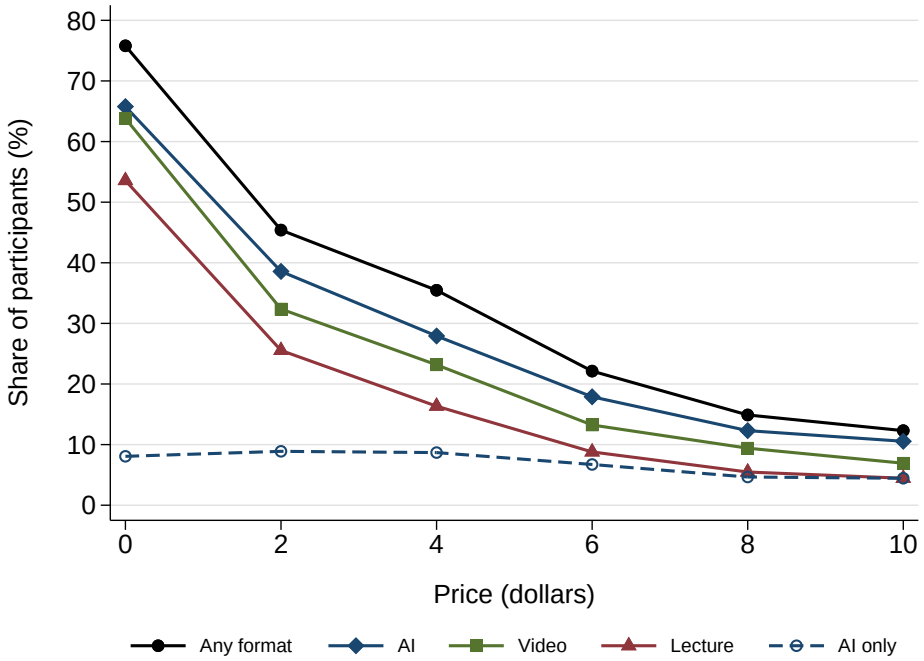


Figure 11: Demand Curves by Format

Notes: Figure 11 plots the share of participants whose willingness to pay for each format is at or above the price on the horizontal axis. The "any format" line is the share with willingness to pay at or above the price for at least one of the three formats. The "AI only" line is the share with willingness to pay at or above the price for the AI format but below the price for both the video and lecture formats. The willingness-to-pay elicitation uses two-dollar increments, so the curves are evaluated at even prices. The sample is the 967 participants whose willingness to pay we observe for all three formats.

5 Conclusion

This paper investigates a central friction in human capital accumulation: the self-selection mechanisms that prevent individuals with high potential returns from seeking education. Standard human capital theory predicts that individuals will invest in education when the expected marginal returns exceed the marginal costs. Our findings challenge this assumption in the context of financial literacy education, documenting a pattern in which the individuals who stand to benefit most from education are no more likely to demand it than those who would benefit least.

Using administrative data from a large university, we first show that enrollment in financial education is characterized by strong positive selection. Students from higher socioeconomic backgrounds and those with stronger quantitative skills effectively select into the course, while underrepresented minorities and those from less privileged backgrounds select out. Our experiment suggests that beliefs are an important part of the explanation. The cost of education implied by participants' choices does not vary with baseline knowledge, whereas the gap between the perceived and the actual gain increases with it. Participants expect roughly the same gain from education regardless of their baseline knowledge, while the actual gain is largest for those who start lowest.

Demand therefore fails to track returns: relative to what they stand to gain, low-ability individuals value education too little compared with high-ability individuals. Low-ability individuals also overestimate their baseline knowledge.

Formats designed to lower the cognitive cost of learning do not appear to correct this pattern. While AI- and video-based tools increase aggregate demand relative to traditional lecture-based learning, they do not change how demand varies with baseline knowledge. Demand does not shift toward lower-ability individuals, suggesting that misperceived returns, and not only the cognitive cost of learning, may be an important constraint on take-up.

These results have important policy implications. Governments and institutions often rely on individuals to "opt-in" to financial education programs to improve decision-making and reduce inequality. Furthermore, even in cases where financial education is required, individuals have to "opt-in" on the intensive margin and choose to allocate attention and effort. Our findings suggest that self-selection patterns may render these education resources less effective, and may even exacerbate inequality by subsidizing the human capital accumulation of individuals who are already financially sophisticated. Lowering the real and cognitive costs of participation does raise take-up: in our experiment, the AI and video formats bring in the 22.2% of participants who would not pay for the lecture but would pay for one of these formats. Because these formats raise demand at every baseline score, however, lowering costs alone is unlikely to redirect education toward the people who stand to gain the most while beliefs about its benefits remain miscalibrated.

When the returns to education are hard to assess, the individuals who would benefit most may be least able to recognize it. Although we study financial literacy, the same friction may arise in other settings where take-up is voluntary and benefits are uncertain, such as preventive health care or job retraining. Recognizing that those most in need of a resource may undervalue it is an important consideration for policy design and a promising direction for future research.

References

Ambuehl, Sandro, B. Douglas Bernheim, and Annamaria Lusardi, "Evaluating Deliberative Competence: A Simple Method with an Application to Financial Choice," *American Economic Review*, November 2022, *112* (11), 3584–3626.

Becker, Gordon M., Morris H. Degroot, and Jacob Marschak, "Measuring Utility by a Single-Response Sequential Method," *Behavioral Science*, 1964, *9* (3), 226–232.

Bernheim, B. Douglas and Dmitry Taubinsky, "Behavioral Public Economics," in "Handbook of Behavioral Economics: Applications and Foundations 1," Vol. 1, Elsevier, 2018, pp. 381–516.

Bhattacharya, Utpal, Andreas Hackethal, Simon Kaesler, Benjamin Loos, and Steffen Meyer, "Is Unbiased Financial Advice to Retail Investors Sufficient? Answers from a Large Field Study," *The Review of Financial Studies*, April 2012, *25* (4), 975–1032.

Bruhn, Miriam, Gabriel Lara Ibarra, and David McKenzie, "The Minimal Impact of a Large-Scale Financial Education Program in Mexico City," *Journal of Development Economics*, May 2014, *108*, 184–189.

Card, David, "The Causal Effect of Education on Earnings," in "Handbook of Labor Economics," Vol. 3, Elsevier, 1999, pp. 1801–1863.

Cunha, Flavio and James J. Heckman, "The Technology of Skill Formation," *American Economic Review*, 2007, *97* (2), 31–47.

Engel, Christoph and Oliver Kirchkamp, "How to Deal with Inconsistent Choices on Multiple Price Lists," *Journal of Economic Behavior & Organization*, April 2019, *160*, 138–157.

Fernandes, Daniel, John G. Lynch, and Richard G. Netemeyer, "Financial Literacy, Financial Education, and Downstream Financial Behaviors," *Management Science*, August 2014, *60* (8), 1861–1883.

Gemmo, Irina, Pierre-Carl Michaud, and Olivia S. Mitchell, "Selection into Financial Education and Effects on Portfolio Choice," September 2023.

Hastings, Justine S., Brigitte C. Madrian, and William L. Skimmyhorn, "Financial Literacy, Financial Education and Economic Outcomes," *Annual Review of Economics*, May 2013, *5*, 347.

Heckman, James J. and Edward Vytlacil, "Structural Equations, Treatment Effects, and Econometric Policy Evaluation," *Econometrica*, 2005, *73* (3), 669–738.

_ , **Sergio Urzua, and Edward Vytlacil**, "Understanding Instrumental Variables in Models with Essential Heterogeneity," *The Review of Economics and Statistics*, 2006, *88* (3), 389–432.

Huston, Sandra J., "Financial Literacy and the Cost of Borrowing," *International Journal of Consumer Studies*, 2012, *36* (5), 566–572.

Ibarra, Gabriel Lara, David McKenzie, and Claudia Ruiz Ortega, "Learning the Impact of Financial Education When Take-Up Is Low," November 2017.

Jack, B. Kelsey, Kathryn McDermott, and Anja Sautmann, "Multiple Price Lists for Willingness to Pay Elicitation," *Journal of Development Economics*, November 2022, *159*, 102977.

Kaiser, Tim, Annamaria Lusardi, Lukas Menkhoff, and Carly Urban, "Financial Education Affects Financial Knowledge and Downstream Behaviors," *Journal of Financial Economics*, August 2022, *145* (2), 255–272.

Lin, Winston, "Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman's Critique," *The Annals of Applied Statistics*, 2013, *7* (1), 295–318.

Luedtke, Allison Oldham and Carly Urban, "High School Financial Education Courses in the United States.: What Is the Importance of Setting State Policies?," *Journal of Financial Literacy and Wellbeing*, 2023, *1* (3), 431–449.

Lusardi, Annamaria and Carlo de Bassa Scheresberg, "Financial Literacy and High-Cost Borrowing in the United States," Technical Report, National Bureau of Economic Research 2013.

_ **and Olivia S. Mitchell**, "Financial Literacy and Planning: Implications for Retirement Wellbeing," May 2011.

_ **and** _ , "The Importance of Financial Literacy: Opening a New Field," *Journal of Economic Perspectives*, November 2023, *37* (4), 137–154.

_ **, Pierre-Carl Michaud, and Olivia S. Mitchell**, "Optimal Financial Knowledge and Wealth Inequality," *Journal of Political Economy*, April 2017, *125* (2), 431–477.

Miller, Margaret, Julia Reichelstein, Christian Salas, and Bilal Zia, "Can You Help Someone Become Financially Capable? A Meta-Analysis of the Literature," *The World Bank Research Observer*, August 2015, *30* (2), 220–246.

Mitchell, Olivia S. and Annamaria Lusardi, "Financial Literacy and Economic Outcomes: Evidence and Policy Implications," *The Journal of Retirement*, June 2015, *3* (1), 107–114.

Moore, Don A. and Paul J. Healy, "The Trouble with Overconfidence.," *Psychological Review*, April 2008, *115* (2), 502–517.

Roy, A. D., "Some Thoughts on the Distribution of Earnings," *Oxford Economic Papers*, 1951, *3* (2), 135–146.

Urban, Carly, Maximilian Schmeiser, J. Michael Collins, and Alexandra Brown, "The Effects of High School Personal Financial Education Policies on Financial Behavior," *Economics of Education Review*, October 2020, *78*, 101786.

van Rooij, Maarten C. J., Annamaria Lusardi, and Rob J. M. Alessie, "Financial Literacy and Retirement Planning in the Netherlands," *Journal of Economic Psychology*, August 2011, *32* (4), 593–608.

Walters, Christopher R., "The Demand for Effective Charter Schools," *Journal of Political Economy*, December 2018, *126* (6), 2179–2223.

Willis, Lauren E., "The Financial Education Fallacy," *American Economic Review*, May 2011, *101* (3), 429–434.

Willis, Robert J. and Sherwin Rosen, "Education and Self-Selection," *Journal of Political Economy*, October 1979, *87* (5, Part 2), S7–S36.

A Appendix

A.1 Conceptual Framework: Lab Special Case and Belief Microfoundation

A.1.1 The experiment as a special case

In the experiment the score is bounded at 10 and the material teaches a single skill, the Rule of 72. Write the gain as the share $\pi_i \in [0,1]$ of the participant's remaining mistakes that the material removes:

$$\tau_i = \pi_i\Big(10 - Y_i(0)\Big). \tag{13}$$

Equation (13) is a substitutes technology: for any $\pi_i > 0$, the gain falls with the score without education, and it is zero for a participant who would answer every question correctly without the material. Writing the perceived gain the same way, $\hat{\tau}_{im} = \hat{\pi}_{im}(10 - \hat{Y}_i(0))$, where $\hat{\pi}_{im} = \hat{\tau}_{im}/(10 - \hat{Y}_i(0))$ is the share of their perceived remaining mistakes the participant expects format m to remove. The belief error in Equation (3) then becomes

$$\hat{\tau}_{im} - \tau_i = \Big(\hat{\pi}_{im} - \pi_i\Big)\Big(10 - \hat{Y}_i(0)\Big) + \pi_i\Big(Y_i(0) - \hat{Y}_i(0)\Big). \tag{14}$$

The first term is an error about how much of the content the participant will absorb. The second is the error in the forecast of the uneducated outcome, scaled by the share the material would remove. The pattern described in Prediction 1, a perceived gain that is flat in the baseline score, corresponds to $\hat{\pi}_{im}$ rising with the baseline score, because the perceived remaining gap $10 - \hat{Y}_i(0)$ shrinks as the score rises. The share $\hat{\pi}_{im}$ is undefined for participants with $\hat{Y}_i(0) = 10$; tests that use it exclude those participants and report their number.

A.1.2 Belief microfoundation: noisy self-assessment

This subsection states the model behind the paragraph on overconfidence in Section 2. Let θ_i be participant i's ability, with population distribution $\theta \sim N(\mu, \sigma_0^2)$, and let the score without education be an increasing function of ability, so statements about θ carry over to $Y_i(0)$. The participant does not observe θ_i. She observes a signal

$$s_i = \theta_i + \varepsilon_i, \qquad \varepsilon_i \sim N\Big(0, \sigma^2(\theta_i)\Big), \qquad \sigma^{2\prime}(\theta) < 0, \tag{15}$$

and forms $\hat{\theta}_i = \mathbb{E}[\theta_i \mid s_i]$ by updating the population prior. The assumption $\sigma^{2\prime}(\theta) < 0$ states that the signal is more precise for participants with higher ability. The reason is that assessing one's own financial decisions uses the skill being assessed: a participant who can price a compound-interest contract can tell which of her past choices were right, while one who cannot is guessing

about her own performance.[8] The posterior mean is

$$\hat{\theta}_i = \omega(\theta_i)\, s_i + \Big(1 - \omega(\theta_i)\Big)\mu, \qquad \omega(\theta) = \frac{\sigma_0^2}{\sigma_0^2 + \sigma^2(\theta)}, \tag{16}$$

where the weight on the own signal, $\omega(\theta) \in (0, 1)$, rises with ability.

Implications. Taking the expectation of Equation (16) given θ, $\mathbb{E}[\hat{\theta} \mid \theta] = \omega(\theta)\theta + (1 - \omega(\theta))\mu$. Three implications follow.

(i) *Forecasts rise less than one for one with the true score.* The slope of $\mathbb{E}[\hat{\theta} \mid \theta]$ in θ is $\omega(\theta) + \omega'(\theta)(\theta - \mu)$. With constant precision the slope is $\omega \in (0, 1)$. With precision rising in ability the slope stays in $(0, 1)$ whenever $\omega'(\theta)|\theta - \mu| < 1 - \omega(\theta)$, which holds for smooth precision schedules. Column (1) of Table 11 estimates this slope.

(ii) *Overconfidence is concentrated at low scores.* The mean error $b(\theta) = \mathbb{E}[\hat{\theta} \mid \theta] - \theta = (1 - \omega(\theta))(\mu - \theta)$ is positive for $\theta < \mu$ and negative for $\theta > \mu$. Its derivative, $b'(\theta) = -(1 - \omega(\theta)) - \omega'(\theta)(\mu - \theta)$, is negative under the condition in (i). Below the mean, the second term adds to the decline: low-ability participants are both far from the prior mean and reliant on it. Column (2) of Table 11 and Figure 8 report this pattern. Note that the slope in Column (2) equals the slope in Column (1) minus one, so the two columns are one estimate reported two ways.

(iii) *Second moments.* The conditional variance of the error $\hat{\theta} - \theta$ is $\omega(\theta)^2\sigma^2(\theta)$ plus the squared mean error, and both fall with ability. Stated confidence, if it reports the posterior precision, rises with ability and is more closely related to accuracy for high scorers. Any source of additional feedback about one's own financial decisions, such as prior coursework or talking about money at home, lowers σ^2 at a given ability and reduces miscalibration.

Implications (i) and (ii) are also produced by regression to the mean with constant precision, so they are consistent with the model rather than evidence for it. Implication (iii) is specific to precision that rises with ability and is the test that distinguishes the mechanism from constant-precision shrinkage.

Tests of implication (iii). Table A1 reports the moments of the estimation error in the baseline score, the estimated baseline score minus the actual one, by baseline score. The mean error falls from +1.23 points at baseline scores of 4 or below to −2.40 points at 10, and the share of participants who overestimate their score falls from 60% to 0%, as implication (ii) states. The standard deviation of the error around its conditional mean falls only slightly, from 2.86 points

[8] The assumption follows the overconfidence-as-inference literature (Moore and Healy, 2008). It requires no self-serving motive; motivated beliefs would reinforce the pattern below, not replace it.

at a baseline score of 5 to 2.47 at 10, and the mean absolute error rises with the baseline score, by 0.07 points per baseline point (s.e. = 0.04; Column (1) of Table A2), because the mean error grows in absolute value faster than the dispersion around it shrinks. The dispersion evidence for precision that rises with ability is therefore weak.

The relationship between stated certainty and accuracy differs sharply by ability. Among participants with baseline scores of 5 or below, moving from certainty of 0 to certainty of 1 is associated with an absolute error that is 1.96 points larger (s.e. = 0.33), holding the baseline score fixed; among participants with scores above 5 the coefficient is -2.25 (s.e. = 0.29), so the same move in certainty is associated with a smaller absolute error. The difference between the two coefficients is statistically significant ($p < 0.001$). Confident low scorers are less accurate than unconfident low scorers, while confident high scorers are more accurate, which is the pattern precision rising with ability implies.

Prior exposure to financial decisions does not reduce miscalibration holding the baseline score fixed. In a regression of the absolute error on indicators for having taken a personal finance course, having talked about finances often or always while growing up, and owning stocks or bonds, with baseline-score fixed effects and demographic controls (Column (6) of Table A2), the three coefficients are 0.04 (s.e. = 0.14), -0.11 (s.e. = 0.16), and -0.12 (s.e. = 0.13), and they are jointly insignificant ($p = 0.714$). In the signed-error regression (Column (5)), participants who talked about finances often or always while growing up overestimate their baseline score by 1.1 points more than other participants with the same score (s.e. = 0.2).

Baseline score	N	Mean error (est. minus actual)	SD of error	Mean absolute error	Share overestimating	Mean certainty
≤ 4	43	1.23	2.65	2.30	0.60	0.48
5	372	0.65	2.86	2.28	0.43	0.55
6	182	-0.81	2.54	2.22	0.27	0.52
7	158	-1.39	2.55	2.42	0.26	0.50
8	98	-2.15	2.54	2.66	0.18	0.56
9	70	-2.59	2.49	2.96	0.19	0.56
10	126	-2.40	2.47	2.40	0.00	0.72

Table A1: Moments of the Estimation Error in the Baseline Score, by Baseline Score

Notes: The estimation error is the participant's estimate of their baseline score minus their actual baseline score (both out of 10). Each row reports, for participants with the given baseline score, the number of participants, the mean error, the standard deviation of the error, the mean absolute error, the share with a positive error, and the mean stated certainty about the estimate (0 to 1).

What the model does not deliver. Shrinkage applied to both forecasts in Equation (2) with the same weight compresses the perceived gain toward the population mean gain. It does not by itself make the mean perceived gain flat in the baseline score while the true gain declines, which is the pattern Prediction 1 requires for a rising belief error. That pattern is an additional

	(1) Absolute error	(2) Squared error	(3) Absolute error	(4) Absolute error	(5) Signed error	(6) Absolute error
Baseline score	0.070*	0.769***				
	(0.037)	(0.256)				
Certainty (0 to 1)			1.960***	-2.248***		
			(0.333)	(0.290)		
Took a personal finance course					0.289	0.035
					(0.183)	(0.136)
Talked about finances growing up					1.115***	-0.113
					(0.212)	(0.160)
Owns stocks or bonds					0.191	-0.116
					(0.180)	(0.130)
Sample	All	All	Score ≤ 5	Score > 5	All	All
Baseline-score fixed effects	No	No	Yes	Yes	Yes	Yes
Observations	1,049	1,049	415	634	1,049	1,049
R^2	0.012	0.020	0.095	0.127	0.224	0.022

Table A2: Dispersion of the Estimation Error, Certainty, and Prior Exposure

Notes: The estimation error is the estimated baseline score minus the actual baseline score. Columns (1) and (2) regress the absolute and squared error on the baseline score. Columns (3) and (4) regress the absolute error on stated certainty (0 to 1) with baseline-score fixed effects, separately for participants with baseline scores of 5 or below and above 5. Columns (5) and (6) regress the signed and absolute error on an indicator for having taken a personal finance course, an indicator for having talked about finances often or always while growing up, and an indicator for owning stocks or bonds, with baseline-score fixed effects. All columns control for college education, gender, race, income, and age. Robust standard errors are reported in parentheses.

empirical statement about how participants forecast the educated outcome, and Section 4.2 tests it directly.

A.2 Administrative Data: Additional Results

Table A4 compares UGBA 135 students who spend more and less time than the median on the course website, our measure of attention described in Section 3.2.

	All Mean (SE)	Fin Lit Class Mean (SE)	No Fin Lit Class Mean (SE)	Difference Δ	p-value
Male	0.45	0.47	0.44	0.03	0.03
	(0.01)	(0.01)	(0.01)		
White	0.26	0.22	0.27	-0.05	0.00
	(0.00)	(0.01)	(0.01)		
Asian	0.39	0.52	0.36	0.16	0.00
	(0.01)	(0.01)	(0.01)		
Underrepresented Minority	0.22	0.12	0.24	-0.11	0.00
	(0.00)	(0.01)	(0.00)		
Quantitative Majors	0.63	0.77	0.60	0.17	0.00
	(0.01)	(0.01)	(0.01)		
Cumulative GPA	3.52	3.58	3.51	0.07	0.00
	(0.00)	(0.01)	(0.00)		
Parents - No College	0.19	0.16	0.20	-0.03	0.00
	(0.00)	(0.01)	(0.00)		
Parents - No 4Yr College	0.29	0.25	0.30	-0.05	0.00
	(0.00)	(0.01)	(0.01)		
Application fee waiver	0.25	0.20	0.27	-0.06	0.00
	(0.00)	(0.01)	(0.01)		
N	8,766	1,444	7,322		

Table A3: UC Berkeley Student Characteristics by Financial Literacy Class - Class of 2023

Notes: Table A3 shows demographics of students from the 2023 graduating class who did and did not take a large personal finance class at UC Berkeley. Each row compares, with a two-sample t-test, students who take a large personal finance class at UC Berkeley (UGBA 135) to students who do not take the class from the same graduating year. Negative numbers and p-values less than 0.10 are highlighted in red. Students who self-selected into the personal finance class tend to have higher cumulative GPAs and are more likely to have quantitative majors, less likely to be underrepresented minorities, less likely to be first-generation college students, and less likely to have qualified for a University of California application fee waiver, which applicants receive based on family size and income (a proxy for low family income).

	All Mean (SE)	High Attention Mean (SE)	Low Attention Mean (SE)	Difference Δ	p-value
UGBA 135 GPA	3.63	3.72	3.53	0.18	0.00
	(0.02)	(0.02)	(0.03)		
Term GPA	3.46	3.54	3.37	0.17	0.00
	(0.02)	(0.03)	(0.03)		
Male	0.51	0.48	0.55	-0.07	0.02
	(0.02)	(0.02)	(0.02)		
White	0.19	0.16	0.22	-0.06	0.03
	(0.01)	(0.02)	(0.02)		
Asian	0.52	0.56	0.48	0.08	0.01
	(0.02)	(0.02)	(0.02)		
Underrepresented Minority	0.17	0.16	0.18	-0.02	0.38
	(0.01)	(0.02)	(0.02)		
Quantitative Majors	0.80	0.81	0.78	0.03	0.38
	(0.01)	(0.02)	(0.02)		
Parents - No College	0.19	0.20	0.17	0.03	0.27
	(0.01)	(0.02)	(0.02)		
Parents - No 4Yr College	0.27	0.30	0.25	0.04	0.15
	(0.01)	(0.02)	(0.02)		
Application fee waiver	0.22	0.25	0.20	0.04	0.13
	(0.01)	(0.02)	(0.02)		
N	977	489	488		

Table A4: Student Characteristics by Attention in UGBA 135 - 2023-2024 Academic Year

Notes: Table A4 shows student characteristics by attention for students who took UGBA 135 during the 2023-2024 academic year. Attention is measured as the amount of time (in minutes) the student spent on the UGBA 135 course website, which includes time spent watching videos and completing comprehension check quizzes, watching lecture videos, and reviewing course notes and other materials. Students are in the high attention category if they are above the median in time spent on the course website. Students in the low attention category are below the median. Note that time spent on the course website may be missing for a small number of students. Negative numbers and p-values less than 0.10 are highlighted in red.

	Takers	Droppers	Difference	
			Graduation-year FE	+ Covariates
Panel A. Credit bureau records, quarter before the class term				
Has a credit file (%)	87.0	85.8	0.90	0.75
			(0.86)	(0.85)
Credit score	692	687	5.3	5.8
			(3.77)	(3.77)
Any bankruptcy (%)	0.20	0.19	−0.01	−0.03
			(0.12)	(0.12)
Any delinquency (%)	0.24	0.19	0.02	0.02
			(0.13)	(0.13)
Number of delinquencies	0.003	0.002	0.001	0.001
			(0.0018)	(0.0018)
Any account in collections (%)	0.60	0.90	−0.32	−0.28
			(0.22)	(0.22)
Total debt ($)	7,620	8,010	−650	−780
			(892)	(907)
Joint test, *p*-value			0.54 (F = 0.86)	
Students linked to the UC-CCP	10,742	1,811		
with a credit file	9,348	1,554		
Panel B. Student characteristics before the class term				
Cumulative GPA	3.45	3.50	−0.05***	−0.04***
			(0.009)	(0.009)
Quantitative major (%)	55.3	56.4	−1.0	−1.2
			(1.08)	(0.90)
Quantitative courses taken	5.97	6.07	0.04	0.05
			(0.09)	(0.09)
Male (%)	48.8	46.6	2.3**	1.7
			(1.09)	(1.10)
Underrepresented minority (%)	13.2	13.4	−0.29	−1.3**
			(0.74)	(0.68)
First-generation college student (%)	16.8	16.8	−0.29	−0.78
			(0.82)	(0.73)
Application fee waiver (%)	13.2	11.9	0.91	0.54
			(0.70)	(0.64)
Age at credit-file entry	18.98	18.94	0.02	0.02
			(0.06)	(0.06)
Joint test, *p*-value			<0.001 (F = 4.50)	
Students in the Registrar data	14,852	2,475		

Table A5: Pre-Class Balance: UGBA 135 Takers vs. Droppers

Notes: This table compares UGBA 135 takers with droppers. Panel A compares credit records in the last quarter before the class term for students linked to the UC-CCP; Panel B compares Registrar records. Columns report group means and taker-indicator coefficients controlling for graduation-year fixed effects, then additionally for baseline covariates excluding the row's outcome. A dropper is a student who was enrolled in UGBA 135 at the start of the term and absent at term end, with no record of completing the course. Credit records are measured in the last quarter before the class term. Age at credit-file entry is measured for students linked to the UC-CCP and is shown in Panel B, but it is included in the Panel A joint test. Droppers have a higher pre-class GPA than takers (3.50 vs. 3.45). Parentheses contain conventional, unclustered OLS standard errors. Stars mark significance of the difference at the 10, 5, and 1 percent levels.

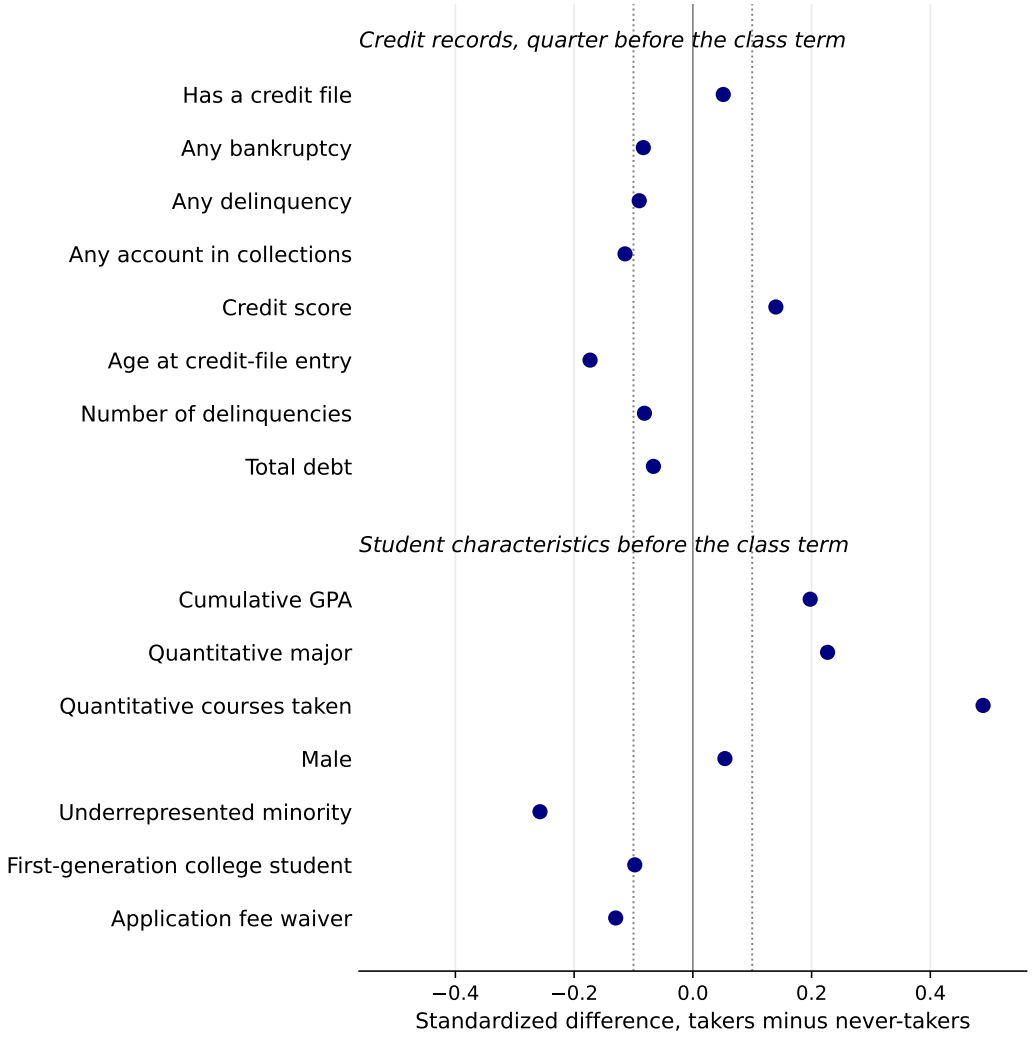


Figure A.1: Standardized Differences Before the Class: Takers vs. Never-Takers

Notes: Each marker is the difference between the taker mean and the never-taker mean, divided by the square root of the average of the two group variances. Credit outcomes are measured in the last quarter before the class term; Registrar characteristics are measured before the class term. Differences are unadjusted. Dotted lines mark ±0.1.

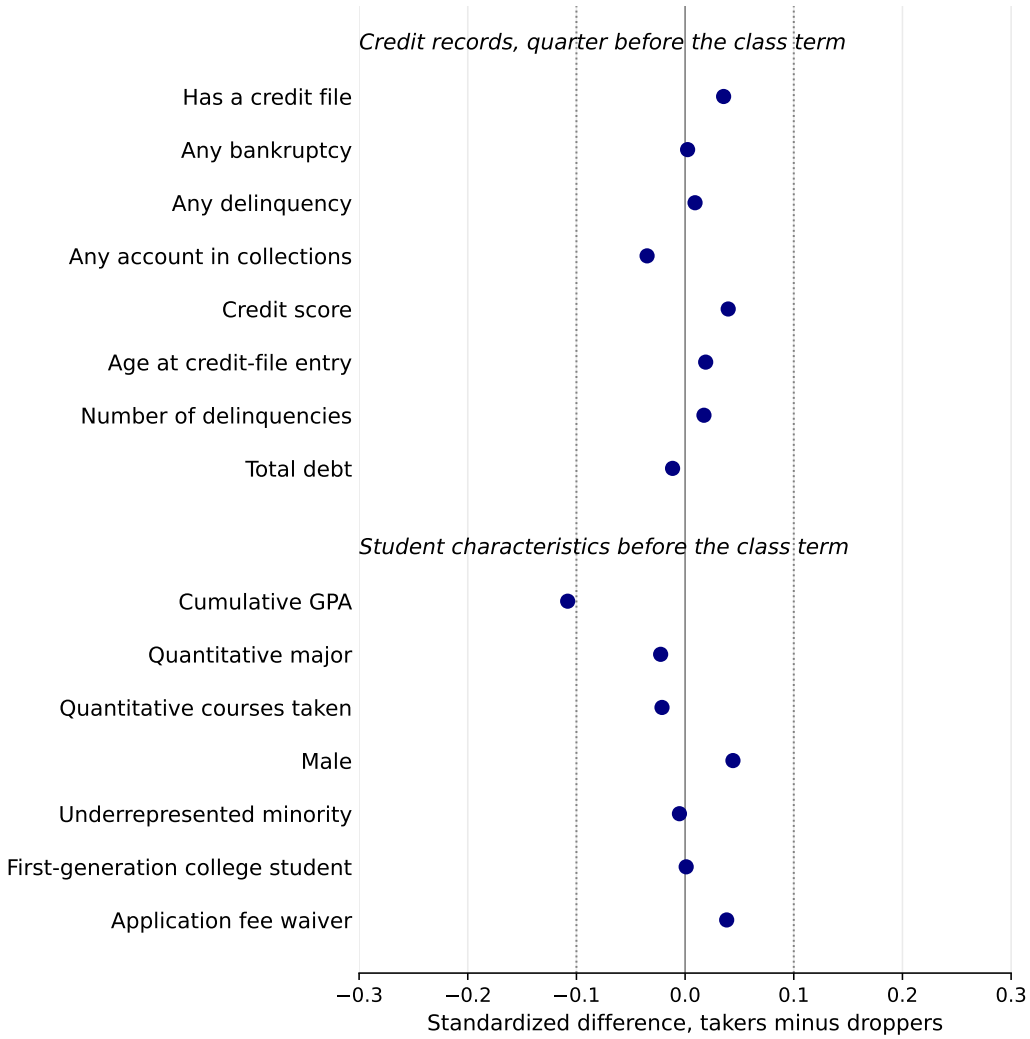


Figure A.2: Standardized Differences Before the Class: Takers vs. Droppers

Notes: Each marker is the difference between the taker mean and the dropper mean, divided by the square root of the average of the two group variances. Credit outcomes are measured in the last quarter before the class term; Registrar characteristics are measured before the class term. Differences are unadjusted. Dotted lines mark ±0.1.

A.3 UC-CCP Estimation Details and Additional Results

Sample for the selection regressions. Table A6 reports regressions of course completion in Panel A and pre-class credit score in Panel B. All six specifications use the same 68,315 students: 8,833 course completers and 59,482 never-takers. Students must have observed pre-class census GPA and major, baseline credit score, and complete information on the other regressors. We exclude imputed GPA, major inferred from application records, and graduation cohorts containing fewer than 10 otherwise eligible students. Standard errors are clustered within 178 graduation-cohort-by-pre-class-major groups. These estimates describe students with the required baseline information, rather than all university students.

Pre-existing credit-score differences. On the common sample of 68,315 students, completers have pre-class credit scores 20.08 points higher than never-takers after controlling for graduation cohort (SE 1.74). Adding pre-class major reduces the gap to 15.77 points (SE 1.83), and adding the remaining student characteristics reduces it to 7.73 points (SE 1.75). Because these comparisons use identical students, the reduction reflects covariate adjustment rather than changes in sample composition. The remaining difference indicates that takers already have higher credit scores before the course, conditional on the included characteristics. Appendix Figure A.3 displays the sequential adjustments; Panel B of Table A6 reports the estimates.

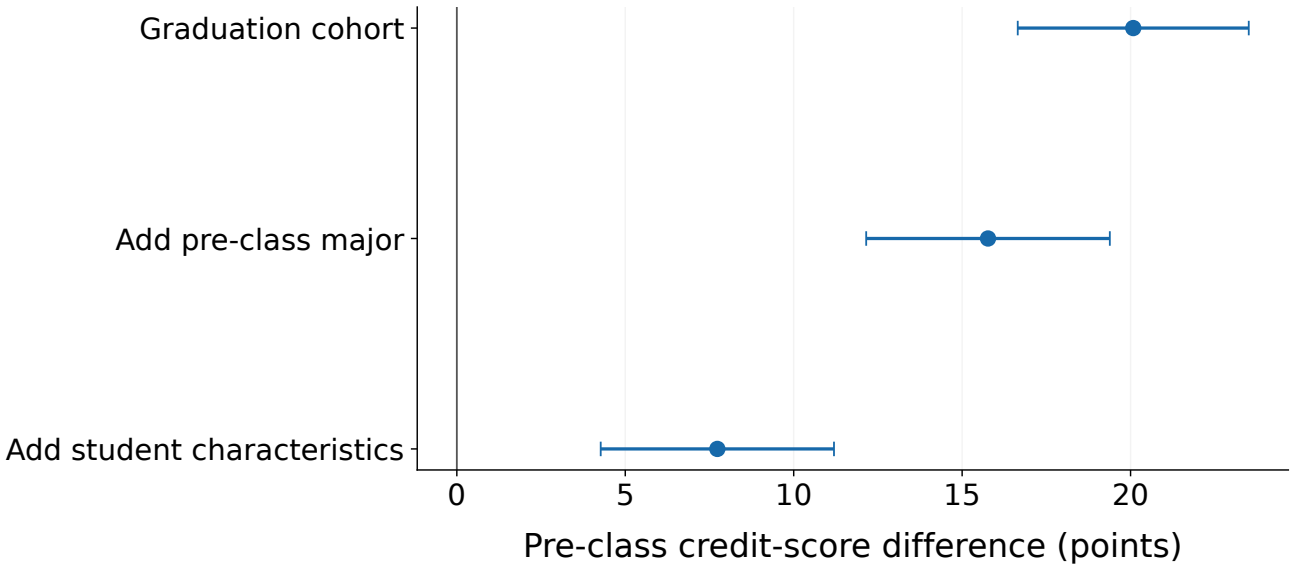


Figure A.3: Pre-Class Credit-Score Differences after Sequential Adjustment

Notes: Points show the coefficient on course completion in regressions of pre-class credit score. The first regression controls for graduation cohort; the second adds pre-class major; the third also adds gender, ethnicity, parental college attendance, fee-waiver status, residency, pre-class GPA, and prior quantitative coursework. All regressions contain the same 68,315 students. Bars are 95% pointwise confidence intervals with standard errors clustered within graduation-cohort-by-pre-class-major groups. Baseline measurement for never-takers uses the assigned comparison dates described in Section 3.3.

	(1)	(2)	(3)
Panel A. Course completion (percentage points)			
Male	−0.10	−0.47	−0.47
	(0.35)	(0.40)	(0.40)
Underrepresented minority	−6.66***	−5.78***	−5.64***
	(0.43)	(0.41)	(0.41)
White/Other	−5.54***	−4.63***	−4.61***
	(0.43)	(0.39)	(0.39)
International ethnicity category	−6.20***	−6.46***	−6.21***
	(0.68)	(0.69)	(0.69)
Neither parent attended college	0.65*	0.71**	0.73**
	(0.34)	(0.33)	(0.33)
Application fee waiver	−0.95**	−0.40	−0.38
	(0.44)	(0.44)	(0.44)
Non-US resident	−2.35***	−2.68***	−2.60***
	(0.52)	(0.51)	(0.51)
Pre-class GPA (one point)	3.51***	3.22***	3.14***
	(0.33)	(0.39)	(0.40)
Prior quantitative department-terms (one)	0.93***	1.00***	0.99***
	(0.07)	(0.06)	(0.06)
Pre-class credit score (100 points)	—	—	0.41***
			(0.09)
Pre-class major (reference: Business)			
Computer science	—	−11.15***	−11.14***
		(1.79)	(1.79)
Economics	—	−11.73***	−11.70***
		(1.86)	(1.85)
Engineering	—	−23.85***	−23.77***
		(1.72)	(1.72)
Life sciences	—	−16.62***	−16.56***
		(1.79)	(1.79)
Literature and arts	—	−17.35***	−17.28***
		(1.79)	(1.78)
Mathematics	—	−21.57***	−21.56***
		(1.86)	(1.87)
Physical sciences	—	−21.79***	−21.72***
		(1.71)	(1.71)
Social sciences	—	−16.31***	−16.25***
		(1.76)	(1.76)
Undeclared	—	−12.63***	−12.47***
		(1.75)	(1.75)
Panel B. Pre-class credit score (points)			
Course completion	20.08***	15.77***	7.73***
	(1.74)	(1.83)	(1.75)
Graduation-cohort indicators	Yes	Yes	Yes
Pre-class major indicators	No	Yes	Yes
Pre-class credit score (Panel A only)	No	No	Yes
Student characteristics (Panel B only)	No	No	Yes
Students	68,315	68,315	68,315

Table A6: Student Characteristics and Selection into the Course

Notes: Entries are coefficients with cluster-robust standard errors in parentheses. Panel A reports linear probability models in percentage points; Panel B reports pre-class credit-score differences in score points. Each column adds additional controls, as indicated. All six models use the same complete-case sample with observed baseline credit score, pre-class census GPA and major, and the other required characteristics, excluding graduation cohorts with fewer than 10 eligible students. Ethnicity coefficients are relative to Asian non-underrepresented students; major coefficients are relative to Business. Binary variables enter as zero/one indicators. In Panel B, other student characteristics are the demographic, residency, GPA, and prior-coursework variables shown in Panel A; baseline credit score is the dependent variable. Standard errors are clustered within 178 graduation-cohort-by-pre-class-major groups. A dash denotes a regressor not included. In Panel B, column (1) controls for graduation cohort, column (2) adds pre-class major, and column (3) adds the student characteristics in Panel A. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Cross-sectional comparisons. Table A7 reports the raw and adjusted credit-score differences shown in Figure A.4. Each row uses the outcome measured at the stated date. Graduation-relative outcomes are measured one, three, or five years after the observed or assigned graduation quarter. Age-relative outcomes are separate snapshots. The pre-class row describes baseline selection. Missing outcomes are not assigned zero, and sample sizes vary across horizons and specifications. The adjusted regressions include the student characteristics, graduation-cohort indicators, and major indicators defined in Equation (8).

Measurement	Raw	Raw N	Adjusted	Adjusted N
Pre-class	19.84***	75,353	6.18***	72,417
	(1.84)		(1.81)	
Graduation +1 year	25.75***	69,830	12.41***	67,456
	(1.28)		(1.05)	
Graduation +3 years	26.96***	58,631	13.55***	56,684
	(1.26)		(1.02)	
Graduation +5 years	25.47***	45,949	12.81***	44,342
	(1.42)		(1.15)	
Age 25	29.99***	63,927	14.02***	61,891
	(1.15)		(0.96)	
Age 30	30.32***	37,287	13.31***	35,997
	(1.65)		(1.18)	

Table A7: Credit-Score Differences by Measurement Date

Notes: All coefficients and standard errors (SE) are in credit-score points. N is the number of students in the regression. Standard errors are clustered by graduation-cohort-by-major group. Raw and adjusted models need not contain the same students. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Weighting for observed characteristics. Let W_i contain gender and ethnicity indicators, parental college attendance, fee-waiver status, residency, pre-class GPA, prior quantitative coursework, pre-class quantitative-major status, transfer status, and baseline credit score, total debt, and credit-file presence. We fit a logistic participation model, replacing missing numeric predictors by their sample medians for this weighting step; outcome values are not imputed. The fitted propensity score $\hat{p}_i = \widehat{\Pr}(D_i = 1 \mid W_i)$ is the estimated probability that student i takes the course, given these observed predictors. To construct a comparison group resembling the takers, takers receive weight one and never-takers receive weight $\hat{w}_i = \hat{p}_i/(1-\hat{p}_i)$. For outcome Y_{ih} at horizon h, the normalized contrast is

$$\widehat{\tau}_h^{\mathrm{ATT}} = \frac{\sum_{i\in\mathcal{O}_h} D_i Y_{ih}}{\sum_{i\in\mathcal{O}_h} D_i} - \frac{\sum_{i\in\mathcal{O}_h}(1-D_i)\hat{w}_i Y_{ih}}{\sum_{i\in\mathcal{O}_h}(1-D_i)\hat{w}_i}. \tag{17}$$

Here $\mathcal{O}_h$ is the set of students with an observed outcome at horizon h and usable weights. The first term is the mean outcome among takers. The second is the weighted mean among never-takers. ATT denotes a comparison targeted to the characteristics of students who took the course.

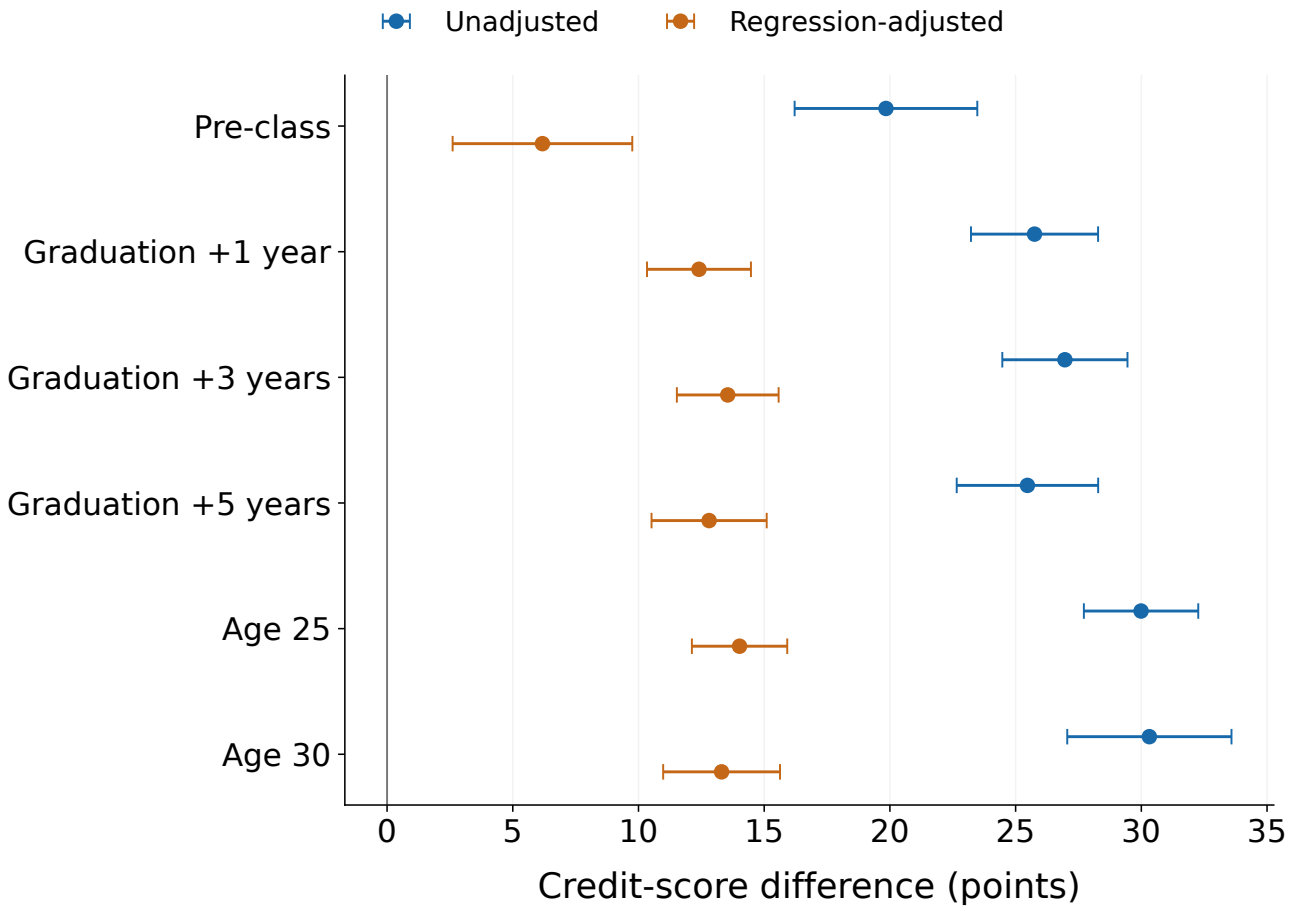


Figure A.4: Raw and Adjusted Credit-Score Differences

Notes: Points show taker–never-taker differences in credit-score points, with 95% pointwise confidence intervals. "Pre-class" denotes the last quarter before the class term. Graduation-relative and age-relative measurements are separate comparisons. Adjusted regressions include student characteristics, graduation-cohort indicators, and major indicators. Standard errors are clustered by graduation-cohort-by-major group.

Interpreting the contrast as their average causal effect additionally requires that the observed predictors account for all differences affecting untreated outcomes and that comparable never-takers exist.

Overlap, effective sample size, and remaining imbalance. Overlap means that takers and never-takers have similar estimated participation probabilities over the range used for comparison. Without overlap, a small number of never-takers can receive disproportionate influence. For weights w_i within a group, the effective sample size is $(\sum_i w_i)^2 / \sum_i w_i^2$: the size of an equally weighted sample with comparable weight concentration, not the actual number of students. Residual balance describes the differences in measured characteristics that remain after weighting, expressed as weighted differences in means divided by a reference standard deviation. The weighting diagnostics use a broader sample of 11,037 takers and 81,276 never-takers. Unlike the complete-case

selection regressions, the propensity model retains observations with missing numeric predictors through median imputation. Taker-targeted weighting yields a never-taker effective sample size of approximately 58,402. Across the 14 predictors included in the propensity model, the largest absolute standardized difference after this weighting is 0.063, for baseline credit score. These baseline diagnostics do not establish balance within every outcome-observed follow-up sample, and they cannot address unobserved differences.

Alternative weighting comparisons. Table A8 reports the credit-score contrasts three years after graduation. Population-targeted weighting emphasizes the covariate distribution of all eligible students; never-taker-targeted weighting emphasizes that of never-takers. Overlap weighting gives takers weight $1-\hat{p}_i$ and never-takers weight $\hat{p}_i$, emphasizing students with characteristics common to both groups. Restricted comparisons retain students whose estimated probabilities lie between the indicated bounds, changing the comparison population. Entropy balancing chooses comparison weights to match selected covariate moments, while augmented inverse-probability weighting also uses an outcome regression. The reported standard errors for these comparisons use analytic inference treating the fitted weights as fixed; they do not include uncertainty from re-estimating the propensity model. Differences across rows reflect different weighting targets, estimators, and, in some cases, samples. Table A10 shows that the credit-score, delinquency, and collections differences are similar across alternative weighting samples; the total-debt difference is larger and marginally significant in the restricted samples.

Specification	Gap	SE	N
Taker-targeted weighting	13.34	0.92	58,631
Population-targeted weighting	15.38	1.50	58,631
Never-taker-targeted weighting	15.64	1.59	58,631
Overlap weighting	13.71	0.95	58,631
Stabilized population weighting	15.38	1.50	58,631
Taker weighting, 5–95% restriction	13.01	0.83	49,616
Taker weighting, 10–90% restriction	10.96	0.79	31,547
Population weighting, 5–95% restriction	15.28	1.00	49,616
Entropy balancing	13.40	0.92	58,631
Augmented inverse-probability weighting	11.74	0.71	44,376

Table A8: Weighted Credit-Score Differences Three Years after Graduation

Notes: Takers are compared with never-takers. Gaps and SE are in credit-score points; N counts students with outcomes used by the estimator. The restrictions refer to the fitted participation probability. Estimates remain observational. Analytic standard errors treat estimated weights as fixed.

Differences across student characteristics. Table A11 reports the coefficient δ from Equation (9) for credit score. A binary-characteristic coefficient compares the adjusted gap for students with that characteristic to the gap for students without it. A continuous-characteristic coefficient

Panel A. One year after graduation

Outcome	Unweighted	ATT	Overlap	N
Credit score (points)	25.75***	10.89***	11.21***	69,830
	(1.28)	(1.05)	(1.07)	
Total debt (dollars)	−2,138***	480	372	68,224
	(637)	(677)	(666)	
Any delinquency (pp)	−0.933***	−0.292***	−0.300***	69,867
	(0.126)	(0.100)	(0.108)	
Any collections (pp)	−1.347***	−0.262*	−0.292**	69,867
	(0.188)	(0.141)	(0.148)	

Panel B. Five years after graduation

Outcome	Unweighted	ATT	Overlap	N
Credit score (points)	25.47***	12.67***	13.00***	45,949
	(1.42)	(0.99)	(1.02)	
Total debt (dollars)	8,768***	9,993***	9,506***	44,924
	(2,804)	(2,909)	(2,869)	
Any delinquency (pp)	−0.958***	−0.372***	−0.396***	46,003
	(0.151)	(0.106)	(0.111)	
Any collections (pp)	−0.989***	−0.172	−0.170	46,003
	(0.203)	(0.160)	(0.168)	

Table A9: Unweighted and Weighted Outcome Differences One and Five Years after Graduation

Notes: Each entry is the difference in the outcome between takers and never-takers, measured one year (Panel A) and five years (Panel B) after graduation, with standard errors in parentheses. The unweighted column compares raw means. ATT reweights never-takers so that their baseline characteristics match those of takers; overlap weighting reweights both groups toward the students most comparable across the two groups. "pp" denotes percentage points. Missing numeric participation-model predictors are median-imputed; outcomes are not imputed. Standard errors are clustered by graduation-cohort-by-major group and treat fitted weights as fixed. Estimates describe observational differences. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

reports the change in the gap for a one-standard-deviation increase. The Benjamini–Hochberg adjusted q values account for the full set of 40 comparisons across four outcomes and ten characteristics. They are not recalculated using only the credit-score rows shown here. Figure 2 and Appendix Figures A.5–A.7 present the four outcomes on their own scales. These regressions describe variation in adjusted associations, not variation in individual causal treatment effects.

Follow-up credit score and baseline characteristics. To illustrate how follow-up credit scores vary with starting credit score, we estimate

$$S_{i,g+3} = \alpha + \tau D_i + \rho b_i + \delta D_i b_i + X_i'\gamma + \lambda_{c(i)} + u_i, \qquad b_i = (B_i - \bar{B})/s_B. \tag{18}$$

Here $S_{i,g+3}$ is credit score three years after graduation; B_i is pre-class credit score; and $\bar{B}$ and s_B are its mean and standard deviation in the estimation sample. Thus b_i measures starting credit

	(1) Broad sample median-filled	(2) Selection sample median-filled	(3) Selection sample complete cases
Credit score (points)	13.34***	12.49***	11.31***
	(0.92)	(0.88)	(0.84)
	[58,631]	[45,020]	[42,771]
Total debt (dollars)	2,009	2,939*	3,223*
	(1,370)	(1,678)	(1,784)
	[57,466]	[44,263]	[42,187]
Any delinquency (pp)	−0.35***	−0.35***	−0.34***
	(0.08)	(0.09)	(0.09)
	[58,682]	[45,054]	[42,805]
Any collections (pp)	−0.63***	−0.58***	−0.53***
	(0.12)	(0.14)	(0.14)
	[58,682]	[45,054]	[42,805]

Table A10: Weighted Outcome Differences Three Years after Graduation: Alternative Weighting Samples

Notes: Each entry is the ATT-weighted taker–never-taker difference, with standard errors in parentheses and the number of students in brackets. Column (1) uses the broad linked sample with missing numeric predictors replaced by sample medians, as in Table 2. Column (2) refits the participation model on the selection-regression sample with median filling. Column (3) restricts that sample to students with all weighting predictors observed. Standard errors treat fitted weights as fixed. "pp" denotes percentage points. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

score in standard-deviation units relative to the sample mean. The intercept is α, τ is the taker–never-taker gap at the mean starting score, ρ is the baseline-score slope among never-takers, and δ is the difference in slopes between takers and never-takers. The vector X_i contains the other student controls, γ their coefficients, $\lambda_{c(i)}$ the graduation-cohort indicators, and u_i the error. Major indicators are not included in this regression. For each baseline value, we average predictions over the same empirical distribution of controls and cohorts for the two groups.

At standardized starting score b, the adjusted gap and its estimated variance are

$$\widehat{\Delta}(b) = \widehat{\tau} + \widehat{\delta} b, \qquad \widehat{\mathrm{Var}}\{\widehat{\Delta}(b)\} = \widehat{\mathrm{Var}}(\widehat{\tau}) + 2b\,\widehat{\mathrm{Cov}}(\widehat{\tau}, \widehat{\delta}) + b^2 \widehat{\mathrm{Var}}(\widehat{\delta}). \tag{19}$$

Hats denote estimated quantities. The variance measures sampling uncertainty in the estimated gap; the covariance term accounts for joint uncertainty in the estimated intercept difference $\widehat{\tau}$ and slope difference $\widehat{\delta}$. Its square root is the standard error. We use the covariance matrix clustered by graduation-cohort-by-major group and a Student-t reference distribution with the number of clusters minus one degrees of freedom to construct pointwise 95% intervals. These intervals apply separately at each displayed value and are not simultaneous confidence bands for the entire curve.

Characteristic	Interaction	N	BH q
Pre-class credit score (per SD)	-6.16^{***}	46,803	<0.0001
	(1.13)		
Pre-class credit file	-3.82^{**}	55,447	0.0899
	(1.82)		
Male	0.21	56,684	0.9799
	(1.51)		
Underrepresented minority	9.52^{***}	56,684	0.0001
	(2.10)		
Neither parent attended college	3.01	56,684	0.1950
	(1.86)		
Application fee waiver	1.21	56,684	0.6664
	(1.94)		
Pre-class quantitative major	-13.94^{***}	56,684	<0.0001
	(1.36)		
Age (per SD)	3.80^{**}	56,684	0.0439
	(1.49)		
Pre-class GPA (per SD)	−2.06	56,684	0.4973
	(2.13)		
Prior quantitative coursework (per SD)	-5.42^{***}	56,684	<0.0001
	(0.78)		

Table A11: Variation in the Adjusted Credit-Score Gap

Notes: Each row is a separate regression. SD denotes a sample standard deviation. Binary characteristics take values zero and one. SE are clustered by graduation-cohort-by-major group. N is the regression sample size. BH q is the Benjamini–Hochberg adjusted p value across all 40 outcome–characteristic interactions. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Academic preparation and common comparison ranges. Parallel regressions replace B_i with pre-class GPA or prior quantitative coursework, remove that characteristic from X_i to avoid entering it twice, and additionally control for baseline credit score. For each characteristic, we restrict the comparison to the intersection of the two groups' 5th–95th percentile ranges. We divide this range into pooled quantile groups; repeated values can yield fewer than five groups, as occurs for the quantitative-coursework count. The regressions use students with observed outcomes and covariates, and the linear and quadratic specifications use the same students. The quadratic specification adds b_i^2 and $D_i b_i^2$, allowing curvature in both groups. Follow-up must occur after the actual or assigned course-completion date. Tests of the interaction terms reject equal slopes for each of the three characteristics after adjustment across the three tests within each functional form (all adjusted $q < 0.001$). These tests concern observational credit-score associations, not directly measured financial literacy.

Stacked event-study specification. We form a separate comparison sample, called a stack, around each course-start quarter. Let s index these samples, q_s denote the course-start quarter

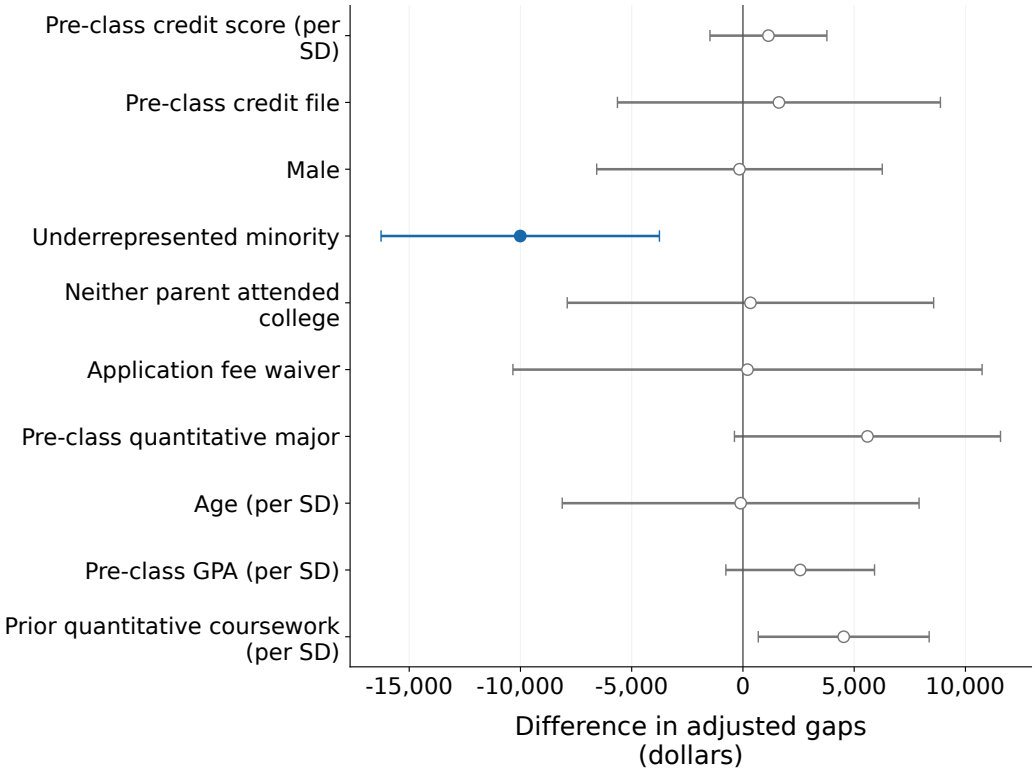


Figure A.5: Adjusted Total-Debt Differences across Student Characteristics

Notes: Each point is the interaction coefficient δ from Equation (9) for total debt, in dollars, three years after graduation: the change in the taker–never-taker gap associated with the indicated characteristic. Continuous characteristics are in standard-deviation units. Bars are 95% pointwise confidence intervals, with standard errors clustered by graduation-cohort-by-major group. Filled markers pass the 5% Benjamini–Hochberg threshold across all 40 outcome–characteristic comparisons.

for sample s, t denote calendar quarter, and D_{is} indicate that student i takes the course in that sample's course-start quarter. We estimate

$$Y_{ist} = \alpha_{is} + \lambda_{ts} + \sum_{\substack{k=-8 \\ k\neq -1}}^{20} \beta_k D_{is} \mathbf{1}\{t - q_s = k\} + \varepsilon_{ist}. \tag{20}$$

The outcome Y_{ist} is measured for student i in sample s and quarter t. The student-by-sample fixed effect α_{is} absorbs time-invariant differences, and the calendar-quarter-by-sample fixed effect λ_{ts} absorbs common quarter-specific changes. The indicator $\mathbf{1}\{t - q_s = k\}$ equals one when the observation is k quarters from the course-start quarter and zero otherwise. The coefficient β_k measures the change in the taker–comparison difference relative to the omitted quarter $k = -1$, the last quarter before the course begins. The error term is ε_{ist}. Eligible comparisons include never-takers and students whose own course starts at least eight quarters later; comparison observations after their own course starts are excluded. Up to 500 eligible comparison students are sampled for each course-start quarter. Standard errors are clustered by student, allowing correlation across

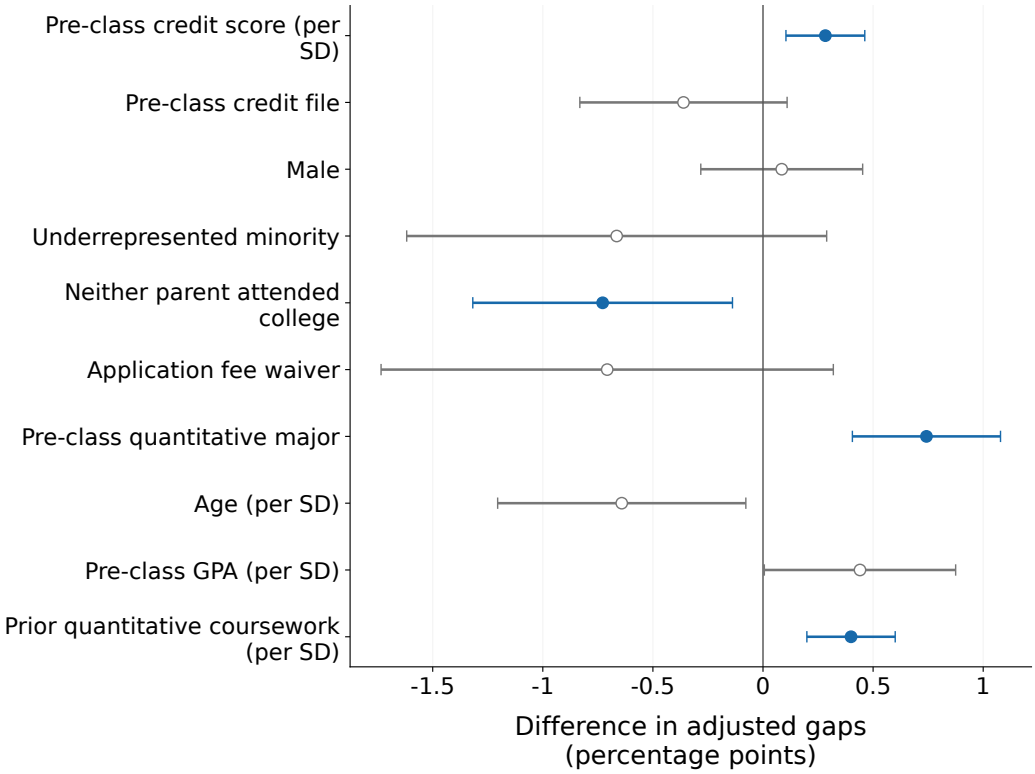


Figure A.6: Adjusted Differences in Any Delinquency across Student Characteristics

Notes: Each point is the interaction coefficient δ from Equation (9) for any delinquency, in percentage points, three years after graduation: the change in the taker–never-taker gap associated with the indicated characteristic. Continuous characteristics are in standard-deviation units. Bars are 95% pointwise confidence intervals, with standard errors clustered by graduation-cohort-by-major group. Filled markers pass the 5% Benjamini–Hochberg threshold across all 40 outcome–characteristic comparisons.

observations and across comparison samples containing the same student.

The pooled specification replaces the event-quarter interactions with separate terms $D_{is}\mathbf{1}\{t - q_s = 0\}$ and $D_{is}\mathbf{1}\{t - q_s \geq 1\}$. The first describes the course-start quarter and the second describes subsequent quarters, using available pre-course quarters as the reference. Because enrollment can span more than one quarter, this post-start period need not begin after course completion. The pooled coefficient is not mechanically the average of the dynamic coefficients and does not automatically weight every student equally. Table A12 reports the pooled estimates. A causal interpretation requires parallel trends in the absence of the course, no anticipatory effects, and suitable comparisons. The joint test of pre-course credit-score coefficients has $p = 0.420$; failure to reject does not establish parallel trends.

Students who dropped the course. Table A5 provides descriptive baseline comparisons between completers and students who enrolled initially but did not complete the course. The numbers of observed students differ across characteristics. Course dropping is not randomly assigned, so

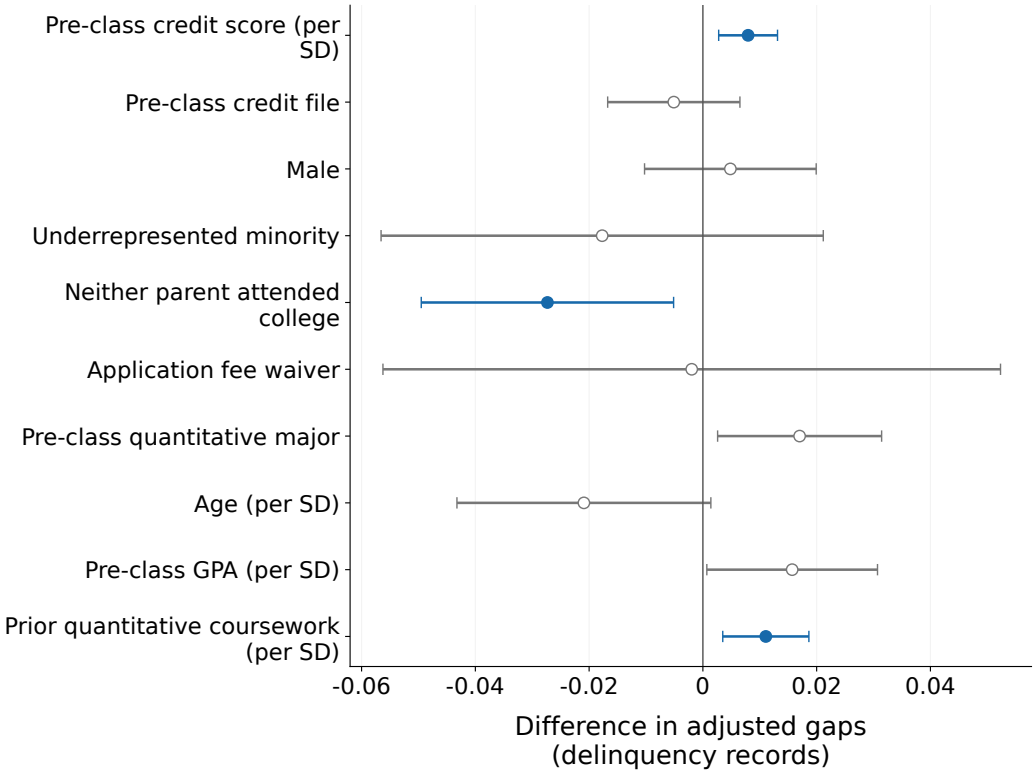


Figure A.7: Adjusted Differences in the Number of Delinquencies across Student Characteristics

Notes: Each point is the interaction coefficient δ from Equation (9) for the number of delinquency records three years after graduation: the change in the taker–never-taker gap associated with the indicated characteristic. Continuous characteristics are in standard-deviation units. Bars are 95% pointwise confidence intervals, with standard errors clustered by graduation-cohort-by-major group. Filled markers pass the 5% Benjamini–Hochberg threshold across all 40 outcome–characteristic comparisons.

Outcome	Post-course	Students	BH q
Credit score (points)	2.83	21,568	0.2574
	(2.28)		
Total debt (dollars)	4,081***	21,342	<0.0001
	(999)		
Any delinquency (pp)	−0.529***	21,568	<0.0001
	(0.100)		
Any collections (pp)	−0.077	21,568	0.6680
	(0.179)		

Table A12: Pooled Changes after the Course-Start Quarter

Notes: Estimates compare quarters after course start with available pre-course quarters, including a separate indicator for the course-start quarter. SE are clustered by student. Students is the number of unique persons, not stacked observations. The BH q values retain the original outcome-family adjustment. Binary outcomes have been converted to percentage points. Credit-file presence has no usable variance in this specification and is not interpreted as a zero effect. Stars denote $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

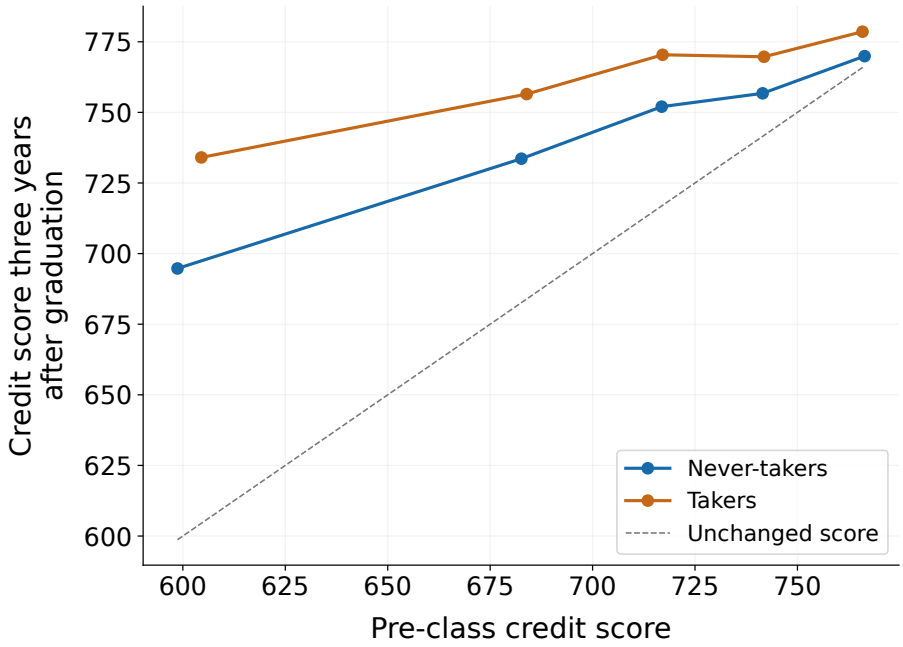


Figure A.8: Observed Follow-Up Credit Score by Pre-Class Credit Score

Notes: The outcome is credit score three years after graduation, measured after the actual or assigned course-completion date. Points are observed means for takers and never-takers within pre-class credit-score groups. The diagonal denotes an unchanged credit score, not a test of parallel trends. These means are descriptive.

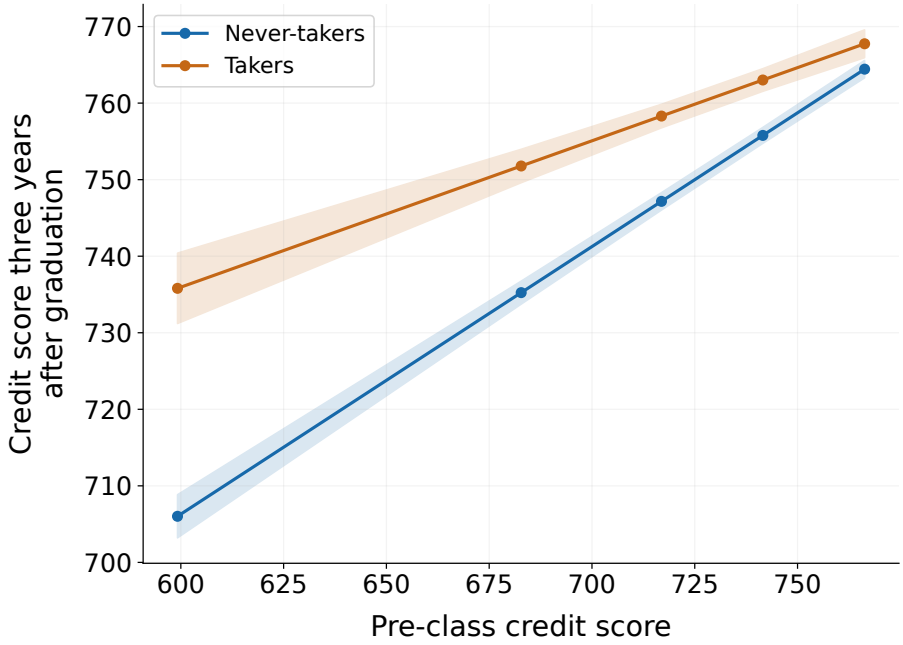
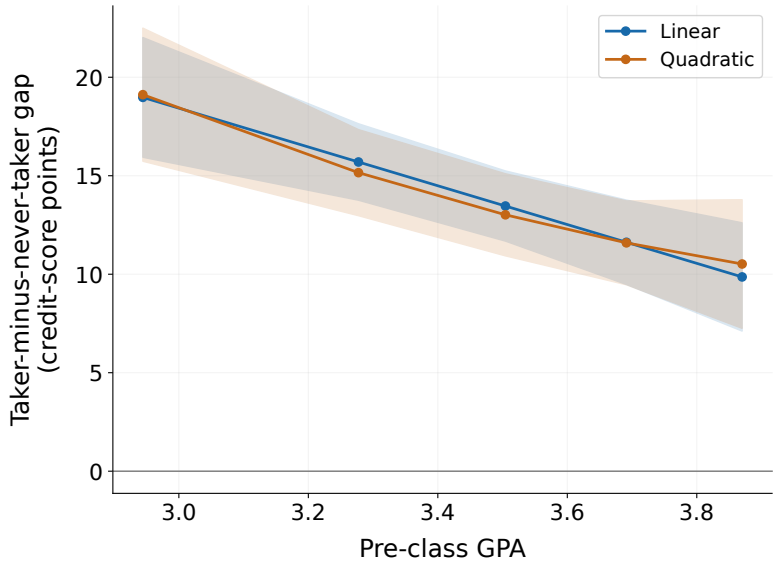
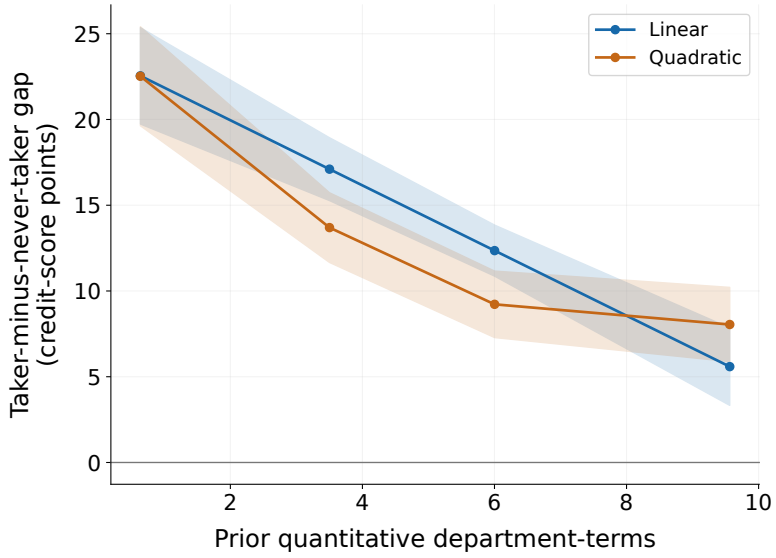


Figure A.9: Adjusted Follow-Up Credit Score by Pre-Class Credit Score

Notes: The outcome is credit score three years after graduation. Lines average predicted scores from Equation (18) over the same covariate and cohort distribution for takers and never-takers. Shaded bands are 95% pointwise confidence intervals clustered by graduation-cohort-by-major group. The displayed baseline range is shared by takers and never-takers.



(a) Pre-class GPA



(b) Prior quantitative coursework

Figure A.10: Adjusted Credit-Score Gap by Academic Preparation

Notes: The outcome is credit score three years after graduation. Lines show the adjusted taker–never-taker difference from a linear interaction and a quadratic sensitivity specification fitted to the same students. Both models control for baseline credit score and the other student characteristics, with graduation-cohort indicators. Panel (a) uses 38,422 students and panel (b) uses 42,153 students. Shaded bands are 95% pointwise intervals clustered by graduation-cohort-by-major group. Quantitative coursework counts distinct quantitative-department-by-term records, not individual catalog courses.

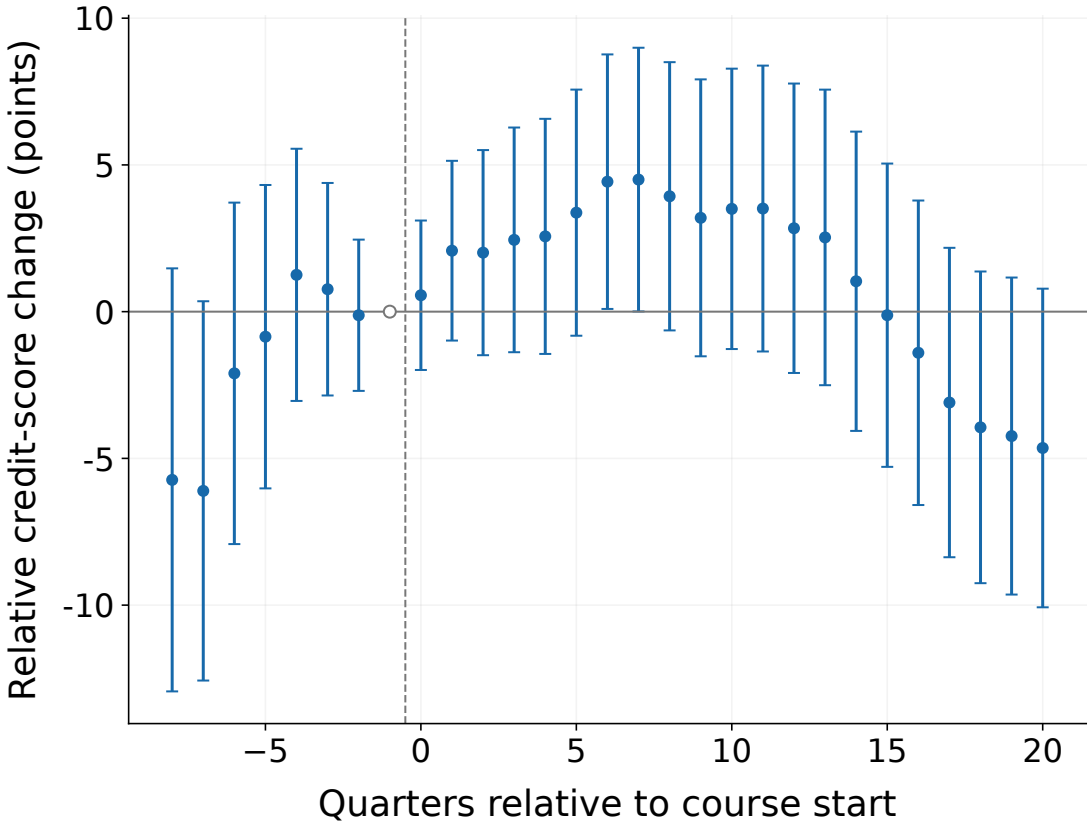


Figure A.11: Credit-Score Changes around Course Enrollment

Notes: Event time is quarters relative to course start; the previous quarter is the omitted reference. Points show coefficients from Equation (20), with 95% pointwise intervals clustered by student. The sample includes 21,568 distinct students. The vertical axis shows the estimated change relative to the omitted quarter; a causal interpretation requires the assumptions stated in the text.

neither small differences nor statistically insignificant differences establish that droppers provide an unbiased counterfactual for completers. The first observed age of entry into credit records, where examined, describes credit-history coverage rather than age at university entry and need not precede the course.

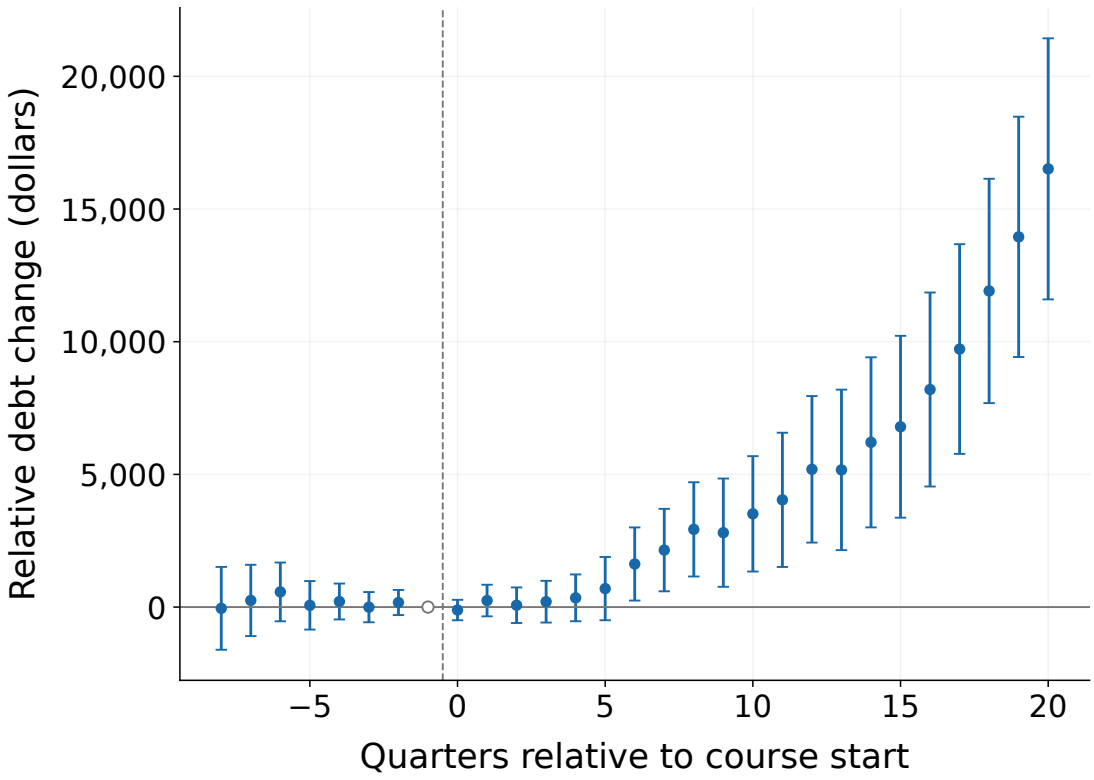


Figure A.12: Debt Changes around Course Enrollment

Notes: Total debt is measured in dollars. Event time is quarters relative to course start; the previous quarter is the omitted reference. Intervals are 95% pointwise intervals clustered by student. The sample includes 21,342 distinct students. Larger debt balances do not, by themselves, establish a welfare improvement or loss.

A.4 Experimental Design Details

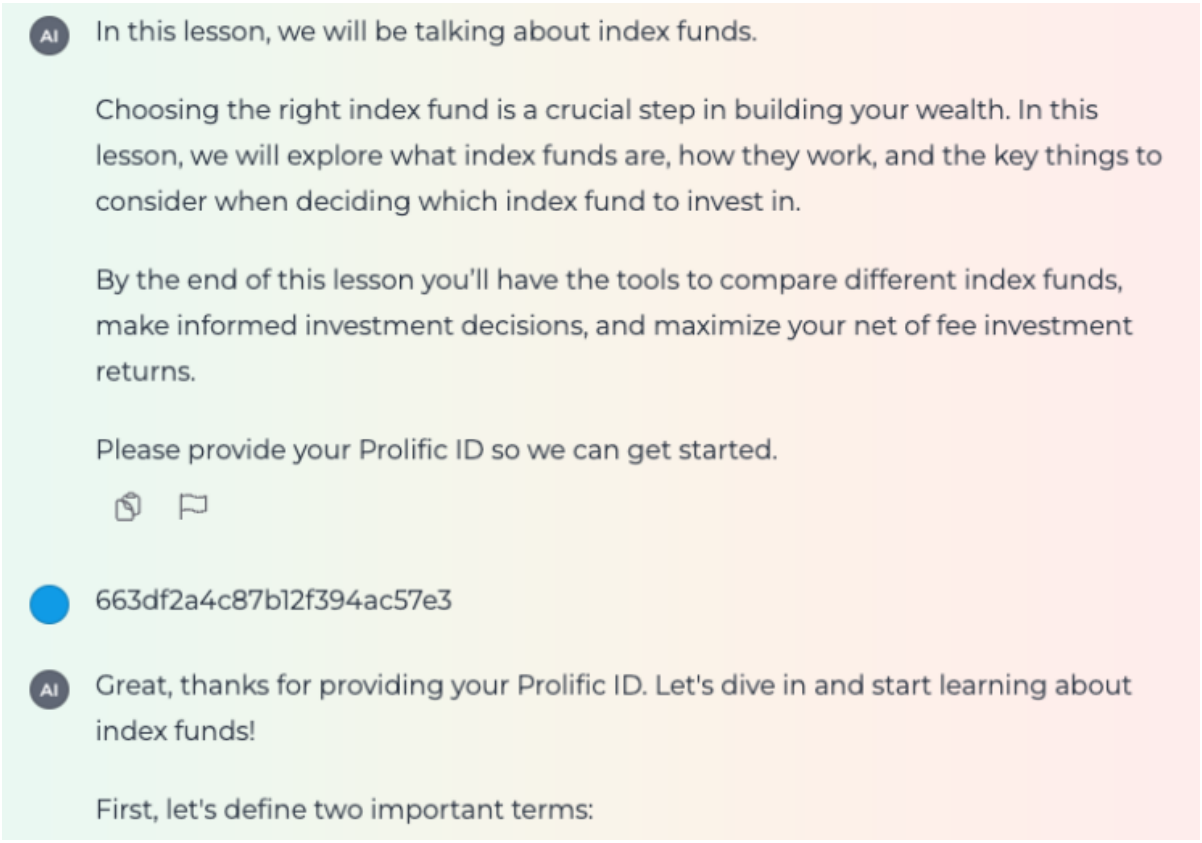


Figure A.13: Example of the AI Education Material

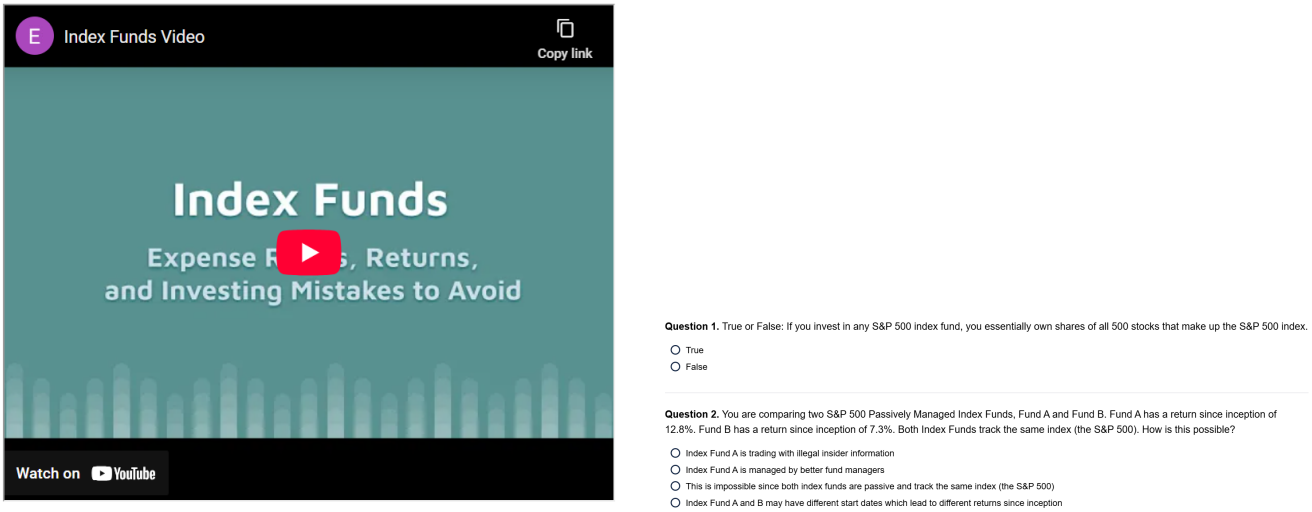


Figure A.14: Example of the Video Education Material

Notes: Screenshots of the video material: a pre-recorded video (left) and multiple-choice questions with correct answers and explanations (right). The material also includes a PDF summary of key definitions.

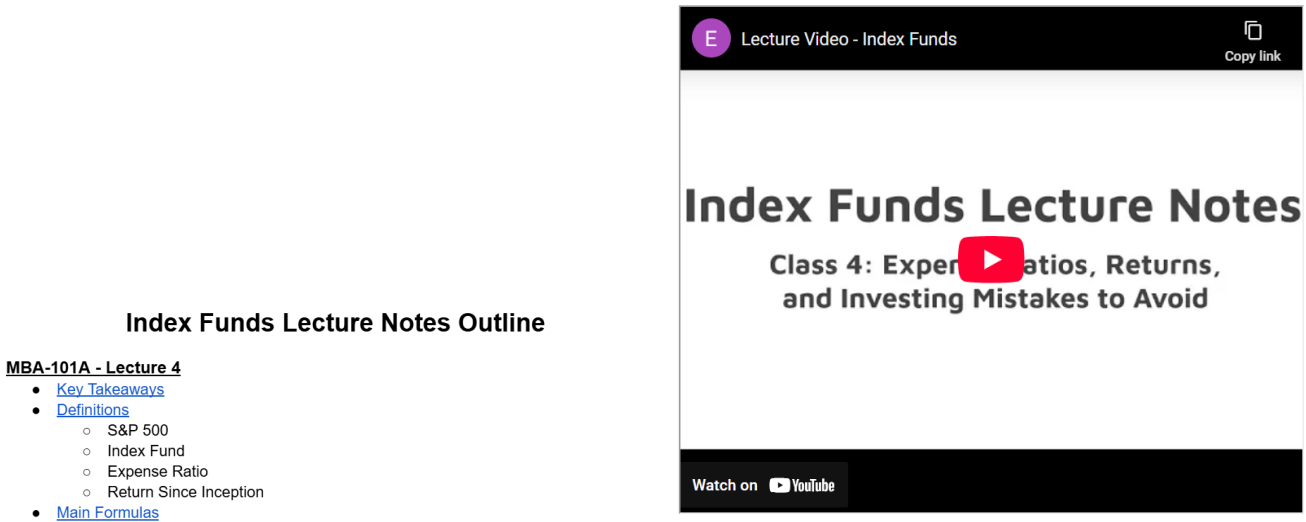


Figure A.15: Example of the Lecture Education Material

Notes: Screenshots of the lecture material: the outline of the lecture notes (left) and the lecture video, recorded in a university personal finance course (right). The material also includes comprehension check questions.

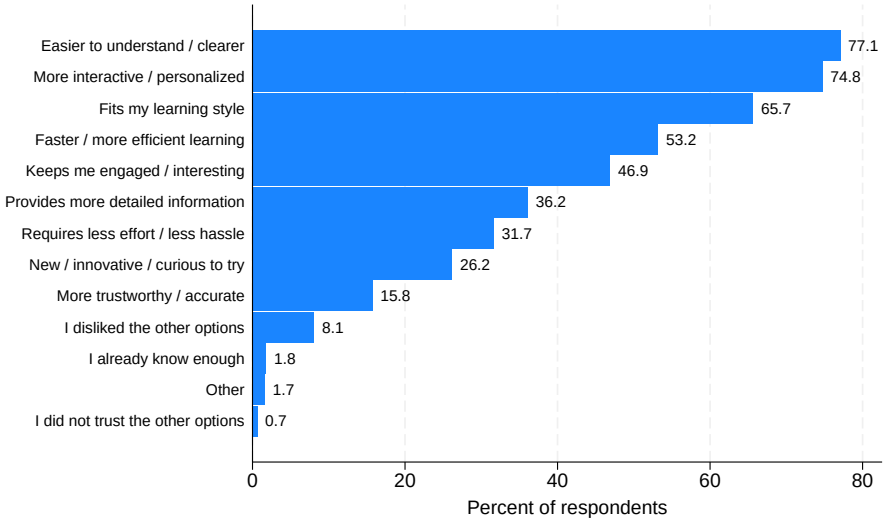


Figure A.16: Reasons for Preferring the AI Material

Notes: Participants who ranked the AI material as their most preferred option explained their choice in a free response and then classified that response into one or more listed reasons. Each bar is the share of these participants who chose the reason; participants could choose more than one reason, so shares sum to more than 100%.

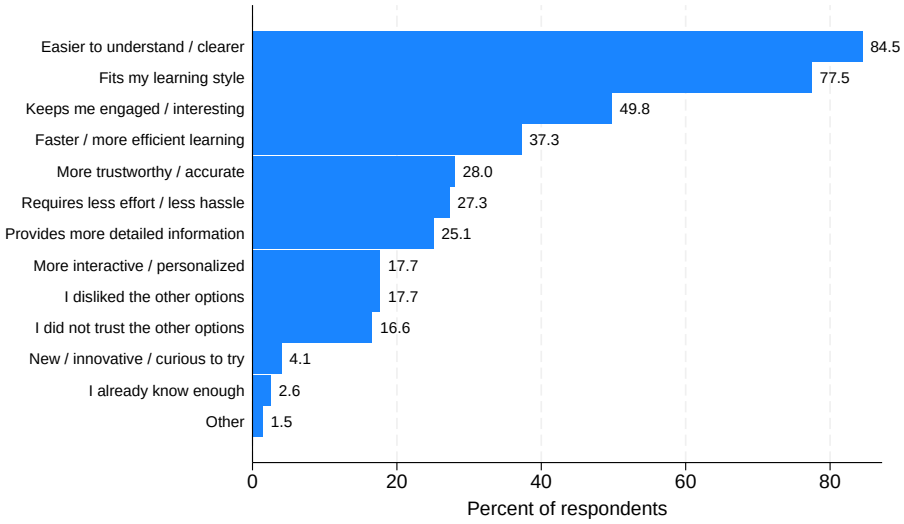


Figure A.17: Reasons for Preferring the Video Material

Notes: Participants who ranked the video material as their most preferred option explained their choice in a free response and then classified that response into one or more listed reasons. Each bar is the share of these participants who chose the reason; participants could choose more than one reason, so shares sum to more than 100%.

A.5 Additional Results

	(1) Estimated baseline score	(2) Estimated minus actual baseline score
Baseline score of 5 or below	-0.418**	2.398***
	(0.176)	(0.176)
Constant	5.585***	-2.138***
	(0.351)	(0.354)
Observations	1,049	1,049
R^2	0.015	0.161

Table A13: Estimated and Actual Baseline Scores, Baseline Scores of 5 or Below versus Above 5

Notes: Each column regresses the column outcome on an indicator for a baseline score of 5 or below. Column (1) uses the estimated baseline score. Column (2) uses the estimated minus the actual baseline score; positive values indicate overestimation. All regressions control for college education, gender, race, income, and age. Robust standard errors are reported in parentheses.

	Average WTP across the three formats (dollars)			
	(1)	(2)	(3)	(4)
Forecast with the material minus forecast without, $\hat{Y}_i(m) - \hat{Y}_i(0)$	0.291***		0.273***	
	(0.059)		(0.060)	
Forecast with the material minus estimated baseline score, $\hat{Y}_i(m) - \hat{Y}_{i,0}$		0.170***	0.096	
		(0.063)	(0.062)	
Forecast with the material, $\hat{Y}_i(m)$				0.466***
				(0.087)
Forecast without the material, $\hat{Y}_i(0)$				-0.247***
				(0.060)
Estimated baseline score, $\hat{Y}_{i,0}$				-0.041
				(0.065)
p-value: depends only on $\hat{Y}_i(m) - \hat{Y}_i(0)$				0.003
p-value: depends only on $\hat{Y}_i(m) - \hat{Y}_{i,0}$				<0.001
Observations	967	967	967	967
R^2	0.039	0.015	0.042	0.052

Table A14: Willingness to Pay and the Three Forecasts

Notes: Table A14 reports OLS regressions of average willingness to pay (WTP, in dollars) across the three formats on forecasts elicited before random assignment. The forecast with the material is averaged across the three formats. Column (1) uses the forecast with the material minus the forecast without the material; Column (2) uses the forecast with the material minus the estimated baseline score; Column (3) uses both; Column (4) uses the three forecasts separately. In Column (4), the first p-value tests the restriction that WTP depends on the forecasts only through the forecast with the material minus the forecast without it: the coefficients on the forecast with and without the material sum to zero, and the coefficient on the estimated baseline score is zero. The second p-value tests the restriction that WTP depends on the forecasts only through the forecast with the material minus the estimated baseline score. The sample is the 967 participants whose three price lists are all consistent. All columns control for college education, gender, race, income, and age. Robust standard errors are reported in parentheses.

	Inconsistent price lists allowed		
	(1) None	(2) At most 1 (main sample)	(3) At most 2
Average WTP on baseline score	0.028	0.051	0.053
	(0.067)	(0.065)	(0.063)
Average WTP on perceived gain	0.291***	0.240***	0.232***
	(0.059)	(0.056)	(0.055)
Endline score on assigned material × average WTP	0.002	0.004	-0.004
	(0.029)	(0.029)	(0.028)
Mean of average WTP (dollars)	1.06	1.10	1.13
Participants	967	1,049	1,075

Table A15: Robustness of the Willingness-to-Pay Results to Inconsistent Price Lists

Notes: Each column uses a different rule for participants with inconsistent price lists. A price list is inconsistent when the participant switches between the two options more than once, or switches only from completing the material to skipping it; that list's WTP is then missing. Column (1) keeps only participants with no inconsistent list; it is the sample for the average-WTP results in the text. Column (2) keeps every participant in the final sample, who have at most one inconsistent list. Column (3) also keeps participants with two. Participants with three inconsistent lists have no WTP, so allowing three gives the same sample as Column (3). Average WTP is the mean of the participant's consistent WTPs (dollars). The first row regresses average WTP on the baseline score. The second row regresses average WTP on the perceived gain, the forecast with the material minus the forecast without it, averaged across formats. The third row reports the coefficient on assigned material × average WTP in Equation (12), with the endline score as the outcome. All regressions control for college education, gender, race, income, and age. Robust standard errors are reported in parentheses.

	(1)	(2)	(3)	(4)
Baseline score	0.113		0.102	0.227
	(0.084)		(0.093)	(0.200)
Actual change in score		-0.058	0.001	0.241
		(0.090)	(0.094)	(0.181)
Received this material		-1.140**	-0.906**	
		(0.456)	(0.437)	
Received this material × actual change		0.157	0.135	
		(0.177)	(0.166)	
Perceived difficulty (0 to 10)			-0.170***	-0.071
			(0.055)	(0.133)
Perceived gain			0.411***	0.354***
			(0.061)	(0.094)
Change in perceived difficulty				-0.220*
				(0.128)
R^2	0.002	0.007	0.087	0.097
Observations	1,016	1,016	1,016	249

Table A16: Mechanisms Behind Willingness to Pay for AI (OLS)

Notes: This table reports OLS regressions of willingness to pay (WTP, in dollars) for the AI format. Column (1) regresses WTP on the baseline number correct. Column (2) regresses it on the change in number correct (endline minus baseline), an indicator for receiving the AI material, and their interaction. Column (3) adds the baseline number correct, perceived difficulty (0–10), and the perceived gain, the forecast of the endline score with the AI material minus the forecast without the material. Columns (1) through (3) use every participant whose price list for the AI format is consistent. Column (4) uses only participants who received the AI material and adds the change in perceived difficulty from before to after the material. Robust standard errors in parentheses.

	(1)	(2)	(3)	(4)
Baseline score	0.035		0.078	0.265
	(0.072)		(0.081)	(0.181)
Actual change in score		0.101	0.116	0.193
		(0.077)	(0.084)	(0.189)
Received this material		-0.094	-0.214	
		(0.418)	(0.409)	
Received this material × actual change		-0.044	-0.015	
		(0.175)	(0.173)	
Perceived difficulty (0 to 10)			-0.167***	-0.178
			(0.055)	(0.154)
Perceived gain			0.275***	0.433***
			(0.058)	(0.121)
Change in perceived difficulty				0.020
				(0.111)
R^2	0.000	0.002	0.041	0.070
Observations	1,021	1,021	1,021	260

Table A17: Mechanisms Behind Willingness to Pay for Video (OLS)

Notes: This table reports OLS regressions of willingness to pay (WTP, in dollars) for the video format. Column (1) regresses WTP on the baseline score. Column (2) regresses it on the actual change in score (endline minus baseline), an indicator for receiving the video material, and their interaction. Column (3) adds the baseline score, perceived difficulty of the video material (0–10), and the perceived gain, the forecast of the endline score with the video material minus the forecast without the material. Columns (1) through (3) use every participant whose price list for the video format is consistent. Column (4) uses only participants who received the video material and adds the change in perceived difficulty from before to after the material. Robust standard errors in parentheses.

	(1)	(2)	(3)	(4)
Baseline score	0.016		0.015	0.082
	(0.081)		(0.090)	(0.190)
Actual change in score		0.074	0.083	-0.008
		(0.088)	(0.096)	(0.173)
Received this material		-0.635	-0.534	
		(0.429)	(0.417)	
Received this material × actual change		-0.185	-0.175	
		(0.173)	(0.168)	
Perceived difficulty (0 to 10)			-0.330***	-0.060
			(0.061)	(0.161)
Perceived gain			0.287***	0.260*
			(0.065)	(0.139)
Change in perceived difficulty				-0.216*
				(0.120)
R^2	0.000	0.008	0.064	0.052
Observations	1,028	1,028	1,028	263

Table A18: Mechanisms Behind Willingness to Pay for Lecture (OLS)

Notes: This table reports OLS regressions of willingness to pay (WTP, in dollars) for the lecture format. Column (1) regresses WTP on the baseline score. Column (2) regresses it on the actual change in score (endline minus baseline), an indicator for receiving the lecture material, and their interaction. Column (3) adds the baseline score, perceived difficulty of the lecture material (0–10), and the perceived gain, the forecast of the endline score with the lecture material minus the forecast without the material. Columns (1) through (3) use every participant whose price list for the lecture format is consistent. Column (4) uses only participants who received the lecture material and adds the change in perceived difficulty from before to after the material. Robust standard errors in parentheses.

	WTP for AI (dollars)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Uses AI in general (z)	0.807***				0.320*				
	(0.166)				(0.165)				
Uses AI for finances (z)		0.860***			0.302*				
		(0.159)			(0.169)				
Believes AI outperforms me (z)			0.488***		0.259				
			(0.173)		(0.162)				
Trusts AI information (z)				1.340***	1.035***				
				(0.168)	(0.188)				
AI usage index (z)						0.874***			
						(0.203)			
AI attitude index (z)						1.214***			
						(0.223)			
AI belief index (z)							2.073***	2.285**	1.937***
							(0.255)	(0.925)	(0.310)
Baseline score								0.121	
								(0.081)	
AI belief index × baseline score								-0.031	
								(0.132)	
Actual change in score									-0.001
									(0.074)
AI belief index × actual change									0.105
									(0.119)
Constant	1.873***	1.866***	1.872***	1.875***	1.879***	1.881***	1.880***	1.082*	1.888***
	(0.154)	(0.154)	(0.155)	(0.150)	(0.150)	(0.150)	(0.150)	(0.560)	(0.183)
R^2	0.026	0.030	0.009	0.072	0.081	0.074	0.073	0.075	0.074
Observations	1,016	1,016	1,016	1,016	1,016	1,016	1,016	1,016	1,016

Table A19: Relationship Between AI Beliefs and WTP for AI

Notes: This table reports regressions of willingness to pay for AI on specific belief components (AI usage, financial AI use, perceived AI superiority, trust in AI), on composite belief indices, and on interactions between beliefs and either baseline performance or actual improvement. Standard errors in parentheses.

	(1) AI belief index (z)	(2) AI belief index (z)
Baseline score	-0.006	
	(0.012)	
Actual change in score		-0.015*
		(0.009)
R^2	0.000	0.003
Observations	1,049	1,049

Table A20: AI Belief Index and Measures of Ability

Notes: This table reports regressions of the AI Belief Index on participants' baseline quiz performance (Column 1) and actual improvement (Column 2). Standard errors in parentheses.

	(1) AI belief index (z)	(2) AI belief index (z)
College degree	0.134***	
	(0.045)	
Household income (10,000s of dollars)		0.009**
		(0.004)
Observations	1,049	1,049
R^2	0.009	0.005
Demographic controls	No	No

Table A21: Demographic Correlates of AI Beliefs

Notes: Table A21 reports regressions of the *AI Belief Index*—a standardized index combining participants' AI usage, attitudes, trust, and perceived comparative advantage—on demographic variables. Column (1) reports a specification with a college indicator; Column (2) uses income (midpoint of self-reported income bin). Higher values of the AI Belief Index indicate more positive attitudes toward using AI tools. Standard errors are reported in parentheses.

	(1) WTP for AI	(2) WTP for Video	(3) WTP for Lecture	(4) Average WTP
Baseline score	0.125	0.073	-0.002	0.046
	(0.105)	(0.091)	(0.102)	(0.079)
Actual change in score	0.056	0.121	0.050	0.056
	(0.096)	(0.087)	(0.095)	(0.076)
Perceived difficulty (0 to 10)	-0.134**	-0.172***	-0.371***	-0.124*
	(0.062)	(0.060)	(0.072)	(0.067)
Perceived gain	0.400***	0.297***	0.332***	0.288***
	(0.072)	(0.065)	(0.075)	(0.063)
AI belief index (z)	2.024***	0.258	0.307	0.830***
	(0.296)	(0.217)	(0.239)	(0.195)
Pseudo-R^2	0.025	0.008	0.012	0.011
Observations	1,016	1,021	1,028	967

Table A22: Determinants of WTP for Educational Materials (Tobit)

Notes: Tobit regressions censor WTP at \$-11 and \$11. Columns correspond to WTP for AI, video, lecture, and average WTP. Coefficients represent latent WTP. The perceived gain is the forecast of the endline score with the format (averaged across formats in the last column) minus the forecast without the material, both elicited before random assignment. All regressions control for college education, gender, race, income, and age. Robust standard errors in parentheses.

	(1) Group mean	(2) Comparison mean	(3) Difference (1) − (2)	(4) p-value
Panel A. Would pay for the AI or video format but not the lecture (1), rest of the sample (2)				
Baseline score (out of 10)	6.67	6.56	0.11	0.422
Female (%)	60.5	54.8	5.7	0.135
Age (years)	43.3	42.2	1.1	0.263
College degree (%)	70.2	68.5	1.7	0.623
Household income (thousands of dollars)	62.9	65.6	-2.7	0.464
White (%)	70.2	66.1	4.1	0.246
Took a financial literacy course (%)	33.0	31.2	1.8	0.625
Own financial ability relative to AI (1–5)	2.43	2.50	-0.07	0.391
Certainty about own baseline estimate (0–100)	55.9	55.8	0.1	0.947
Overestimated own baseline score (%)	28.8	29.1	-0.3	0.935
Actual change in score, assigned to a material	1.66	1.58	0.08	0.664
Participants	215	752		
Participants assigned to education	167	563		
Panel B. Would not pay for the lecture: AI format only (1), video format only (2)				
Baseline score (out of 10)	6.47	6.89	-0.42	0.215
Female (%)	64.1	55.3	8.8	0.369
Age (years)	41.8	41.7	0.1	0.974
College degree (%)	67.9	76.3	-8.4	0.342
Household income (thousands of dollars)	65.3	58.0	7.3	0.391
White (%)	66.7	76.3	-9.6	0.275
Took a financial literacy course (%)	38.5	34.2	4.3	0.657
Own financial ability relative to AI (1–5)	2.28	2.84	-0.56	0.009
Certainty about own baseline estimate (0–100)	49.0	62.9	-13.9	0.007
Overestimated own baseline score (%)	29.5	26.3	3.2	0.722
Actual change in score, assigned to a material	1.57	1.48	0.09	0.835
Participants	78	38		
Participants assigned to education	54	31		

Table A23: Characteristics of the Participants the AI and Video Formats Bring In

Notes: Table A23 compares the participants the AI and video formats bring in with other participants. The sample is the 967 participants whose willingness to pay we observe for all three formats. Panel A compares participants with a negative willingness to pay for the lecture and a positive willingness to pay for the AI or video format (column 1) with all other participants (column 2). Panel B compares two groups of participants with a negative willingness to pay for the lecture: those with a positive willingness to pay for the AI format and a negative one for the video format (column 1), and those with a positive willingness to pay for the video format and a negative one for the AI format (column 2). Column 3 reports the difference in means, and column 4 reports its p-value from a regression of the row variable on a group indicator with heteroskedasticity-robust standard errors. Own financial ability relative to AI is the answer to "How would you rate your financial decision-making ability compared to AI?" on a scale from 1 (much worse than AI) to 5 (much better than AI). Certainty is the answer, from 0 to 100, to how certain the participant is about their estimate of their own baseline bonus, which equals their baseline number correct. Overestimated own baseline score indicates an estimated baseline bonus above the actual one. The change in number correct (post minus pre) is measured for participants randomly assigned to an education format; the last row of each panel reports their number.